



BASE24 PLUS ISO Interface Manual

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Document History

12JUN2001	chh/174
Symptom:	Release 6.0 Enhancements.
Problem:	None.
Fix:	Created Release 6.0 manual for PLUS ISO Interface.
Dependency:	Restore PISOMAN.
Reference:	WO #000101-1 (General Release 6.0)
24JUN2002	DOO/1693
Symptom:	Transaction Security Services for DKM Enhancement.
Problem:	None.
Fix:	Added verbiage to support Transaction Security Services for DKM Enhancement.
Dependency:	Restore release 6.0 files, modify the appropriate CUSTMACS flags, and run MAKE.
Reference:	WO #020217-03
07FEB2003	TMH/173
Symptom:	Plus Shared Deposit Enhancement
Problem:	This enhancement adds support for ATM deposit, ATM debit adjustment of deposit, ATM credit adjustment of deposit, POS debit adjustment of return and POS credit adjustment of return transactions.
Fix:	1. Added verbiage to support the new processing codes and correction reason codes. Sections modified: Supported Transaction Types PLUS Adjustment/Chargeback Screen 2. Removed all verbiage for PLUS ISO Visa Cash as this bindable extension is no longer supported.
Dependency:	Apply fixes to RP01IPLS, RP09IPLS, RQIPACS, PISIS, PISOG and PISOS and run MAKE. Replace PISOMNWD and PISOUPTD.
Reference:	WO #021211-01

14OCT2003 cal/283
Symptom: PLUS ISO October 2003 Mandates Enhancement
Problem: None.
Fix: Updated Correction Codes list to include new values 2401, 2402 and 2403.
Dependency: Apply fixes to RQIPACS and PISIS, run MAKE. Replace PISOMNWD and PISOUPDT.
Reference: WO #037025-01

15MAR2004 HUAC
Symptom: PLUS ISO 3DES/Triple-Single DES Enhancements
Problem: None.
Fix: Added 3DES/Triple-Single DES supports.
Modified Section:
INSTALLATION STEPS
LCONF ASSIGNS AND PARAMS
PIN ENCRYPTION
New Section:
DYNAMIC KEY MANAGEMENT
KEY6 SETTING
Dependency: Apply fixes to PISODDLs, PISOG and PISOS and run MAKE. Replace PISOLOGM, PISOMNWD and PISOUPDT.
Reference: WO #020614-01.

29NOV2004 HAYEST
Symptom: PLUS ISO 04.3 Mandates and Miscellaneous Enhancements
Problem: None.
Fix: 1) Modified the reason code values on the PAC Screen to Reflect the current valid values for chargebacks and adjustments. Added 3 additional fields to the PAC screen.
2) Removed support for administrative types 1, 2, 3, and 4 from section PAM Screen. Removed menu entries 1, 2, 3, and 4 from the PAM Menu Screen. Deleted Message Type 1 Screen, Message Type 2 Screen, Message Type 3 Screen, and Message Type 4 Screen from the PAM Screen section.
3) Modified section Installation Steps, replacing "BOJ" with "AUTOMATIC".
4) Removed assign PLF from section LCONF Assigns and Params.
5) Removed PCS support from section Initialization.
6) Modified section Switch Token to reflect the current definition of the Switch Token.
Dependency: Apply fixes to PISIFM, PISIM, PISIMM, PISIS, PISOD, PISODDLM, PISODDLs, PISO FM, PISOG, PISOM, PISOMM, PISOS, RP01IPLM, RP09IPLM, RQIPACM, RQIPACS, RQIPAMM, RQIPAMS, RQIPPFM, RQIPPFs, RQPISFM, RQPISFS, RQPISOM, RQPISOS, RQPMLFM, RQPMLFS, SCRNPAC, SCRNPAM, SCRNPFF, SCRNPISF, SCRNPISO, SCRNPMLF, SVIPAMM, SVIPAMS, SVIPPFM, SVIPPFs, SVPISFM, SVPISFS, SVPMLFM, SVPMLFS, SWPISIM, and SWPISOM and run MAKE. Replace PISOLOGM, PISOMNWD, and PISOUPDT.
Reference: WO #040728-01

03JAN2005 HAYEST
Symptom: PLUS ISO 05.1 Mandates
Problem: None.
Fix: Removed support for the VISA DEBIT ACQUIRER INST field on the PLUS Proprietary Member Center File (PPF) screen. Renamed the INTERLINK ACQUIRER INST field to UNIQUE ACQUIRER ID on the PPF screen. Added quasi-cash POS transaction support. Removed chargeback reason code 2681 from the list of valid correction reason codes for chargeback on the IPAC screen. Added isa-fee and fee-pgm to PISO-SWI-TKN-DATA definition. Modified section: PPF Screen 1
 SWITCH TOKEN
 SUPPORTED TRANSACTION TYPES
 PLUS ADJUSTMENT/CHARGEBACK SCREEN
Dependency: Apply fixes to PISIS, PISODDLS, PISOS, RQIPACS and SCRNIPPF and run MAKE. Replace PISOMNWD and PISOUPDT.
Reference: WO #0401122-01

21SEP2005 linkec
Symptom: PLUS ISO October 2005 Mandates.
Problem: None.
Fix: Updated Appendix B with tokens added for this enhancement. Added mCCR-fee and crd-lvl-prod-id-val to PISO-SWI-TKN-DATA definition.
Dependency: Apply fixes to PISIS, PISODDLS, PISOG and PISOS. Replace PISOLOGM, PISOMNWD and PISOUPDT.
Reference: WO #050627-06

26SEP2005 hayest
Symptom: Visa/Plus ATM Interchange Reimbursement Fees Support
Problem: Visa is modifying the Visa and Plus ATM Interchange Reimbursement Fee structure to provide tiered reimbursement levels. Each ATM Acquirer Member will be assigned a unique Acquirer ID to designate the appropriate Tier (1 or 2). This Acquirer ID will be mapped to data element 125.1 in all external messages outbound from the interface to the network.
Fix: The Acquirer Information File (AIF) has been created to support network supplied acquirer data that will be mapped from the file to the external message by the interchange interfaces. An AIF requester/server has been created on new subvolume SW60AIF in order to enable file maintenance functions on the AIF. The interface has been modified to create and initialize an AIF extended memory table and to map data from records within the extended memory table to data elements 32 and 125.1, if configured to do so. The interface has been modified to support the AIF instead of the PPF. The PPF will no longer be supported. A new utility has been created in HISWUTLS to retrieve a record from the AIF extended memory table for the BASE24 interchange interfaces. Modified sections: CONTROL FILES, PLUS ISO INTERFACE PROCESS FILES, INSTALLATION STEPS, LCONF ASSIGNS AND PARAMS, INITIALIZATION and PLUS OPERATIONS SCREENS, deleted section PPF SETTINGS and added section AIF SETTINGS.
Dependency: Restore new subvolume SW60AIF. Apply fix to BACOUTLS, CUSTMACS, FLGSDOC, HISWUTLS, SWIFM, SWIM, SWIMM, PISIM,

PISIS, PISODDLM, PISODDLS, PISOG, PISOM, PISOMM, PISOS, RQPISOS, SCRNPISO and SWPISOM and run MAKE. Replace PISOLOGM, PISOMNWD and PISOUPDT. Purge files RQIPPFM, RQIPFFO, RQIPFFS, SCRNIPIPF, SVIPPF, SVIPPFM and SVIPFFS from subvolume SW60PISO. Restore new PPF conversion files CNVPPFM, CNVPPFR and CNVPPFS. MAKE the PPF conversion program using CNVPPFM. Run the PPF conversion using CNVPPFR.

Reference: WO #050830-01

06OCT2005 gruberg

Symptom: AIF records cannot be retrieved using a prikey of term-ln and other prikey fields wildcarded.

Problem: Customer requirements call for an AIF record to be retrieved by term-ln.

Fix: Added searches to retrieve and AIF record with any combination prikey fields wildcarded.

Section modified: AIF Settings

Dependency: Apply fixes to HISWUTLS and RQAIFS and run MAKE. Replace PISOMNWD and PISOUPDT.

Reference: Case #401407

15DEC2005 hayest

Symptom: Issuer and Acquirer Discretionary Data Suppression Enhancement.

Problem: None.

Fix: Added support for LCONF parameters SWI-ACQ-DISCR-DATA-SUPPRS and SWI-ISS-DISCR-DATA-SUPPRS and SWI-ACQ-CVD2-DATA-SUPPRS.

Section modified: LCONF ASSIGNS AND PARAMS

Dependency: Apply fixes to PISOG and PISOS. Replace PISOMNWD and PISOUPDT. Run MAKE.

Reference: WO #050305-10

04MAY2006 hayest

Symptom: Some customers support devices that participate in several networks. These devices require a different Sponsor ID for each network.

Problem: The AIF does not support the switch FIID and therefore is unable to support devices that participate in several networks.

Fix: Modified documentation to support the Switch FIID as part of the primary key of the AIF.

Section modified: AIF SETTINGS

Dependency: Apply fixes to HISWUTLS, AIFDDL, RQAIFS, SCRNAIF, SVAIFS, PISIS and PISOS. Run MAKE. Replace PISOMNWD and PISOUPDT. Restore new AIF conversion files AIF0506M, AIF0506R and AIF0506S. MAKE the AIF conversion programs using AIF0506M. Run the AIF conversion using AIF0506R.

Reference: WO #060107-03.

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PREFACE

This manual describes the PLUS ISO Interface process and PLUS ISO Interlink enhancement.

The PLUS ISO Interface process is used to establish a communications link between a BASE24 network and the PLUS Interchange.

The PLUS ISO Interlink Gateway enhancement enables members to participate in a gateway to Interlink. This gateway allows PLUS members to acquire transactions for Interlink issuers and authorize transactions from Interlink acquirers.

The Release 6.0 PLUS ISO Interface has been enhanced so that it may support dynamic key management via Transaction Security Services in a Release 6.0 network environment. Refer to the "BASE24 Release 6.0 Implementation Guide" and the "BASE24 6.0 Classic Install Guide" for details on implementing and installing Transaction Security Services.

This manual is intended as a reference source for persons responsible for the interface between BASE24 and PLUS. It provides a comprehensive description of all the functions performed by the PLUS ISO Interface and PLUS ISO Interlink Gateway process involving incoming and outgoing message traffic for both acquirer and issuer processors.

This document is located on the SW60PISO subvolume in a file called PISOMNWD. It may be printed using and copied as needed by ACI customers who have purchased ACI's PLUS ISO Interface. The document may also be ordered, at a cost, through ACI Publications.

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Section 1 INTRODUCTION

PLUS ISO INTERFACE PROCESS

This section introduces the PLUS ISO Interface process and provides an overview of PLUS ISO Interface functionality. This section also identifies the BASE24 files used by the PLUS ISO Interface process.

THE PLUS INTERCHANGE

The PLUS Interchange serves as a link between two networks. It allows a cardholder on one network to initiate a transaction at a terminal belonging to the other network.

PLUS connects an acquirer with an issuer. Both the acquirer and the issuer must conform to PLUS standards. These standards include sending and accepting the appropriate form of the PLUS external message and providing the correct message flows.

THE PLUS ISO INTERFACE PROCESS FUNCTIONS

The PLUS ISO Interface process is responsible for maintaining the relationship between BASE24 and PLUS. In order to maintain this relationship, the PLUS ISO Interface process must perform a variety of functions. These functions include:

- Facilitating message exchange between BASE24 and PLUS.
When BASE24 receives an inbound transaction from PLUS, the PLUS ISO Interface process translates the message into the appropriate internal message and sends the message to a BASE24 Authorization process. When BASE24 receives a transaction and sends it to PLUS, the PLUS ISO Interface process translates the message from the internal message format to the external message format and forwards the message to PLUS.
- Providing store-and-forward capabilities.
When the link between BASE24 and PLUS is down, the PLUS ISO Interface process stores non-interactive messages--messages not requiring an immediate response in order to complete a transaction--intended for PLUS in its Store-and-Forward File (SAF). When the link is restored, the messages are retrieved from the file and sent to PLUS.
- Facilitating settlement between BASE24 and PLUS.
The PLUS ISO Interface processes transactions based on the PLUS Interchange's reporting day. Since PLUS' cutover rarely occurs at the same time as the BASE24-atm cutover, the reporting day and the transactions to be settled will differ between the two. The PLUS ISO Interface process produces a set of interchange reports to reconcile these points of view. These reports allow the BASE24 participants to reconcile the settlement positions of PLUS and BASE24.
- Managing communications between BASE24 and PLUS.
The PLUS ISO Interface process sends and receives network management messages to determine the status of the link between BASE24 and PLUS. Network management messages include logon, logoff, echo test, and key management messages.

THE PLUS ISO INTERLINK GATEWAY PROCESS FUNCTIONS

Basic POS support has been added to the PLUS ISO Interface. The PLUS ISO Interlink Gateway Extension is a bindable module for the PLUS ISO Interface that provides Interlink support to the PLUS ISO Interface. The PLUS ISO/Interlink Gateway module is bound together with the PLUS ISO Interface to create the PLUS ISO/Interlink Interface.

Only one process is required to process PLUS transactions, all existing gateway traffic, and Interlink transactions. This means that customers do not have to set up a separate process specifically to handle gateway traffic to Interlink. Acquirers may initiate Interlink's Resubmissions in two ways. One method is where the merchant forwards the paper draft to the acquirer and the acquirer manually enters the transaction into the interchange. This is done via the PLUS Adjustment/Chargeback (PAC) Screen. The second method is that the merchant sends the resubmission online. This method is not fully supported by the PLUS ISO/Interlink Gateway. The RSAF has been added to store these transactions for resubmission, but an RPQ is required to initiate the resubmission process.

MERCHANDISE CREDIT TRANSACTIONS

Merchandise credit transactions facilitate the crediting of a cardholder's account when merchandise is returned. This transaction will reverse all or a portion of the original purchase amount as well as the interchange fees paid in connection with the purchase.

The transaction is initiated at the point of sale, although not necessarily at the same location, and may occur several days after the date of original purchase. The card must be read electronically (i.e., track2) and the PIN must be entered at the time of the transaction. The merchandise credit amount must be equal to or less than the amount of the original purchase. The merchandise credit transaction must also contain the date of the purchase transaction being credited.

Merchandise credit transactions include online merchandise credits, cancellation of merchandise credits, reversals of merchandise credits, and chargebacks and representations of merchandise credits. Store-and-forward merchandise credits are also supported by Interlink but are not currently supported by BASE24 acquirers.

Use of the merchandise credit transaction is optional for acquirers. All issuers are required to support the new transaction by October 1, 1994. All issuers and participating acquirers must certify to support the merchandise credit enhancement.

GOOD FAITH COLLECTION GUIDELINES

Good faith collection guidelines have been developed to offer members a consistent approach to processing their good faith collections. Good faith collections typically involve transactions that cannot be processed according to standard Interlink processing procedures (e.g., time frames have expired). The existing adjustment message will be used in the adjustment message to identify it as a good faith collection.

Issuers must support the settlement and processing of good faith collection through the Interlink switch. Issuers are not required to certify for this change, but online test transactions will be available. Issuers must ensure that edits currently applied to adjustments (e.g., allowable time frame between original and adjustment) only apply if appropriate. Processing good faith collection is optional for acquirers. Again, acquirers are not required for this change, but online test transactions will be available.

DAILY COLLECTION OF INTERCHANGE FEES

Interlink interchange fees will be included in the current daily settlement and reporting process. In support of this enhancement, an optional service will be introduced to provide online funds transfer advice messages to V.I.P. ISO format issuers and acquirers on a daily basis. PLUS does not pass these messages on to processors going through PLUS but does provide the information in their reports.

SUPPORTED TRANSACTION TYPES

The PLUS ISO Interface process provides support for BASE24-atm transactions. These include:

- Withdrawals
- Transfers
- Balance inquiries
- Deposits

The PLUS ISO Interface process also provides support for BASE24-pos transactions. These include:

- Purchases
- Debit Adjustments
- Purchase Returns
- Mail/Phone Orders
- Balance Inquiries
- Quasi-Cash (issuer only)

Any of the above transactions can be received from or sent to PLUS for authorization. The PLUS ISO Interface process can also function on behalf of both issuers and acquirers.

An issuer processes transactions on behalf of the card-issuing financial institution. In the case of PLUS, this means that the PLUS ISO Interface process receives transactions from PLUS that require authorization by a BASE24 Authorization process.

An acquirer accepts transaction requests that must be sent to another network for authorization. In the case of PLUS, this means that the PLUS ISO Interface process forwards transactions initiated on the BASE24 network to PLUS for authorization.

TRANSACTION PROCESSING

The following paragraphs provide an overview of transaction flows for normal and link-down processing. They also identify the message set (different than the transaction set) supported by PLUS.

NORMAL MESSAGE FLOW

When a message from PLUS is received by the PLUS ISO Interface process, it is translated from the external message into the appropriate internal message format. The internal message is then sent to the appropriate BASE24 Authorization process. When the message has been processed, the authorizing entity sends an internal response to the PLUS ISO Interface process. Using information from the internal message, the PLUS ISO Interface process formats an external response message and sends it to PLUS.

When a message from BASE24 is received by the PLUS ISO Interface process, it is translated from the internal message into the external message format. The external message is then sent to the interchange for authorization. When the message has been processed, PLUS sends an external response to the PLUS ISO Interface process. Using information from the external message, the PLUS ISO Interface process formats an internal response message and sends it to BASE24.

LINK-DOWN MESSAGE FLOW

If the link to PLUS is down and an alternate destination is available, the link-down situation is handled by using alternate routing to send the transaction to the alternate destination. Otherwise, the PLUS ISO Interface process responds to a link-down situation by denying all interactive transactions. All non-interactive messages (i.e. reversals) will be logged to the Store-and-Forward File (SAF).

SUPPORTED MESSAGES TYPES

The PLUS ISO Interface process supports the following internal and external message types:

Type	Description	Internal	External
0100	Authorization Request		X
0110	Authorization Response		X
0120	Authorization Advice		X
0130	Authorization Advice Acknowledgement		X
0200	Financial Request	X	X
0210	Financial Response	X	X
0220	Financial Advice		X
0230	Financial Advice Acknowledgement		X
0400	Acquirer Adjustment Request		X
0402	Issuer Chargeback Request		X
0410	Acquirer Adjustment Response		X
0412	Issuer Chargeback Response		X
0420	Acquirer Reversal/Override Advice	X	X
0420	Acquirer Adjustment Advice		X
0422	Issuer Chargeback Advice		X
0430	Acq. Reversal/Override/Adjustment Ack.		X
0432	Issuer Chargeback Acknowledgement		X
0480	Acquirer Adjustment Reject Advice		X
0482	Issuer Chargeback Reject Advice		X
0490	Acquirer Adjustment Reject Ack.		X
0492	Issuer Chargeback Reject Advice Ack.		X
0600	Administrative Message Request		X
0610	Administrative Message Response		X
0620	Administrative Message Advice		X
0630	Administrative Message Acknowledgement		X

Type	Description	Internal	External
0720	Acquirer Fee Advice		X
0722	Issuer Fee Advice		X
0730	Acquirer Fee Acknowledgement		X
0732	Issuer Fee Acknowledgement		X
0800	Network Management Request		X
0810	Network Management Response		X
5400	Internal Adjustment	X	

NETWORK MANAGEMENT MESSAGES

Network management messages are used to manage the operational status of the communications lines between BASE24 and PLUS. These messages are sent as 0800 messages and require 0810 messages in response. BASE24 supports the following types of network management messages.

- Logon messages
- Echo test messages
- Logoff messages
- Key management messages

BASE24 FILES

The PLUS ISO Interface process uses several files. These files can be divided into control files and transaction files and are identified below.

CONTROL FILES

Control files contain configuration information and parameters that determine how the PLUS ISO Interface process performs certain functions. The control files used by the PLUS ISO Interface process include:

- Acquirer Processing Code File (APCF)
- Enhanced Interchange Configuration File (ICFE)
- Issuer Processing Code File (IPCF)
- Key File (KEYF)
- Key6 File (KEY6)
- Logical Network Configuration File (LCONF)
- Network Environment Source File (NEFS)
- Acquirer Information File (AIF)
- Token File (TKN)

For more information on these files and how they are used by the PLUS ISO Interface process, see section 2.

TRANSACTION FILES

Transaction files are used to store information about individual transactions processed through the PLUS ISO Interface process. The transaction files used by the PLUS ISO Interface process include:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)
- PLUS Administrative Message File (PAM)

For more information on the ILF and SAF files, see the "BASE24 Files and Record Structures Manual." For more information on the PAM, see section 6.

DYNAMIC KEY MANAGEMENT

PLUS generates new keys following successful processor signon at startup and after scheduled outages. At a minimum, key changes occur for every processor once every 24 hours.

If LCONF parameter DES-TRIPLE-SINGLE is 'Y', the interchange sends a double length PIN encryption key in the external message. The KEY6 File (KEY6) is used instead of the Key File (KEYF).

If LCONF parameter SECURE-DEV-TYP is 'TSS', the interchange may send a single or double length PIN encryption key in the external message. The Key File (KEYF) is used.

If LCONF parameter DES-TRIPLE-SINGLE is 'N' or does not exist and LCONF parameter SECURE-DEV-TYP does not contain the value 'TSS', the interchange sends a single length PIN encryption key in the external message. The KEYF is used.

The Key Exchange Key (KEK) in the KEYF or KEY6 must contain the security module-encrypted form of the Key Exchange Key (KEK) used when inbound and outbound PIN keys are exchanged using Dynamic Key Management (DKM). The PLUS ISO Interface does not initiate key changes automatically. The processor is allowed to initiate a key change request to the interchange by issuing a NEW KEY command from BASE24's Network Control Supervisor (NCS) screen or from the NCPCOM prompt. Refer to the NETWORK CONTROL COMMANDS section of this manual for more details. After the interchange responds to the key change request, it starts the key change process to establish the new keys for the processor.

PLUS sends a key change to the PLUS ISO Interface with the new Pin Encryption Key (KPE) encrypted under the first variant of the KEK.

After receiving the new PIN Encryption Key (KPE) the PLUS ISO Interface begins using the new PIN Encryption Key (KPE) for the encryption/decryption of the cardholder's PIN.

PIN ENCRYPTION

The PLUS ISO Interface process supports three PIN encryption options:

- Clear PIN processing. PINs are handled in the clear. They are not encrypted.
- Hardware encrypted PINs. A security module is accessed for PIN processing. PINs are encrypted under the secure keys of a security module. A PIN is encrypted at the point of entry and remains encrypted until it is received by the issuer, (the final destination), for verification. The PIN never appears in clear text, except at the point of entry and at the final destination. The security modules that can be used include the Atalla Security Module, the Atalla High Performance Security Module, and the Racal Guardata Security Module.
- Software encrypted PINs. PINs are encrypted via software code, using a DES (Data Encryption Standard) algorithm. No security module is in use. The PINs are in clear text within the processor itself. The PLUS ISO Interface process can be set up to use software encryption/decryption even when PLUS is set up to use hardware on their end.

TRIPLE TO SINGLE DES ENCRYPTION

The PLUS ISO Interface supports triple to single DES encryption where PINs may be translated from encryption under a double length key (3DEA) to encryption under a single length key (DEA) or translated from DEA to 3DEA. Triple to single DES encryption is supported when translating PINs in the following situations:

- Internal clear PIN to external PIN encrypted in hardware
- Internal PIN encrypted in hardware to external PIN encrypted in hardware
- External PIN encrypted in hardware to internal PIN encrypted in hardware

If LCONF parameter DES-TRIPLE-SINGLE is 'Y', this indicates that the interface translates PINs from being single DES encrypted inside BASE24 to triple DES encrypted when sent to the interchange and translates PINs from being triple DES encrypted when received from an interchange to being single DES encrypted inside BASE24. The KEY6

File (KEY6) is used instead of the Key File (KEYF) in order to store and process the double length PIN encryption keys. The KEY6 supports double length PIN encryption and key encryption keys. The intermediate keys remain single length within the KEY6.

TRIPLE DES ENCRYPTION VIA SECURITY SERVICES

The PLUS ISO Interface supports triple DES encryption using Transaction Security Services only in a Release 6.0 network environment. PINs may be translated from encryption under a single or double length key to encryption under a single or double length key.

Triple DES encryption using Transaction Security Services is supported when translating PINs in the following situations:

- Internal clear PIN to external PIN encrypted in hardware
- Internal PIN encrypted in hardware to external PIN encrypted in hardware

- External PIN encrypted in hardware to internal PIN encrypted in hardware

The interface translates PINs from being single or triple DES encrypted inside BASE24 to single or triple DES encrypted when sent to the interchange and translates PINs from being single or triple DES encrypted when received from an interchange to being single or triple encrypted inside BASE24.

LCONF parameter DES-TRIPLE-SINGLE must not exist or may exist with the value 'N' if the PLUS ISO Interface is to support triple DES encryption using Transaction Security Services. The Key File (KEYF) is utilized. LCONF parameter SECURE-DEV-TYP must have the value 'TSS'. LCONF assign SECURE-DEV-1 must specify the name of the primary IAS process and may be process specific. LCONF assign SECURE-DEV-2 may specify the name of the secondary IAS process and may be process specific.

SINGLE DES ENCRYPTION

If LCONF parameter DES-TRIPLE-SINGLE does not exist or has the value 'N' and LCONF parameter SECURE-DEV-TYP does not have the value 'TSS', single DES encryption is supported by the PLUS ISO Interface and the KEYF is used instead of the KEY6.

MESSAGE FORMATS

The PLUS ISO Interface process routes messages using two different formats-- internal or external.

The internal message format is used whenever information is routed between the PLUS ISO Interface process and the rest of BASE24. The internal message is called the BASE24-atm Standard Internal Message (STM). For more information on the STM, see the "BASE24-atm System Description Manual."

The external message format is used whenever information is routed between the PLUS ISO Interface process and the Interchange. A message sent in the external message format is called a Standard External Message (SEM).

An internal PATHWAY message is used whenever information is routed between the BASE24 CRT operations screens and the PLUS ISO Interface process. The following types of messages are generated from BASE24 CRT operations screens:

- Acquirer adjustments
- Issuer chargebacks
- Administrative messages

DATA COMMUNICATION LINES

The PLUS ISO Interface process supports bisynchronous multipoint (3270) and bisynchronous point-to-point (2780/3780) data communications lines between BASE24 and PLUS (SNA is not currently supported). For more information about configuring the data communications lines used by the PLUS ISO Interface process, see section 2.

TRANSACTIONAL INTERFACE MONITORING (T.I.M.)

This Interchange Interface supports the Transactional Interface Monitoring (T.I.M.) Enhancement. This enhancement provides the ability to monitor, in real-time, the actual transaction health between BASE24 and the interchange. T.I.M. provides tracking of more than a dozen transaction conditions/counters at the process level. Depending upon the particular interface, this enhancement supports issuer and acquirer traffic for both BASE24-atm and BASE24-pos transactions. This enhancement requires ACI's NET24-OM (XN-NT100) and Compaq's Measure product.

TRANSMISSION DATE AND TIME CONSIDERATIONS

The PLUS ISO Interface uses the GUARDIAN call to JULIANTIMESTAMP to fill in the transmission date and time (field 7) in request messages outbound to PLUS. The transmission date and time is the date and time assigned by the originating institution, expressed in Greenwich Mean Time (GMT). An incorrect GUARDIAN GMT offset will cause the transmission date and time to be set incorrectly in outbound requests.

The GUARDIAN GMT offset value is the difference in hours between the location of the Tandem processor and Greenwich and is configured in the ALLPROCESSORS section of Tandem's CONFTEXT. The CONFTEXT file is used as input to the sysgen program. If the offset is not set correctly, the call to JULIANTIMESTAMP will result in an incorrect time setting in field 7.

A Tandem SETTIME command will NOT adjust the GUARDIAN GMT offset, only the system time. The use of Tandem's USA66 parameter will adjust both the system time and the GMT offset automatically during the fall and spring time changes. ACI recommends the verification of the GUARDIAN GMT offset value and the use of the USA66 parameter. On more current releases of GUARDIAN, C20 or higher, the GMT offset value may be modified using COUP. However, after the change is made, the system must be cold loaded to incorporate the new value.

LOG MESSAGE DOCUMENTATION

Log messages generated by the PLUS ISO Interface process are documented in the SW60PISO subvolume in a file named PISOLOGM.

Section 2 INSTALLATION AND CONFIGURATION

INTRODUCTION

This section lists the files that reside on the SW60PISO subvolume. This section also lists the steps required to install the PLUS ISO Interface process. In addition, it provides guidelines for configuring the required control files.

PLUS ISO INTERFACE PROCESS FILES

The following is a list of all files on the SW60PISO subvolume and a brief description of each file.

PISO	- EXECUTABLE OBJECT FOR PLUS ISO INTERFACE
PISOCOB	- DDL COB FILE USED BY PLUS ISO REQUESTERS, SERVERS, AND REPORTS
PISOD	- DUMMY PROCEDURE FILE
PISODDLM	- MAKE FILE FOR PLUS ISO DDL'S
PISODDLS	- DDL SOURCE FILE USED TO CREATE DDLCOB, DDLFUP,
PISOE -	- EXTERNALS FOR PISOS SOURCE PISOFM MAKE CONFIGURATION FILE FOR PLUS ISO INTERFACE
PISOFUP	- DDL FUP FILE USED TO CREATE PLUS ISO FILES: PAM, AIF, and SAF AND DDLTAL
PISOG	- GLOBALS FOR PLUS ISO INTERFACE
PISOLOGM	- LOG MESSAGE MANUAL FOR PLUS ISO INTERFACE
PISOM	- MAKE FILE FOR PLUS ISO INTERFACE
PISOMNWD	- WORD DOCUMENT FOR THIS MANUAL
PISOMM	- MAKE MACRO DOCUMENTATION FILE
PISOO	- NON-EXECUTABLE OBJECT FROM PISOS COMPILE
PISOS	- SOURCE FILE FOR PLUS ISO INTERFACE
PISOTAL	- DDL TAL FILE SOURCED IN BY PISOS
*PISOUPTD	- ENHANCEMENT UPDATE DOCUMENTATION
RP01IPL	- EXECUTABLE OBJECT FOR PLUS ISO REPORT 1
RP01IPLM	- MAKE FILE FOR PLUS ISO REPORT 1
RP01IPLS	- SOURCE FILE FOR PLUS ISO REPORT 1
RP09IPL	- EXECUTABLE OBJECT FOR PLUS ISO REPORT 9
RP09IPLM	- MAKE FILE FOR PLUS ISO REPORT 9
RP09IPLS	- SOURCE FILE FOR PLUS ISO REPORT 9
RQIPACM	- MAKE FILE FOR PLUS ISO ADJUSTMENT/CHARGEBACK REQUESTER
RQIPACO	- PSEUDO TARGET FOR RQIPACS
RQIPACS	- SOURCE FILE FOR PLUS ISO ADJUSTMENT/CHARGEBACK REQUESTER
RQIPAMM	- MAKE FILE FOR PLUS ISO ADMINISTRATIVE MESSAGE REQUESTER
RQIPAMO	- PSEUDO TARGET FOR RQIPAMS
RQIPAMS	- SOURCE FILE FOR PLUS ISO ADMINISTRATIVE MESSAGE REQUESTER
RQPMLFM	- MAKE FILE FOR PLUS MAINTENANCE LOG FILE REQUESTER
RQPMLFS	- PLUS MAINTENANCE LOG FILE REQUESTER
RQPISOM	- MAKE FILE FOR PLUS ISO ICFE SWITCH DEPENDENT SCREEN REQUESTER
RQPISOO	- PSEUDO TARGET FOR RQPISOS

RQPISOS - SOURCE FILE FOR PISO ICFE SWITCH DEPENDENT SCREEN
 REQUESTER
SCRNIPAC - PATHWAY SCREEN FOR IPAC FILE
SCRNIPAM - PATHWAY SCREEN FOR IPAM FILE
SCRNPISO - PATHWAY SCREEN SOURCE FILE
SCRNPMLF - PATHWAY SCREEN FOR PLUS FILE MAINTENANCE LOG FILE
SVIPAM - OBJECT FILE FOR PLUS ISO ADMINISTRATIVE MESSAGE SERVER
SVIPAMM - MAKE FILE FOR PLUS ISO ADMINISTRATIVE MESSAGE SERVER
SVIPAMS - SOURCE FILE FOR PLUS ISO ADMINISTRATIVE MESSAGE SERVER
SVPMLF - OBJECT FILE FOR PLUS MAINTENANCE LOG FILE SERVER
SVPMLFM - MAKE FILE FOR PLUS MAINTENANCE LOG FILE SERVER
SVPMLFS - SOURCE FILE FOR PLUS MAINTENANCE LOG FILE SERVER
SWPISOM - MAKE FILE FOR THE SW60PISO SUBVOLUME

* This file will only exist when enhancements or major fixes have been made to the PLUS ISO Interface.

PLUS ISO INTERLINK GATEWAY PROCESS FILES

The following is a list of all files on the SW60PISI subvolume and a brief description of each file.

PISIE	- EXTERNALS FOR PISI SOURCE FILE
PISIFM	- MAKE CONFIGURATION FILE FOR PLUS ISO INTERLINK GATEWAY PROCESS
PISIM	- MAKE FILE FOR PLUS ISO INTERLINK GATEWAY PROCESS
PISIMM	- MAKE MACRO DOCUMENTATION FILE FOR PLUS ISO INTERLINK GATEWAY PROCESS
PISIO	- NON-EXECUTABLE OBJECT FOR PLUS ISO INTERLINK GATEWAY PROCESS
PISIS	- SOURCE FILE FOR PLUS ISO INTERLINK GATEWAY PROCESS
RQPISFM	- MAKE CONFIGURATION FILE FOR PLUS ISO INTERLINK GATEWAY REQUESTORS
RQPISFO	- PSEUDO TARGET FOR RQPISFS
RQPISFS	- SOURCE FILE FOR PLUS INTERLINK SERVICES FILE REQUESTER
SCRNPISF	- SOURCE FILE FOR PLUS INTERLINK SERVICES FILE SCREENS
SVPISF	- EXECUTABLE OBJECT FOR PLUS INTERLINK SERVICES FILE SERVER
SVPISFM	- MAKE CONFIGURATION FILE FOR PLUS INTERLINK SERVICES FILE SERVER
SVPISFS	- SOURCE FILE FOR PLUS INTERLINK SERVICES FILE SERVER
SWPISIM	- MAKE FILE FOR THE SW60PISI SUBVOLUME

INSTALLATION STEPS

The steps required to install the PLUS ISO Interface along with the PLUS ISO Interlink Extension are described below and are listed in the order in which they should be performed. Examples follow each step of the procedure to clarify the function the procedure is attempting to perform.

During the course of the installation, it may be necessary to refer to the following manuals.

- ACI MAKE User Guide (Preliminary Version)
 - BASE24-atm Transaction Processing Manual
 - BASE24-pos Transaction Processing Manual
 - BASE24 Base Files Maintenance Manual
 - BASE24 Device Control Manual
 - BASE24 Logical Network Configuration File Manual
 - BASE24 Text Command Reference Manual
 - BASE24 Tokens Manual
 - BASE24 Transaction Security Manual
 - NET24-XPNET Network Control Command Reference Manual
 - NET24-XPNET Network Control Operations Guide
 - Tandem's Communication Management Interface Operator Reference Manual
 - Tandem's Subsystem Control Facility (SCF) Reference Manual
 - Tandem's System Management Manual
1. Restore the software tape to the appropriate volume and subvolume by entering the following command at the TACL prompt:

```
RESTORE /OUT $S.#PISO/ $TAPE, *.*.*.LISTALL,TAPEDATE,  
OPEN,MYID,VOL <$your vol>
```
 2. Configure a Tandem port and line via a network control facility. If a Tandem communication subsystem controller (CSS 6100) is used, the line should be configured using the EBCDIC character set qualifier.

3. Volume to \$vol.BA60MAKE and edit CUSTMACS
 - a) Set CLASSIC_ON to true.
 - b) Set SWITCH_ON to true.
 - c) If the PLUS ISO Interface will be run in the release 6.0 network environment, set SW PRE 60 LCONF ON to false. If the PLUS ISO Interface will be run in a pre-release 6.0 network environment, set SW PRE 60 LCONF ON to true.
 - d) Set SWI_RPT_ON to true if intending to use the PLUS ISO Interface ILF reports.
 - e) Set PLUS_ISO_ON to true.
 - f) If the PLUS INTERLINK GATEWAY has been purchased, Set PLUS_INTERLINK_ON to true.
 - g) Set PLUS_ISO_ENHANCED_MODE_ON to true.
 - h) Set AIF_ON to true.
 - i) Set SNF_ON to true.

NOTE: It is recommended that you use B24C/B24M files so all necessary modules are rebuilt. If using B24C/B24M files, part b in steps 4 thru 6 may be skipped.
4. Volume \$vol.SW60SNF and perform the following:
 - a) Edit the macro Make file SNFMM and ensure that the system, volume and subvolume references are correct.
 - b) Make the SW60SNF subvolume by entering the following command at the TACL prompt:
RUN \$vol.subvol.makevXXX/OUT \$S.#SNF/SWSNFM -FORCE where: XXX is the Make version.
5. Volume \$vol.SW60AIF and perform the following:
 - b) Edit the macro Make file AIFMM and ensure that the system, volume and subvolume references are correct.
 - b) Make the SW60AIF subvolume by entering the following command at the TACL prompt:
RUN \$vol.subvol.makevXXX/OUT \$S.#AIF/SWAIFM -FORCE where: XXX is the Make version.
6. Volume to \$vol.SW60PISO and perform the following:
 - a) Edit the macro Make file PISOMM and ensure that the system, volume and subvolume references are correct.
 - b) Make the SW60PISO subvolume by entering the following command at the TACL prompt:
RUN \$vol.subvol.makevXXX/OUT \$S.#PISO/SWPISOM -FORCE where: XXX is the Make version.

7. If part b in steps 4 thru 6 was skipped in order to run the B24C/B24M files, perform the following:
 - a) Volume to \$vol.SW60SRC.
 - b) Edit configuration Make files SWICSEFM and SWIFM and ensure that the system, volume and subvolume references are correct.
 - c) Volume to \$vol.BA60MAKE
 - d) Edit B24C and ensure that the -force option is NOT included in the Make command unless this is a first time install.
 - e) Make all necessary modules on your BASE24 system by entering the following command at the TACL prompt:
/IN B24C/
8. Make will move the object files to the locations specified in the configuration Make file if the move_obj_subvolume_on define is set to TRUE. Whoever runs the Make must have security to write to the subvolume where Make attempts to place the objects.
9. Add the following servers to the PATHCONF file.
 - a) For the PLUS ISO Interface only, add the ICFE, PPATH, and PISO-ADM servers to the PATHCONF. Add the SNF and/or AIF servers if using the Station Name File (SNF) and/or Acquirer Information File (AIF).The following are server definition examples. Different values may be used for various fields. Use previous server definitions in the PATHCONF to assist in determining values.

```
RESET SERVER
SET      SERVER      CPUS          x:y
SET      SERVER      PROGRAM       \sys.$vol.xxxxOBJ.SVICFE
SET      SERVER      DELETEDELAY 10 MINS
SET      SERVER      PRI           145
SET      SERVER      NUMSTATIC     0
SET      SERVER      HOMETERM      \sys.$PLOG
SET      SERVER      TIMEOUT       60 SECS
ADD SERVER SERVER-ICFE
```

```
( This PPATH server definition example is for XPNET 2.1 or )
( above. For additional configuration information regarding )
)
( SERVER-PPATH for XPNET 2.1 or above, refer to the XPNET )
( CTS/E PWS XN-PL320-01 Implementation Guide. )
)
```

```
RESET SERVER
SET      SERVER      AUTORESTART 0
SET      SERVER      CPUS          x:y (must be valid CPUs, e.g.,
0:1)
SET      SERVER      CREATEDELAY 30 SECS
SET      SERVER      DEBUG          OFF
SET      SERVER      DELETEDELAY 1 MINS
SET      SERVER      HOMETERM       \sys.$PLOG
SET      SERVER      LINKDEPTH      1
SET      SERVER      MAXLINKS       1
SET      SERVER      MAXSERVERS     1
SET      SERVER      NUMSTATIC      1
SET      SERVER      PRI             145
SET      SERVER      PARAM EXT      "yes"
SET      SERVER      PARAM GENERIC   "no"
SET      SERVER      PARAM MODE     "path"
SET      SERVER      PROCESS        $S1AP.#PISOxxx (ASSOCIATIVE
ON)
SET      SERVER      PROGRAM        \sys.$SYSACI.XPNET.PATH
SET      SERVER      TIMEOUT        60 SECS
SET      SERVER      TMF            OFF
SET      SERVER      VOLUME         \sys.$volume.LNETEXEC
ADD SERVER SERVER-PPATHxxx
```

* Where \$S1AP is any valid PPD name and xxx is the PLUS process number. The Server Process and the port attribute in the line declaration in the Network Environment File (NEF) must be the same. xxx is optional. If used, the process number field on the screen must contain the same value.

```
RESET SERVER
SET      SERVER      CPUS      x:y  (must be valid CPUs, e.g.,
0:1)
SET      SERVER      PROGRAM    $vol.xxxxOBJ.SVIPAM
SET      SERVER      DELETEDELAY 10 MINS
SET      SERVER      PRI        145
SET      SERVER      NUMSTATIC  0
SET      SERVER      HOMETERM   $PLOG
SET      SERVER      TIMEOUT    60 SECS
ADD SERVER SERVER-PISO-ADM
```

```
RESET SERVER
SET      SERVER      CPUS      x:y
SET      SERVER      PROGRAM    $vol.xxxxOBJ.SVAIF
SET      SERVER      DELETEDELAY 10 MINS
SET      SERVER      PRI        145
SET      SERVER      NUMSTATIC  0
SET      SERVER      HOMETERM   $PLOG
SET      SERVER      TIMEOUT    60 SECS
ADD SERVER SERVER-AIF
```

```
RESET SERVER
SET  SERVER      AUTORESTART 0
SET  SERVER      CPUS          (5:6)
SET  SERVER      CREATEDELAY 1 MINS
SET  SERVER      DEBUG         OFF
SET  SERVER      DELETEDELAY 10 MINS
SET  SERVER      HOMETERM      \sys.$PLOG
SET  SERVER      LINKDEPTH     1
SET  SERVER      MAXSERVERS    1
SET  SERVER      NUMSTATIC      0
SET  SERVER      PRI           145
SET  SERVER      PROGRAM       $vol.xxxxOBJ.SVSNF
SET  SERVER      TIMEOUT       60 SECS
SET  SERVER      TMF           OFF
SET  SERVER      VOLUME        \sys.$volume.LNETEXEC
ADD SERVER SERVER-SNF
```

- b) If the PLUS INTERLINK GATEWAY has been purchased, add the PLUS-ISF and PISO-MLF servers to the PATHCONF.

```
RESET SERVER
SET  SERVER      CPUS          x:y
SET  SERVER      PROGRAM       $vol.xxxxOBJ.SVPISF
SET  SERVER      DELETEDELAY 10 MINS
SET  SERVER      PRI           145
SET  SERVER      NUMSTATIC      0
SET  SERVER      HOMETERM      $PLOG
SET  SERVER      TIMEOUT       60 SECS
ADD SERVER      SERVER-PLUS-ISF
```

```
RESET SERVER
SET      SERVER      CPUS          x:y
SET      SERVER      PROGRAM       $vol.xxxxOBJ.SVPMLF
SET      SERVER      DELETEDELAY 10 MINS
SET      SERVER      PRI           145
SET      SERVER      NUMSTATIC     0
SET      SERVER      HOMETERM      $PLOG
SET      SERVER      TIMEOUT       60 SECS
ADD SERVER SERVER-PISO-MLF
```

10. Move to the location of the template files used by the network. Create the template files by entering the following commands:
FUP /IN \$vol.SW60PISO.PISOFUP, OUT \$S.#PISOFUP /
FUP /IN \$vol.SW60SNF.SNFFUP, OUT \$S.#SNFFUP/
11. Add an ICFE record for the PLUS ISO Interface process using the information provided later in this document and in the "BASE24 Base Files Maintenance Manual".
12. Add a KEYF or KEY6 record for the PLUS ISO Interface process using the information provided later in this document and in the "BASE24 Base Files Maintenance Manual" and "BASE24 Transaction Security Manual".
13. Add the necessary assigns and params to the Logical Network Configuration File for the PLUS ISO Interface process using the information provided later in this document and in the "BASE24 Logical Network Configuration File Manual".

14. Ensure that the following are defined in the Network Environment File (NEF). If the following are not defined, add the declarations via a network control facility.
 - a) A PLUS ISO Interface process declaration
 - b) A PLUS station declaration (PLUS can support sixteen pairs of stations.)
 - c) A PLUS device declarationWhen using a Tandem communication subsystem controller (CSS 6100) the TRANSMIT TRANSPARENT ENABLED declaration must be set.
 - d) A PLUS line declaration (PLUS can support multiple lines.)
 - e) A Server Path station declaration (XPNET 2.1 or greater)
 - f) A Server Path line declaration (XPNET 2.1 or greater)For details on the previous declarations, refer to the "NET24-XPNET Network Control Command Reference Manual".
15. XPNET 3.0
The following commands are used to add a process, line, device and station. They are sent to the XPNET 3.0 Network from an NCPCOM prompt. These are only examples and many of the input values must be determined based on your own network. If necessary, refer to the "NET24-XPNET Network Control Operations Guide."

```
RESET PROCESS
SET PROCESS BCPU 3
SET PROCESS PROGRAM $vol.xxxxobj.piso
SET PROCESS LIBRARY $SYSACI.XPNET.SKELB
SET PROCESS PPD $PISO
SET PROCESS PRIORITY 135
SET PROCESS CPU 2
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 100
SET PROCESS QMT 150
SET PROCESS SAVEABEND OFF
SET PROCESS QMI 90
ADD PROCESS P1A^PISO
```

This following device, line and station samples are for non-SNA protocols

```
RESET DEVICE
SET  DEVICE      RECVLEN 4000
SET  DEVICE      XMITLEN 1000
ADD DEVICE D1A^PISO

RESET LINE
SET  LINE  ATIMER 0
SET  LINE  BTIMER 0
SET  LINE  ERANALYSIS 30
SET  LINE  PORT $PISO
SET  LINE  LCLLUNAME APCLU1
SET  LINE  PRTLUNAME PARTLU1
SET  LINE  UCRATE LU62LINRATE
SET  LINE  UCSTATE LU62LINSTATE
SET  LINE  UCQUEUE LU62LINQUEUE
SET  LINE  PROTOCOL LU6.2
ADD LINE L1A^PISO

RESET STATION
SET  STATION      DESTINATION P1A^PISO
SET  STATION      DEVICE PISO^DEV
SET  STATION      LINENAME \SYS1.P1A^NODE.L1A^PISO
ADD STATION S1PISO01
```

This following device, line and station samples are for the SNA protocol.

```
RESET DEVICE
SET  DEVICE      ACLength 2
SET  DEVICE      INPUTFORMAT TXT
SET  DEVICE      MACRO SLU
SET  DEVICE      OUTPUTFORMAT TXT
SET  DEVICE      POLLFORMAT SP
SET  DEVICE      PROFILE PPISOLU2
SET  DEVICE      RECVLEN 4095
SET  DEVICE      SELECTFORMAT SP
SET  DEVICE      TYPE 0
SET  DEVICE      XMITLEN 4095
ADD DEVICE D1A^PISO^DV, UNDER SYSNAME \sys, UNDER NODE P1A^NODE
```

```
RESET LINE
SET  LINE  ERANALYSIS 30
SET  LINE  PORT $HLSX.#SNAP1.LU2
SET  LINE  CMETHOD ACCEPT
SET  LINE  EXPDATA ROUTE
SET  LINE  FMM STATUS
SET  LINE  PROTOCOL SLU
SET  LINE  MODE ASYNC
ADD LINE L1A^PISO1 , UNDER SYSNAME \sys, UNDER NODE P1A^NODE
```

The port name on the above line definition must match the PPD name used in the HLSOBJ run command (\$HLSX), followed by the line added using SCF (#SNAP1) and lastly, an appropriate LU number.

```
RESET STATION
SET  STATION      APPLREJ OFF
SET  STATION      DESTINATION P1A^PISO01
SET  STATION      ENDBRACKET ON
SET  STATION      ERRORACT REINEVERY
SET  STATION      PASSFMH OFF
SET  STATION      REINSTATE 60
SET  STATION      RETRY 1
SET  STATION      CLASS SRV
SET  STATION      DEVICE D1A^PISODV
SET  STATION      LINENAME \sys.P1A^NODE.L1A^PISO1
ADD STATION S1PISO01 , UNDER SYSNAME \sys, UNDER NODE P1A^NODE
```

LCONF ASSIGNS AND PARAMS

The Logical Network Configuration File (LCONF) assigns and params required by the PLUS ISO Interface process are described below.

Refer to the "BASE24 Logical Network Configuration File (LCONF) Manual" for instructions on adding them to the LCONF.

NOTE: Some of these assigns and params are used throughout BASE24. Therefore, they may already be defined in the LCONF.

ASSIGNS

The following assigns are used by the PLUS ISO Interface process. The assigns marked with asterisks (**) are required if reports are to be generated.

AIF

The fully-qualified file name of the Acquirer Information File. This assign is optional.

APCFEMT

The physical file name of the Acquirer Processing Code File Extended Memory Table.
Required.

ICF

The physical file name of the Interchange Configuration File. Required if the BASE24-REL param does not contain the value of 60.

ICFE

The physical file name of the Enhanced Interchange Configuration File. Required if the BASE24-REL param is not present or if the param is present and contains the value of 60.

ILF

The fully-qualified file name of the Interchange Log File. This assign is required.

Each interface process must have its own ILF assign defined in the LCONF. ILF assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

ILF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. BASE24 uses the specified template file to create ILFs.

IPCFEMT

The physical file name of the Issuer Processing Code File Extended Memory Table. Required.

KEYF

The fully-qualified file name of the Key File. This assign is required if the BASE24-REL param is not present, also if the param is present and contains the value of 50 or greater. In a Release 5.0 or above system this assign is used by the interface to identify the Key File.

KEY5

The fully-qualified file name of the Key File. This assign is required if the BASE24-REL param is present and does not contain the value of 50 or greater. In a Release 3.4/4.0 system this assign is used by the interface to identify the Key File.

KEY6

The physical file name of the KEY6 File. Required if the BASE24-REL param is present and contains the value of 50 or greater and LCONF parameter DES-TRIPLE-SINGLE has the value 'Y'.

LMCF

The fully-qualified file name of the Log Message Configuration File. This assign is required.

PAM

The fully-qualified file name of the PLUS Administrative Message File. This assign is required.

PAM assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. The PLUS ISO Interface process uses the specified template file to create PAMs.

SAF

The physical file name of the Store-and-Forward File. This assign is required.

Each interface process must have its own SAF assign defined in the LCONF. SAF assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

SAF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. BASE24 uses the specified template file to create SAFs.

SECURE-DEV1

The Tandem SYSGEN device name of the primary security module. However, if the interface is running in a Release 6.0 network environment using Transaction Security Services, this parameter must contain the PPD name of the primary IAS process and may be process specific. Required.

SECURE-DEV2

The Tandem SYSGEN device name of the secondary security module. However, if the interface is running in a Release 6.0 network environment using Transaction Security Services, this parameter must contain the PPD name of the secondary IAS process and may be process specific. Optional.

SNF

The fully-qualified file name of the Station Name File. This assign is required to use send/receive processing.

TKN

The fully-qualified file name of the Token File. This assign is optional and is only used if the BASE24-REL param is not present, also if the param is present and contains the value of 50 or greater. This assign is only used in a Release 5.0 or above systems.

RPT **

The fully-qualified file name of the Report Perusal File used by BASE24-atm.

RPT assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. BASE24 uses the specified template file to create Report Perusal Files.

RPT-ERROR-FILE **

The spooler location for log messages generated by the Interchange Interface report programs.

SW-RPT-OBEY-FILE **

The fully-qualified file name of the obey file used to start the interchange reports. The interface process creates its report obey files using this name.

Each interface process must have its own SW-RPT-OBEY-FILE assign defined in the LCONF. SW-RPT-OBEY-FILE assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

SW-RPT01-PROG-NAME **

The fully-qualified file name of the executable object code that generates the interface's SETL01 report program.

The PLUS ISO Interface has its own RPT01 report program and does not use the standard BASE24 interface report. Therefore, each interface process must have its own SW-RPT09-PROG-NAME assign defined in the LCONF. SW-RPT09-PROG-NAME assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

SW-RPT09-PROG-NAME **

The fully-qualified file name of the executable object code that generates the interface's STAT09 report program.

The PLUS ISO Interface has its own RPT09 report program and does not use the standard BASE24 interface report. Therefore, each interface process must have its own SW-RPT09-PROG-NAME assign defined in the LCONF. SW-RPT09-PROG-NAME assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LCONF assign screen.

If the PLUS INTERLINK GATEWAY has been purchased, the following assigns are also required.

PISO-ISF

The fully-qualified file name of the PLUS ISO/Interlink Services File. This assign is required.

PISF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. The PLUS ISO Interface process uses the specified template to create PISFs.

PISO-MLF

The fully-qualified file name of the PLUS ISO Proprietary Member Center file. This assign is required.

PISO-MLF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. The PLUS ISO Interface process uses the specified template file to create PMLFs.

RSAF

The fully-qualified file name of the PLUS ISO Administrative File. This assign is required

RSFAF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LCONF assign screen. The PLUS ISO Interface process uses the specified template file to create RSAFs.

PARAMS

The following params are used by the PLUS ISO Interface process. The params marked with asterisks (**) are required if reports are to be generated.

ACCT-NUM-TYP

A value (B or D) indicating whether account numbers are stored in binary (B) or decimal (D) format. Currently, only decimal format is supported, allowing for account numbers of up to 19 digits in length.

ATM-LN-CUTOVER

The time (HH:MM) at which all BASE24-atm processes in the logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59 (no default value). This param is required if processing BASE24-atm transactions.

BASE24-REL

A value indicating the release level of BASE24. Valid values are:
40 = BASE24-atm Release 4.0 and/or BASE24-pos Release 3.4
50 = BASE24 Release 5.0/5.1.2/5.3
60 = BASE24 Release 6.0 or above (default)

DATE-DISPLAY-TYPE

A value indicating the format of the month and day portion of the dates on reports. When dates include the year, this param does not affect the year portion of the date. Valid values are:
1 = DDMM
2 = MMDD (default)

DEFAULT-MEMBER-NUMBER

The default member number to be used for inbound POS transaction were no track data is present. Valid values are 0 through 32766 (default 0).

DES-TRIPLE-SINGLE

A value indicating if triple to single DES encryption is required. Valid values are:

Y = Triple to single DES encryption is required (KEY6 assign must exist)
N = Triple to single DES encryption is not required (KEYF assign must exist)

NOTE: If triple to single DES encryption is being supported in the interface via Transaction Security Services in Release 6.0, DES-TRIPLE-SINGLE must not exist or may exist with the value 'N'.

ILF-NOTIFY-INTERVAL

The number of writes made to the Interchange Log File (ILF) between calculating the percentage of file full. Each time a write is made to the ILF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the ILF is. The counter is then reset to zero. When the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

ILF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the ILF is becoming full. The interface process continues to log messages at the interval indicated in the ILF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

KEK-VARIANT-USED

An indicator of whether variant 1 of the key exchange key should be used for encryption and decryption of the interface's working keys. Valid values are:

Y = Yes, use variant 1 (default)

N = No, do not use variant 1

MLF-NOTIFY-INTERVAL

The number of writes made to the Maintenance Log File (MLF) between calculating the percentage of file full. Each time a write is made to the MLF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the MLF is. The counter is then reset to zero. When the file reaches the percent full specified in the MLF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

MLF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the MLF is becoming full. The interface process continues to log messages at the interval indicated in the MLF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

NCD-DFLT-HOPR-CONTENTS

Parameter used by the PLUS ISO Interface process to add default hopper contents to Non Currency Dispense transactions. The value of this param should match a value in the NCD-CONTENTS table in COBNAMES. If the NCD-

DFLT-HOPR-CONTENTS param is not present, it will default to "10".

PLUS-ADMIN-NOTIFY-INTERVAL

The number of writes made to the PLUS Administrative Message file (PAM) between calculating the percentage of file full. Each time a write is made to the PAM, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the PAM is. The counter is then reset to zero. When the file reaches the percent full specified in the PLUS-ADMIN-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

PLUS-ADMIN-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the PAM is becoming full. The interface process continues to log messages at the interval indicated in the PLUS-ADMIN-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

POS-LN-CUTOVER

The time (HH:MM) at which all BASE24-pos processes in the logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59 (no default value). This param is required if processing BASE24-pos transactions.

RESUB-CNFG

A flag indicating whether resubmissions are to be processed in the BASE24 environment. Valid values are:

N = Resubmissions will not be processed (default)

Y = Resubmissions will be processed

```
*****
*
*                      *****  WARNING  *****
*
* The RESUB-CNFG parameter should only be set to Y when
* the BASE24-pos PLUS ISO Interface, Device Handler, and
* Authorization processes have been modified to fully support
* Resubmissions. Otherwise, transactions which did not
* originate as Stand-ins can get added to the Resubmittal
* Store-and-Forward File (RSAF).
*
*****
```

RESUB-LOG-PTLF

A flag indicating whether resubmissions are to be logged to the PTLF upon final submittal to the interchange. The initial submittal is always logged to the PTLF.

N = Do not log the final submittal to the PTLF (default)

Y = Log the final submittal to the PTLF

RESUB-MAX-LIMIT

The maximum number of times a transaction can be resubmitted before it is deleted from the Resubmittal Store-and-Forward file (RSAF). Valid values are 1 through 30 (default 30).

RESUB-TIME

The time (HH:MM) at which the PLUS ISO Interface process is to start sending resubmittal transactions to the interchange. Valid values, in military time, are from 00:00 to 23:59 (no default value). This param is required if BASE24 is configured to process resubmissions.

RSAF-NOTIFY-INTERVAL

The number of writes made to the Resubmittal Store-and-Forward File (RSAF) between calculating the percentage of file full. Each time a write is made to the RSAF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the RSAF is. The counter is then reset to zero. When the file reaches the percent full specified in the RSAF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

RSAF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the RSAF is becoming full. The interface process continues to log messages at the interval indicated in the RSAF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

SAF-NOTIFY-INTERVAL

The number of writes made to the Store-and-Forward File (SAF) between calculating the percentage of file full. Each time a write is made to the SAF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the SAF is. The counter is then reset to zero. When the file reaches the percent full specified in the SAF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

SAF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the SAF is becoming full. The interface process continues to log messages at the interval indicated in the SAF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

SECURE-DEV-CONF-CMD100

The command text the interface process is to send to an Atalla Network Security Processor (NSP), model A5000 or A6000, in a 100 command during initialization or warmboot. The 100 command text re-enables optional features of the security device. The text contained in the 100 command must be obtained directly from Atalla. This param is only required if the user wants BASE24 to issue these commands to the Atalla NSP. (NOTE: features enabled for a security device via 100 commands are not process-specific; they affect all processes interacting with the device.)

SECURE-DEV-CONF-TXT

The configuration text the interface is to send to an Atalla Network Security Processor (NSP), model A5000 or A6000, in a 101 command during initialization or warmboot. The configuration text can consist of one or more of the four-character option words described below. If more than one option word is defined in the configuration text, the words must be separated by commas. Each option word contains a three-digit option ID followed by a one-character action flag. An option ID (i.e., 023, 024, 040) can be used only once in each SECURE-DEV-CONF-TXT param.

- 023E - Enables the NSP's function to remove carriage return and line feed characters from all response messages returned to BASE24.
- 024E - Enables the NSP's function to identify PIN length errors by returning a unique code in response messages to BASE24. This causes the NSP to return L error responses for PIN length errors and S error responses for sanity check errors.
- 024D - Disables the NSP's function to identify PIN length errors by returning a unique code in response messages to BASE24. This causes the NSP to return S error responses for both PIN length and sanity check errors.
- 040x - Specifies that the NSP should check PIN lengths and also sets the minimum valid PIN length for performing the check. The "x" must be a value of 0-9 or A-C to indicate the minimum valid PIN length (A = 10; B = 11; C = 12). A value of 0 effectively disables this check because any PIN would pass.

This param is only required if the user wants BASE24 to issue these 101 commands to the Atalla NSP. (NOTE: features enabled or disabled for a security device via 101 commands are not process-specific; they affect all processes interacting with the device.)

SECURE-DEV-FAIL-NOTIFY

The interval at which messages are logged for denied transactions due to security device failures. For example, when the SECURE-DEV-FAIL-NOTIFY param is set to 20, every twentieth transaction denied because of a security device failure causes a message to be logged. The SEC-DEV-FAIL-NOTIFY param sets the interval after the first security device failure. A message is always logged immediately following the first security device failure. Valid values are 10 through 1000 (default 20).

SECURE-DEV-MONITOR-THRESHOLD

The interval, in number of transactions performed by the security device defined in the SECURE-DEV1 assign, at which the security device defined in the SECURE-DEV2 assign is checked. The security device is checked by sending it a transaction. Valid values are 0 through 32000 (default 0). A value of 0 indicates the security device defined in the SECURE-DEV2 assign is not checked.

SECURE-DEV-TIM-LMT

The maximum amount of time, in hundredths of a second, that BASE24 waits for a response from the security device before timing out the message. Valid values are 1 through 6000 (default 500). Example: 6000 equals 60 seconds.

SECURE-DEV-TYP

A code identifying the type of security module specified in the SECURE-DEV1 and SECURE-DEV2 assigns. Valid values are:

ASM = Atalla Security Module or Atalla Network Security Processor
RSM = Racal-Guardata Host Security Module (same as VSM)
VSM = Racal-Guardata Host Security Module (same as RSM)
TSS = Transaction Security Services
N/A = A security module is not required (default)

SEND-ETX

This param will let the interface know whether to send the end_of_text (ETX) character or not. The SNA protocol does not handle the ETX character. Valid values include the following:

Y = Send the ETX. Used for non SNA protocols.
N = Do not send the ETX (default). Used for SNA protocol.

SPAN-FLAG

A value indicating whether totals are to be written to the Span Clearings File by the interface process. Valid values are:

Y = Yes, write totals to the Span Clearings File.
N = No, do not write totals to the Span Clearings File (default).

STATION-TYP

This LCONF param is used by the PLUS ISO Interface to determine whether to use the station names from the ICFE or the SNF. Valid values are as follows:

0 = Normal Processing; The interface uses the stations from the ICFE with station 1 as the issuer station and station 2 as the acquirer station.
1 = Send/Receive Processing; The interface uses the station pairs as listed in the SNF.

The PLUS ISO Interface will default this param to 0 if it is missing or has an incorrect value.

SWI-ACQ-CVD2-DATA-SUPPRS

A value indicating whether card verification data will be suppressed before logging to the ILF. Valid values are:
Y = Suppress card verification data before logging
N = Do not suppress card verification data (default)

SW-ACQ-DISCR-DATA-SUPPRS

A value indicating whether discretionary data from track 1 and/or will be suppressed before logging to the ILF. Valid values are:
Y = Suppress discretionary data before logging
N = Do not suppress discretionary data (default)

SW-ILF-FILE-RETENTION

The number of days the ILF is to remain on the system. After the specified number of days, the ILF is purged from the system by the interface process. Valid values are 3 through 99 (default 7).

SW-ILF-FILE-OPEN

The number of days the ILF is to remain open. After the specified number of days, the ILF is closed by the interface process. Only customers who do not process Interlink transactions with their PLUS ISO process should use this param. If the param SW-ILF-FILE-OPEN is not in the LCONF, The value will default to the value in param SW-ILF-FILE-RETENTION. This keeps customers from having to add param SW-ILF-FILE-OPEN to their LCONF unless they want to keep fewer ILF's open than they retain on the system. The value of SW-ILF-FILE-OPEN must be equal to or less than the value in SW-ILF-FILE-RETENTION. Valid values are 1 through 5. For example, if you set the value to 2, today's ILF and two old ILF's will remain open. If neither the param SW-ILF-FILE-OPEN or the param SW-ILF-FILE-RETENTION are in the LCONF, they will both default to 7.

SW-ISS-DISCR-DATA-SUPPRS

A value indicating whether discretionary data from track 1 and/or will be suppressed before logging to the ILF. Valid values are:
Y = Suppress discretionary data before logging
N = Do not suppress discretionary data (default)

SW-RPT-STARTUP **

The time (HH:MM) at which the interface process is to run its report obey file. Valid values, in military time, are from 00:00 to 23:59 (no default value).
If the PLUS INTERLINK GATEWAY has been purchased the following params are also required.

ICFE SETTINGS

The Enhanced Interchange Configuration File (ICFE) contains parameters relevant to interchange transaction processing. It is used by the financial institution to define institution and terminal interchange sharing, settlement handling, timer limits, and processing options. The options controlled via the ICFE are described below and on the following pages. For more information and screen formats, refer to the "BASE24 Base Files Maintenance Manual". The ICFE fields are described in the order they appear on the screens.

ICFE SCREEN 1 SETTINGS

The following fields must be set as indicated on ICFE screen 1:

INTERCHANGE FIID

The FIID uniquely identifying the PLUS Interchange. This field must match the logical network ID given to the PLUS Interchange in the BASE24 system. This field and the PROCESS field combine to form the primary key to the ICFE.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PROCESS

The name of the PLUS ISO Interface process as defined in your network environment. This is the process that receives and sends transaction messages from and to PLUS. This field and the INTERCHANGE FIID field combine to form the primary key to the ICFE.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

SWITCH TYPE

A code identifying the interchange. This code must match the last four characters of the subvolume name containing the proper interface. For the PLUS ISO Interface process, this value must be "PISO".

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

INTERCHANGE LOGICAL NET

The logical network identifier given to the interchange. This value must match the value in the INTERCHANGE FIID field.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REPORTING NAME

The name of the interchange as it appears in the headings of BASE24 reports.

FIELD LENGTH: 1-20 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

INSTITUTION ID

The institution identifier assigned by the interchange. For the PLUS interchange is the Proprietary Member Center (PMC) number.

FIELD LENGTH: 1-15 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SWITCH ID

The identification of the interchange center. Not used by the PLUS ISO Interface process.

FIELD LENGTH: 1-15 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

STATION 1

The symbolic name of a PLUS station that sends acquired transaction requests to and receives responses from the switch. This name must match the name of a station defined in your network environment for the PLUS ISO Interface process.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

STATION 2

The symbolic name of a second PLUS station that sends acquired transaction requests to and receives responses from the switch. This name must match the name of a station defined in your network environment for the PLUS ISO Interface process. If only one station is defined, this field must either be blank or must match the station name in the STATION 1 field.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SIC CODE

The Standard Industrial Classification (SIC) code applicable to the institution operating the BASE24-atm or BASE24-pos network. This field is the default SIC code if an incoming POS message does not contain a SIC code.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CURRENCY CODE

The acquirer's currency code for transactions sent to PLUS. Valid values are listed in the following publication: "Codes for the Representation of Currencies and Funds," ISO 4217-1981.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 840 (US dollars)

DEFAULT TERM NUM

A default terminal number used to fill the STM/PSTM terminal number field if an incoming transaction from the interchange does not contain a terminal number. Not used by the PLUS ISO Interface process.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

DEFAULT ACQUIRER ID NUM

A default acquirer ID number used if an incoming transaction from the interchange does not contain an acquirer's institution identification number. Not used by the PLUS ISO Interface process.

FIELD LENGTH: 1-11 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: 00000000000

CUSTOMER BALANCE DISPLAY

A value indicating the method in which PLUS card issuers wish balances to be presented at BASE24 terminals. Valid values are:

0 = Do not display or print

1 = Display only

2 = Print only

3 = Display and print

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: 0 (Do not display or print)

ICFE SCREEN 2 SETTINGS

The following fields on ICFE screen 2 must be set as indicated.

SETTLEMENT INFORMATION

The following fields contain interchange settlement information.

SETTLEMENT HOUR

The current hour of settlement (local time) for this interchange. Valid entries include 00 through 23. For example, if the PLUS Interchange settles at 6:00 a.m., GMT, this field should contain the value 06.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

SETTLEMENT MINUTE

The current minute of settlement (local time) for this interchange. Valid entries include 00 through 59. For example, if the PLUS Interchange settles at 6:00 a.m., GMT, this field should contain the value 00.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

SETTLEMENT DAYS

A value indicating the processing schedule for the interchange. PLUS Interchange settles only 7 days a week. Valid values are:

1 = Processes 7 days a week

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1 (PROCESS 7 DAYS).

REPORT PRIORITY

The priority at which the PLUS ISO Interface process runs its reports. The value entered in this field must be equal to or less than 255.

FIELD LENGTH: 1-3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 100

REPORT CPU

The Central Processing Unit (CPU) in which the report programs are started.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

SWITCH POSTING DATE (YYMMDD)

The current posting date (YYMMDD) for PLUS. The Interchange Interface process updates this date at daily settlement time and upon initialization.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

HOLIDAY DATES (YYMMDD)

The interchange's defined holiday dates (YYMMDD). The interchange does not settle on these holidays and the interface process does not create an Interchange Log File (ILF) for these dates. Not used by the PLUS ISO Interface process, since this field is valid only when 5-day settlement is used.

FIELD LENGTH: 16 fields of 6 numeric characters each
REQUIRED FIELD: No
DEFAULT VALUE: None

ICFE SCREEN 3 SETTINGS

The following fields must be set on ICFE screen 3 as indicated:

TIMER LIMITS

The following timer value fields are used to control specific transaction message activity between the BASE24 network and PLUS. The values entered in these fields must be checked against PLUS timer values to ensure the integrity of message traffic.

NETWORK MANAGEMENT

The number of seconds BASE24 waits for a response to a network management message (sign on, sign off, etc.).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

EXTENDED NETWORK

The number of seconds BASE24 waits before attempting to retry a network management message during periods when the line to the interchange is down.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

WAIT FOR TRAFFIC

The number of seconds BASE24 waits for traffic on the line from PLUS before initiating a network management message.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

PERFORMANCE PERIOD

The number of minutes the PLUS ISO Interface process uses as interval for tracking performance of the DPC.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 20 (MIN)

MAXIMUM OUTSTANDING TRANSACTIONS

The following fields specify the maximum number of outstanding inbound and outbound transactions. The combined value is used to determine the number of extended memory pages required for the transactions in process queue.

OUTBOUND

The maximum number of outstanding outbound transactions.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

INBOUND

The maximum number of outstanding inbound transactions.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

PROCESSING OPTIONS

The following fields define the processing options for PLUS.

ACQUIRER

A value indicating whether the institution acts as an acquirer of PLUS transactions. Valid values are:

Y = Yes, send transactions to PLUS.

N = No, do not send transactions to PLUS.

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

ISSUER

A value indicating whether the stations declared in the ICFE are to be used in the send/receive logic.

Y = Yes, stations are to be used for normal processing with no send/receive logic.

N = Stations are to be used for send/receive logic.

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

PROCESSING MODE

The processing mode of the Interchange Interface process. Not used by the PLUS ISO Interface process.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

AUTO SIGNON START

A value indicating whether the PLUS ISO Interface process is set to programmatically logon to the interchange at startup without operator intervention. Valid values are:

Y = Yes, programmatically logon to PLUS at startup.

N = No, do not programmatically logon to PLUS at startup.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

MAXIMUM TIMEOUTS

The maximum number of consecutive timeouts allowed before network management measures are taken to determine the status of the link. Valid values are 1 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 2

MAX SAF RETRY

The maximum number of times a transaction from the Store-and-Forward File (SAF) is transmitted to PLUS before the record is dumped to the hard-copy log and deleted from the file. Valid values are 0 - 9999. A value of zero indicates that the SAF will be retransmitted indefinitely or until the message completes satisfactorily. A value other than zero indicates the number of retries.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ACK TO SWITCH

A value indicating whether an acknowledgement is required to the switch. Not used by the PLUS ISO Interface process.
Valid values are:

Y = Yes
N = No

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: None

ACK FROM SWITCH

A value indicating whether an acknowledgement is required from the switch. Not used by the PLUS ISO Interface process.
Valid values are:

Y = Yes
N = No

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

NETWORK MANAGEMENT MESSAGE ENABLED

A value indicating whether network management messages are enabled or disabled. Not used by the PLUS ISO Interface process.
Valid values are:

Y = Yes, send network management messages to the switch.
N = No, do not send network management messages to the switch.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

TYPE OF INTERCHANGE REPORTS

The type of reports this interchange requires. Not used by the PLUS ISO Interface.

ILF EXTRACT NUMBER

The number of ILF files to be extracted for host reporting. This field is used in conjunction with the RPT-EXTRACT field in the Extract Configuration File (ECF). For a description of the ECF, refer to the "BASE24 Base Files Maintenance Manual". Valid values are 3 through 9.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 3

ICFE SCREEN 8 SETTINGS

The following fields must be set on ICFE screen 8 as indicated:

ATM ICFE DATA

The following fields contain information relevant to BASE24-atm.

AUTH PROCESS

The authorization process the PLUS ISO Interface process routes ATM transaction requests to if it is servicing only one BASE24 node. If PLUS is servicing more than one logical net-work (through SPAN90), this field must contain the word "GATEWAY", so the PLUS ISO Interface process knows to route the transaction through and to use prefix routing.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

DEFAULT ROUTING GROUP

The terminal routing group used as a default for transactions from PLUS.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00000000000

ACQUIRER TXN PROFILE

A code identifying a group of BASE24-atm transaction processing codes allowed from PLUS. The value of this field is part of the key used to read the Acquirer Processing Code File (APCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

ISSUER TXN PROFILE

A code identifying a group of BASE24-atm transaction processing codes allowed to be sent to PLUS. The value of this field is part of the key used to read the Issuer Processing Code File (IPCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

DEFAULT MERCHANT TYPE

The default merchant type that is sent in a purchase transaction.

FIELD LENGTH: 4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-atm transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from PLUS for an ATM transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

OUTBOUND

The time limit associated with transaction responses from PLUS for a BASE24-atm transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

INBOUND

The time limit associated with a transaction response from BASE24-atm. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

COMPLETION

The time limit associated with completion messages in either direction for BASE24-atm transactions. Not used by PLUS ISO Interface Process. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

COMPLETION ACK

The time limit associated with completion acknowledgement messages from PLUS. BASE24 does not require completions nor does it provide completion acknowledgements. Not used by the PLUS ISO Interface process. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

SHARING GROUPS

A maximum of 24 1-character codes assigned to identify PLUS sharing groups for both incoming and outgoing transactions. Valid values are 1 - 9 and A - Z. Codes cannot be separated by blank spaces, but unused portions to the right can remain blank.

FIELD LENGTH: 24 fields of 1 alphanumeric character each
REQUIRED FIELD: No
DEFAULT VALUE: None

ICFE SCREEN 10 SETTINGS

The following fields must be set on ICFE screen 10 as indicated:

POS ICFE DATA

The following fields contain information relevant to BASE24-pos.

AUTH PROCESS

The Authorization process the PLUS ISO Interface process routes POS transaction requests to if it is servicing only one BASE24 node. If PLUS is servicing more than one logical network (through SPAN90), this field must contain the word "GATEWAY", so the PLUS ISO Interface process knows to route the transaction through and to use prefix routing.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REFERRAL PHONE NUMBER

The phone number that the PLUS ISO Interface process places in all internal responses. This phone number can be called for an authorization.

FIELD LENGTH: 1-18 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

RETAILER ID DEFAULT

The default retailer ID that the PLUS ISO Interface process places in all internal requests when the interchange does not send a value. Not used by the PLUS ISO Interface process.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

TIMEOUT ACTION

A value indicating the action the PLUS ISO Interface process takes when an outgoing request times out at the interchange. Valid values are:

0 = Return a decline response
1 = Default action
2 = Alternate route

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (Return a decline response)

SETTLEMENT ENTITY

Used to determine the value of the draft capture field in the PSTM. Valid values are:

0 = No draft capture
1 = Draft capture

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ACQUIRER TXN PROFILE

A code identifying a group of BASE24-pos transaction processing codes allowed from PLUS. The value of this field is part of the key used to read the Acquirer Processing Code File (APCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

ISSUER TXN PROFILE

A code identifying a group of BASE24-pos transaction processing codes allowed to be sent to PLUS. The value of this field is part of the key used to read the Issuer Processing Code File (IPCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

ADJUSTMENT FLAG

A flag indicating whether the interchange allows an adjustment transaction when the new transaction amount (amount 2) is greater than the original transaction amount (amount 1). The PLUS ISO Interface process checks the value in this field to determine whether processing of an adjustment transaction can continue when the new transaction amount (amount 2) is greater than the original transaction amount (amount 1). Valid values are as follows:

Y = Yes, allow adjustments where the new amount is greater than the Original amount.
N = No, do not allow adjustments where the new amount is greater than the original amount.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

CHARGEBACK FLAG

A flag indicating whether the interchange allows chargeback transactions. The PLUS ISO Interface process checks the value in this field to determine whether processing of a chargeback transaction can continue. Valid values are as follows:

Y = Yes, allow chargeback transactions.
N = No, do not allow chargeback transactions.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-pos transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from PLUS for a BASE24-pos transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

OUTBOUND

The time limit associated with transaction responses from PLUS for a BASE24-pos transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

INBOUND

The time limit associated with a transaction response from BASE24-pos. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

COMPLETION

The time limit associated with completion messages in either direction for BASE24-pos transactions. Used by PLUS as timer for repeat requests. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

COMPLETION ACK

The time limit associated with completion acknowledgement messages from PLUS. BASE24 does not require completions nor does it provide completion acknowledgements. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

ICFE SCREEN 11 SETTINGS

The following fields must be set on ICFE screen 11 as indicated:

POS PRODUCT DATA

The following fields contain information relevant to BASE24-pos.

DEFAULT PRE-AUTH AMOUNT

The default preauthorization amount used in an internal request when the external message does not contain an amount.

FIELD LENGTH: 1-5 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

APPROVAL CODE LENGTH

The number of digits that PLUS requires in an approval code. Valid values are 2 through 6. For PLUS, this field must contain a 5.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 6

PRE-AUTH HOLD INCREMENT

A value indicating the time increment to hold preauthorized funds. This field is used in conjunction with the PRE-AUTH HOLD TIME field. Valid values are:

0 = Minutes
1 = Hours
2 = Days

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (Minutes)

PRE-AUTH HOLD TIME

The length of time to hold the preauthorized funds. This field is used in conjunction with the PRE-AUTH HOLD INCREMENT field. Valid values are 0 through 99.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ALLOWED SERVICES

Codes that identify the types of cards the PLUS Interchange allows. This field is used for both inbound and outbound BASE24-pos messages. The BASE24-defined codes are:

AD = Administrative	AX = American Express
BD = Business deposit-only	C = Private label credit
CB = Carte Blanche	D = Demonstration
DC = Diners Club	M = MasterCard credit
MD = MasterCard debit	MM = MasterCard dual
P = Proprietary debit	V = VISA credit
VD = VISA debit	VV = VISA dual
** = all	

Institutions can also enter additional codes not listed above, as long as the first position of the code is an alphabetic character other than M or V (M and V are reserved for MasterCard and Visa).

ICFE SCREEN 12 SETTINGS

The following fields on ICFE screen 12 (PLUS ISO specific information) must be set as indicated.

BASE24-SWI ICFE FILE MAINTENANCE LNET yy/mm/dd hh:mm 12 OF 13
INTERCHANGE FIID: PISO PROCESS: P1A^PISO

PRODUCT: _ PROTOCOL: _
ILF MAX TRAN UPDATES: ____

NETWORK PARTICIPATION
VISA: _ AFFN: _ CIRRUS: _ MASTERCARD: _ DISCOVER: _ AMEX: _
INTERLINK: VISA DEBIT:

VISA AFFN CIRRUS MASTERCARD DISCOVER AMEX
SHARING: _ SHARING: _ SHARING: _ SHARING: _ SHARING: _ SHARING: _

STANDARD INTERFACE: _ (Y/N) GMT OFFSET: ____ CNTRY-CDE: ____

FOREIGN CURRENCY PARAMETERS
IN-CODE: ____ BUY RATE: ____ SELL RATE: ____

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F8-GO TO PLUS PATHWAY MENU F12-HELP

PRODUCT

A value indicating which product(s) PLUS supports (required, default is 0). Valid values are:

0 = ATM only
1 = POS only
2 = ATM/POS

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

PROTOCOL

A value indicating the protocol supported by the PLUS ISO Interface process. Valid values are:

0 = 2780/3780
1 = 3270 BISYNC
2 = SNA/SDLC

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ILF MAX TRAN UPDATES

The number of BASE24-initiated transactions logged to the ILF before the trace number in the ILF header record is updated.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 200

NETWORK PARTICIPATION

VISA

A value indicating whether VISA is a participating network (required, default is N). Valid values are:

Y = Yes, VISA is a participating network.
N = No, VISA is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

AFFN

A value indicating whether AFFN is a participating network (required, default is N). Valid values are:

Y = Yes, AFFN is a participating network.
N = No, AFFN is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

CIRRUS

A value indicating whether CIRRUS is a participating network (required, default is N). Valid values are:

Y = Yes, CIRRUS is a participating network.
N = No, CIRRUS is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

MASTERCARD

A value indicating whether MASTERCARD is a participating network (required, default is N). Valid values are:

Y = Yes, MASTERCARD is a participating network.
N = No, MASTERCARD is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

DISCOVER

A value indicating whether DISCOVER is a participating network (required, default is N). Valid values are:

Y = Yes, DISCOVER is a participating network.
N = No, DISCOVER is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

AMEX

A value indicating whether AMERICAN EXPRESS is a participating network (required, default is N). Valid values are:

Y = Yes, AMERICAN EXPRESS is a participating network.
N = No, AMERICAN EXPRESS is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

INTERLINK

A value indicating whether Interlink is a participating network (required, default is N). Valid values are:

Y = Yes, Interlink is a participating network.
N = No, Interlink is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

VISA DEBIT

A value indicating whether Visa Debit is a participating network (required, default is N). Valid values are:

Y = Yes, Visa Debit is a participating network.

N = No, Visa Debit is not a participating network.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

VISA SHARING

A value identifying the VISA sharing group used to check against the Institution Definition File (IDF) sharing group values. If a match exists, VISA is considered a network participant (required only if Y is entered for VISA above).

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

AFFN SHARING

A value identifying the AFFN sharing group used to check against the Institution Definition File (IDF) sharing group values. If a match exists, AFFN is considered a network participant (required only if Y is entered for AFFN above).

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

CIRRUS SHARING

A value identifying the CIRRUS sharing group used to check against the Institution Definition File (IDF) sharing group values. If a match exists, CIRRUS is considered a network participant (required only if Y is entered for CIRRUS above).

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

MASTERCARD SHARING

A value identifying the MASTERCARD sharing group used to check against the Institution Definition File (IDF) sharing group values. If a match exists, MASTERCARD is considered a network participant (required only if Y is entered for MASTERCARD above).

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

DISCOVER SHARING

A value identifying the DISCOVER sharing group used to check against the Institution Definition File (IDF) sharing group values. If a match exists, DISCOVER is considered a network participant (required only if Y is entered for DISCOVER above).

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

AMEX SHARING

A value identifying the AMERICAN EXPRESS sharing group used to check against the Institution Definition File (IDF) sharing group values. If a match exists, AMERICAN EXPRESS is considered a network participant (required only if Y is entered for AMERICAN EXPRESS above).

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

STANDARD INTERFACE

A value indicating whether the PLUS ISO Interface process is directly connected to PLUS or is connected through the Series 1 Interface. Valid values are:

Y = Yes, the interface is connected directly to PLUS.

N = No, the interface is connected to PLUS through the Series 1.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: Y

GMT OFFSET

The number of minutes that local time varies from Greenwich Mean Time (GMT). The plus (+) and minus (-) signs are used in the first byte to indicate whether local time is later or earlier than GMT.

FIELD LENGTH: 1-3 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

COUNTRY CODE

The ISO country code of the acquirer.

FIELD LENGTH: 3 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

FOREIGN CURRENCY PARAMETERS

IN-CODE

The currency code of the issuer if it is different from 840 (U.S. currency).

FIELD LENGTH: 3 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 840

BUY RATE

The factor used to convert the transaction amount to the settlement amount. The leftmost digit denotes the number of positions the decimal is moved from the right (e.g., 69972522 = 9.972522).

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 61000000

SELL RATE

The factor used to convert the settlement amount to the cardholder amount. The leftmost digit denotes the number of positions the decimal is moved from the right (e.g., 69972522 = 9.972522).

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 61000000

PLUS ISO/Interlink Services File (PISF)

This file will only be present if the PLUS INTERLINK GATEWAY has been purchased.

PLUS users must create a PLUS ISO/Interlink Services File (PISF) record containing valid services for transactions that are to be sent to Interlink for authorization. This is done via the PISF screen. To display this screen, the PISF function must be selected from the PLUS ISO Menu screen, which is accessed from Screen 12 of the ICFE.

The screen is shown below, followed by descriptions of its fields:

BASE24-PISO	PLUS ISO/INTERLINK	<ln>	<date>	<time>	01 OF 01
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INTERLINK SERVICES

—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—
—	—	—	—	—	—

***** BASE24 *****

FILE DESTINATION:	NEW LOGICAL NETWORK ID:
F12-HELP	

INTERLINK SERVICES

The services that are eligible for Interlink. This field occurs up to thirty times and consists of two parts. Part one is a 2-character service field; part two is a 16-character description field containing a text description of the service. For valid service codes, refer to the "BASE24 Base Files Maintenance Manual." A value of ** in the 2-character service portion of this field indicates that all services are eligible for Interlink.

Part 1

FIELD LENGTH:	2 alphanumeric characters
REQUIRED FIELD:	Yes, if processing for Interlink
DEFAULT VALUE:	None

Part 2

FIELD LENGTH: 16 alphabetic characters
REQUIRED FIELD: System-generated
DEFAULT VALUE: None

KEYF SETTINGS

Each interface process must have its own Key File (KEYF) record. The following describes how the KEYF must be configured for the PLUS ISO Interface.

The KEYF screens are not depicted; however, the fields are described as they actually appear on the Release 5.0 or above KEYF screens. Note that the PLUS ISO Interface only requires settings on screens 1 and 2.

KEYF SCREEN 1 SETTINGS

The following fields on KEYF screen 1 must be set as indicated.

DPC/FIID

The FIID of the interchange using this KEYF record. The value in this field must match the value in the INTERCHANGE FIID field on ICFE screen 1.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

INTERFACE PROCESS

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field must match the value in the PROCESS field on ICFE screen 1.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

ENCRYPT TYPE

Specifies the type of PIN management used by the interchange. Valid values are:

0 = Clear PINs
1 = Security module PIN management
2 = Software DES PIN management

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

BASE24 ENCRYPT TYPE

Specifies the type of PIN management used by BASE24. Valid values are:

0 = Clear PINs
1 = Security module PIN management

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

PIN BLOCK FORMAT

Specifies the PIN block format of the PIN in the inbound and outbound external messages. Currently, PLUS uses the ANSI PIN block format for PIN encryption. This field should be set to 1. Valid values are:

0 = Clear PINs
1 = ANSI (PIN/PAN) PIN block
3 = PIN/PAD PIN block

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ANSI PAN FORMAT

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the external message's PIN/PAN PIN block. Currently, PLUS either the 12 rightmost digits, excluding the check digit or the 12 leftmost digits in the formation of an ANSI PIN block. This field should be set to 0 or 2. Valid values are:

0 = 12 right-most digits of the PAN, excluding the check digit (ANSI standard)
1 = 12 right-most digits of the PAN, including the check digit
2 = 12 left-most digits of the PAN

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes, this field is required regardless of the value in the PIN BLOCK FORMAT field.
DEFAULT VALUE: 0

MAC ENCRYPT TYPE

A code specifying how message authentication will be performed for messages between the interchange and the interface. Not used by the PLUS ISO Interface process.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

PIN PAD CHARACTER

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. Valid values are A through F.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes, this field is required regardless of the value in the PIN BLOCK FORMAT field.
DEFAULT VALUE: F

MAC DATA TYPE

A code specifying the character set being used for the data being authenticated. Not used by the PLUS ISO Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

NUMBER OF KEYS

A code identifying whether the inbound and outbound keys are combined or separate. The value in this field applies to the PIN and MAC keys defined on KEYF screen 2. Not used by the PLUS ISO Interface process. Valid values are:

1 =Combined--The inbound and outbound PIN keys are the same and the

inbound and outbound MAC keys are the same.
2 =Separate--The inbound and outbound PIN keys are different and the inbound and outbound MAC keys are different.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

KEY LENGTH

A code identifying whether single- or double-length Key Exchange Keys (KEKs) are being used with this interface. Not used by the PLUS ISO Interface. Valid values are:

1 = Single-length KEKs
2 = Double-length KEKs

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

FULL MESSAGE MAC

A code specifying whether selected data elements or the entire message will be considered when computing the MAC. Not used by the PLUS ISO Interface.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

INTERMEDIATE KEYS

Intermediate keys are used whenever a PIN which is encrypted under a secure key (security module PIN management) must be decrypted into the clear.

CLEAR

The clear version of the intermediate key. A PIN that is encrypted under the intermediate key at the security module will be decrypted in software using the value in this field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS

The check digits corresponding to the value in the ENCRYPTED INTERMEDIATE KEY field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

ENCRYPTED

Security module encrypted intermediate key. A PIN which has been encrypted under a secure key will be translated to be encrypted under the intermediate key at the security module. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 1-16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under this key for the interchange. The exchange key must be manually exchanged between the two entities.

PIN KEY

The security module encrypted form of the key used to exchange PIN keys. There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros.

FIELD LENGTH: 2 sets of 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS

The check digits corresponding to the value in the PIN KEY field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

MAC KEY

The security module encrypted form of the key used to exchange MAC keys. Not used by the PLUS ISO Interface.

FIELD LENGTH: 2 sets of 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS

The check digits corresponding to the value in the MAC KEY field. Not used by the PLUS ISO Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

KEYF SCREEN 2 SETTINGS

The following fields on KEYF screen 2 must be set as indicated.

PIN KEY INFORMATION

OUTBOUND KEYS

The following fields contain the PIN keys used for transactions sent from the interface to the interchange. If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which outbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

PIN KEY2

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the outbound PIN key has been changed by dynamic key management processing.

This counter is incremented whenever the outbound PIN key is changed by BASE24 or the co-processor. This value should require adjustment via file maintenance only if the co-processor expects a value other than zero when dynamic key management is established. Otherwise, the BASE24 Interchange (BIC) Interface process increments it automatically. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters. Not used by the PLUS ISO interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE 000000

INBOUND KEYS

The following fields contain the PIN keys used in the transactions sent by the interchange to the interface. If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

Key used by the interchange to encrypt PINs in the transactions sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which inbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

PIN KEY2

Key used by the interchange to encrypt PINs in the transactions sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the inbound PIN key has been changed by dynamic key management processing.

This counter is incremented whenever the inbound PIN key is changed by BASE24 or the co-processor. This value should require adjustment via file maintenance only if the co-processor expects a value other than zero when dynamic key management is established. Otherwise, the BASE24 Interchange (BIC) Interface process increments it automatically. Valid values for each character in this field are 0 through 9 and A through F. This field must contain six valid characters. Not used by the PLUS ISO Interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

MAC KEY INFORMATION

OUTBOUND KEYS

The following fields contain the MAC keys utilized for transactions sent from the interface to the co-network. These fields are not used by the PLUS ISO Interface.

MAC KEY1

Key used by the BASE24 interface to generate MACs in messages from BASE24 to the co-processor. Not used by the PLUS ISO Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the MAC KEY1 field. Not used by the PLUS ISO Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which outbound MAC key is currently being used by the interface. Not used by the PLUS ISO Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 1

MAC KEY2

Key used by the BASE24 interface to generate MACs in messages from BASE24 to the co-processor. Not used by the PLUS ISO Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the MAC KEY2 field. Not used by the PLUS ISO Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the outbound MAC key has been changed by dynamic key management processing. Not used by the PLUS ISO Interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

INBOUND KEYS

The following fields contain the MAC keys used in the transactions sent by the co-processor to the interface. These fields are not used by the PLUS ISO Interface.

MAC KEY1

Key used by the BASE24 interface to verify MACs generated by the interchange. Not used by the PLUS ISO Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the MAC KEY1 field. Not used by the PLUS ISO Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which inbound MAC key is currently being used by the interface. Not used by the PLUS ISO Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 1

MAC KEY2

Key used by the BASE24 interface to verify MACs generated by the interchange. Not used by the PLUS ISO Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the MAC KEY2 field. Not used by the PLUS ISO Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the inbound MAC key has been changed by dynamic key management processing. Not used by the PLUS ISO Interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 000000

KEYF SCREEN 3 SETTINGS

KEYF screen 3 contains dynamic key management parameters. The following fields on KEYF screen 3 must contain the values indicated below.

BASE24-BASE KEY FILE LLLL YY/MM/DD HH:MM 03 OF 03
DPC/FIID: INTERFACE PROCESS:

PIN KEY VARIANT:	0	MAC KEY VARIANT:	0
LIMITS			
PIN KEY TIMER VALUE:	0	MAC KEY TIMER VALUE:	0
PIN KEY TIMER INTERVAL:	M (MINS)	MAC KEY TIMER INTERVAL:	M (MINS)
PIN KEY TRAN:	0	MAC KEY TRAN:	0
PIN KEY ERROR:	0	MAC KEY ERROR:	0
CONSECUTIVE PIN KEY ERROR:	0	CONSECUTIVE MAC KEY ERROR:	0
		KMAC SYNCHRONIZATION ERROR:	0

CLEAR OLD KEY TIMER VALUE: 0 (SECONDS)

ORIGINATING ID:
RECEIVING ID:

KEY PROCESSING TYPE: N (NONE)
NOTARIZATION SUPPORTED: 0 (NOT SUPPORTED)

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

PIN KEY VARIANT

A code that specifies the key variant to be applied to the PIN key exchange key (KEK) when translating PIN keys. Valid values are:

0 =Variant 0--Translate the PIN key directly from the PIN KEK parts.
1 =Variant 1--Include the variant 1 constant with the PIN KEK parts when translating the PIN key. The Atalla variant 1 constant is hexadecimal 0800 0000 0000 0000.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

MAC KEY VARIANT

A code that specifies the key variant to be applied to the MAC key exchange key (KEK) when translating MAC keys. Valid values are: Not used by the PLUS ISO Interface.

0 =Variant 0--Translate the MAC key directly from the MAC KEK parts.
3 =Variant 3--Include the variant 3 constant with the MAC KEK parts when translating the MAC key. The Atalla variant 3 constant is hexadecimal 1800 0000 0000 0000.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

LIMITS

The following fields are used to set the parameters for dynamic key management. They are not used by the PLUS ISO Interface. One set of parameters determines when the key used for PIN encryption and translation should be changed and a second set determines when the key used for message authentication should be changed. Both types of keys can be changed based on time intervals, transaction activity, the number of errors since the last key change, or the number of consecutive errors. The MAC key can also be changed based on the number of key synchronization errors since the last key change. These limits can be used individually or in any combination. The counters and timers for the key are reset when the key is changed. If multiple limits are used, the interface will initiate a key exchange message and reset all of the counters and timers whenever any limit is exceeded. The value in the KEY PROCESSING TYPE field determines the type of key exchange message the interface will initiate.

The PIN key parameters on screen 3 must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption). The MAC key parameters on screen 3 must be set to 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

PIN KEY TIMER VALUE

The maximum length of time that a PIN key should be used.

The time period can be expressed in minutes, hours, or days, depending on the value in the PIN KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are:

- 0, 5-1500 if PIN KEY TIMER INTERVAL is M (minutes)
- 0-1000 if PIN KEY TIMER INTERVAL is H (hours)
- 0-180 if PIN KEY TIMER INTERVAL is D (days)

BASE24 checks the values in the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 changes the value in the PIN KEY TIMER INTERVAL field to the longer interval and redisplay the value in the PIN KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60. This parameter is not used if the value in this field is set to 0. This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

MAC KEY TIMER VALUE

The maximum length of time that a MAC key should be used.

The time period can be expressed in minutes, hours, or days, depending on the value in the MAC KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are:

- 0, 5-1500 if MAC KEY TIMER INTERVAL is M (minutes)
- 0-1000 if MAC KEY TIMER INTERVAL is H (hours)
- 0-180 if MAC KEY TIMER INTERVAL is D (days)

BASE24 checks the values in the MAC KEY TIMER VALUE and MAC KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 changes the value in the MAC KEY TIMER INTERVAL field to the longer interval and redisplay the value in the MAC KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60. This parameter is not used if the value in this field is set to 0. This parameter must be set to 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

PIN KEY TIMER INTERVAL

A code specifying the unit of time used with the value in the PIN KEY TIMER VALUE field to set the maximum length of time that a PIN key should be used. BASE24 changes the value in the PIN KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF record is added or updated. Refer to the PIN KEY TIMER VALUE field for additional information. Valid values are:

D = Days
H = Hours
M = Minutes

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: M

MAC KEY TIMER INTERVAL

A code specifying the unit of time used with the value in the MAC KEY TIMER VALUE field to set the maximum length of time that a MAC key should be used. BASE24 changes the value in the MAC KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF record is added or updated. Refer to the MAC KEY TIMER VALUE field for additional information. Valid values are:

D = Days
H = Hours
M = Minutes

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: M

PIN KEY TRAN

The maximum number of transactions that can be performed before changing the PIN key. Valid values are:

0 = This parameter is not used.
50-100000 = The maximum number of transactions.

This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

FIELD LENGTH: 1-6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

MAC KEY TRAN

The maximum number of transactions that can be performed before changing the MAC key. Valid values are:

0 = This parameter is not used.
50-100000 = The maximum number of transactions.

This parameter must be set to 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH: 1-6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

PIN KEY ERROR

The maximum number of PIN translation errors that can occur with the present PIN key before changing the key. Valid values are:

0 = This parameter is not used.
5-9999 = The maximum number of errors.

This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

MAC KEY ERROR

The maximum number of MAC verification errors that can occur with the present MAC key before changing the key. Valid values are:

0 = This parameter is not used.
5-9999 = The maximum number of errors.

This parameter must be set to 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CONSECUTIVE PIN KEY ERROR

The maximum number of consecutive PIN translation errors that can occur with the present PIN key before changing the key. A successful PIN translation, as well as a change in the PIN key, resets the counter associated with this limit. Valid values are:

0 = This parameter is not used.
5-9999 = The maximum number of errors.

This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CONSECUTIVE MAC KEY ERROR

The maximum number of consecutive MAC verification errors that can occur with the present MAC key before changing the key. A successful MAC verification, as well as a change in the MAC key, resets the counter associated with this limit. Valid values are:

0 = This parameter is not used.
5-9999 = The maximum number of errors.

This parameter must be set to 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

KMAC SYNCHRONIZATION ERROR

The maximum number of key synchronization errors that can occur with the present MAC key before changing the key.
Key synchronization is comparing the check digits returned from a security module for the present MAC key with the check digits stored in the KEYF record for the present MAC key. These KEYF MAC key check digit values appear in the CHECK DIGITS1 and CHECK DIGITS2 fields on KEYF screen 2. Valid values are:

0 = This parameter is not used.
3-9999 = The maximum number of errors.

This parameter must be set to 0 unless the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CLEAR OLD KEY TIMER VALUE

The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface process.

The INBOUND PIN KEY, OUTBOUND PIN KEY, INBOUND MAC KEY, and OUTBOUND MAC KEY fields on KEYF screen 2 can contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key (1 or 2) is currently in use. At the time a new key is placed in service, the other key becomes the old key. For example, if Inbound PIN Key number 1 is currently in use when BASE24 receives a new key, the new key is stored as Inbound PIN Key number 2. Inbound PIN Key number 2 becomes the current key and Inbound PIN Key number 1 becomes the old key. (If Inbound PIN Key number 2 is the current key, it becomes the old key and the new key is stored as Inbound PIN Key number 1.)

When processing a transaction, BASE24 uses the new key for PIN processing or message authentication. If the security device used by BASE24 for PIN processing or message authentication detects an error because the interchange is still using the old key, it can repeat the PIN processing or message authentication using the old key before BASE24 declines the transaction. The value in this field controls how long the old key is available. Valid values are:

0	= The old key is not cleared.
1-9999	= The maximum length of time, in seconds, that an old key can be used before being cleared.

This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption) or the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH:	1-4 numeric characters
REQUIRED FIELD:	No
DEFAULT VALUE:	0

ORIGINATING ID

The value in this field is used to determine the sender of dynamic key management messages. The values in the RECEIVING ID and ORIGINATING ID fields need to be agreed upon by the sender and receiver. They do not need to match any specific values in the BASE24 database.

FIELD LENGTH:	4-16 alphanumeric characters
REQUIRED FIELD:	Yes, if the KEY PROCESSING TYPE field contains a value other than N.
DEFAULT VALUE:	No default value

RECEIVING ID

The value in this field is used to determine the receiver of dynamic key management messages. The values in the RECEIVING ID and ORIGINATING ID fields need to be agreed upon by the sender and receiver. They do not need to match any specific values in the BASE24 database.

FIELD LENGTH: 4-16 alphanumeric characters
REQUIRED FIELD: Yes, if the KEY PROCESSING TYPE field contains a value other than N.
DEFAULT VALUE: No default value

KEY PROCESSING TYPE

A code identifying the type of dynamic key management processing this interface can perform. Valid values are:

C =Co-network--Each interface is only responsible for exchanging its outbound key.

M =Master--This interface is responsible for all key generation.

N =None--This interface does not generate or receive keys.

S =Slave--This interface receives keys, but does not generate them.

Dynamic key management requires hardware security devices for PIN management and message authentication. If the ENCRYPT TYPE and MAC ENCRYPT TYPE fields on KEYF screen 1 both are set to values other than 1, meaning security devices are not being used, the value in this field must be set to N (none).

Dynamic key management in a co-network configuration requires the use of separate keys for inbound and outbound messages. This is because each co-network process must generate the key used for its outbound messages and two processes cannot share the responsibility for generating one key. If the NUMBER OF KEYS field on KEYF screen 1 is set to 1, the value in this field cannot be set to C (co-network).

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: N

NOTARIZATION SUPPORTED

A code identifying whether this interface supports Key Notarization. BASE24 does not currently support Key Notarization, so the only valid value is 0.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

KEY6 SETTINGS

Each interface process must have its own KEY6 File (KEY6) record if Triple to Single DES encryption is required. Otherwise, the Key File (KEYF) is used. The following describes how the KEY6 must be configured for the PLUS ISO Interface.

The fields on the screens of the KEY6 are almost identical to those on the KEYF. Therefore, only the screen fields on the KEY6 that differ from those on the KEYF are described below.

KEY6 SCREEN 1 SETTINGS

MAC KEY LENGTH

A code identifying whether single- or double-length Key MAC keys are being used with this interface. Not used by the PLUS ISO Interface. Valid values are:

- 1 = Single-length MAC keys
- 2 = Double-length MAC keys

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

KEY6 SCREEN 2 SETTINGS

PIN KEY INFORMATION

OUTBOUND KEYS

The following fields contain the PIN keys used for outbound transactions (that is, transactions sent by the interface from BASE24 transaction processing system to the DPC or the interchange).

PIN KEY1

A double-length key used for the translation of PINs in transactions sent to the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

PIN KEY2

A double-length key used for the translation of PINs in transactions sent to the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These

fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is, transactions sent by the DPS or interchange to the BASE24 processing system interface).

PIN KEY1

A double-length key used by the DPC or interchange to translate PINs in the transactions sent to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

PIN KEY2

A double-length key used by the DPC or interchange to translate PINs in the transactions sent to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

KEY6 SCREEN 3 SETTINGS

MAC KEY INFORMATION

OUTBOUND KEYS

The following fields contain the MAC keys used for outbound transactions (that is, transactions sent by the interface from BASE24 transaction processing system to the DPC or co-network).

MAC KEY1

A key used by the BASE24 interface to generate MACs in messages from the BASE24 transaction processing system to the DPC or co-network. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these

fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the PLUS ISO Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

MAC KEY2

A key used by the BASE24 interface to generate MACs in messages from the BASE24 transaction processing system to the DPC or co-network. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the PLUS ISO Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

INBOUND KEYS

The following fields contain the MAC keys used in inbound transactions (that is, transactions sent by the DPC or co-network to the BASE24 interface).

MAC KEY1

A key used by the BASE24 interface to verify MACs generated by the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the PLUS ISO Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

MAC KEY2

A key used by the BASE24 interface to verify MACs generated by the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the PLUS ISO Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

AIF SETTINGS

The Acquirer Information File (AIF) contains network assigned acquirer data. One record must be placed in the AIF for each switch FIID, terminal logical network, terminal FIID and/or acquirer institution identification number that can be sent by an acquirer host in PLUS transactions to BASE24.

The AIF allows BASE24 to associate external institutions forwarding transactions to BASE24 through acquirer hosts with their PLUS-assigned PMC Ids and Acquirer Identifications. When a PLUS-bound transaction is received by BASE24 from an acquirer host, the PLUS ISO Interface process uses the switch FIID, terminal logical network, terminal FIID and acquirer institution identification number provided in the transaction to locate a corresponding record in the AIF. Data from the AIF is mapped to data elements 32 and 125.1 to be sent in the message to PLUS.

File maintenance operations on the AIF are performed using the standard BASE24 function keys for files maintenance.

ACCESSING THE AIF SCREEN

The PLUS Menu screen is accessed from screen 12 of the Interchange Configuration File (ICFE). From this screen, the AIF screen can be accessed. Provided an operator is assigned the necessary security clearance for the ICFE, the following procedures can be followed to access the AIF screen. Please refer to the "BASE24 CRT Access Manual" for an explanation of the BASE24 CRT Access system and a complete set of procedures for accessing it.

1. Complete a successful logon to BASE24 via the BASE24 Logon screen by completing the fields displayed on the screen and pressing F1. The Virtual Menu screen is then displayed.
2. Select the SWITCH FUNCTIONS option from the Virtual Menu screen and press F1.
3. Select ICFE from the Switch Functions Menu screen and press F1. Interchange Configuration File (ICFE) screen 1 is then displayed. Access to the ICFE maintenance screens can also be gained by using the FILE DESTINATION field located at the bottom of all BASE24 screens.
4. Enter PISO in the SWITCH TYPE field on ICFE screen 1 or read an existing record that has PISO as its SWITCH TYPE.
5. Go to ICFE screen 12 by pressing F11. Access to ICFE screen 12 can also be gained by entering 12 in the NEW PAGE field located at the bottom of the BASE24 screen and pressing F7.
6. Press F8 to access the PLUS Menu.
7. Position the cursor next to AIF entry using the arrow keys, the return key, or the tab key. Press F1 to select this option.

AIF SCREEN 1

The AIF screen is accessed from Screen 12 of the ICFE.

AIF screen 1 contains the switch FIID, terminal logical network, the terminal FIID, acquirer institution identification number, acquirer identification, proprietary member center identification code and sponsor identification.

If the interface is configured to utilize the AIF and an AIF record is retrieved with a non-blank PMC IDENTIFICATION CODE, the PMC IDENTIFICATION CODE of the AIF record will be mapped to data element 32 (Acquirer Institution ID) for transactions sent outbound from the interface to the network. If the interface is not configured to utilize the AIF, or an AIF record is not found or an AIF record is found with a blank PMC IDENTIFICATION CODE, the value entered in the INSTITUTION ID field on screen 1 of the ICF/ICFE will be mapped to data element 32.

If the interface is configured to utilize the AIF and an AIF record is retrieved with a non-blank ACQUIRER IDENTIFICATION, the ACQUIRER IDENTIFICATION of the AIF record will be mapped to data element 125.1 (Unique Acquirer ID) for transactions sent outbound from the interface to the network. If the interface is not configured to utilize the AIF, or an AIF record is not found or an AIF record is found with a blank ACQUIRER ID, the value in the INSTITUTION ID field on screen 1 of the ICF/ICFE will be mapped to data element 125.1.

The screen is shown below, followed by a description of its fields.

BASE24-SWI ACQ INFORMATION FILE LGNT YY/MM/DD HH:MM 01 OF 01

SWITCH FIID: _____
TERMINAL LOGICAL NETWORK: _____
TERMINAL FIID: _____
ACQUIRER INSTITUTION IDENTIFICATION NUMBER: _____

ACQUIRER IDENTIFICATION: _____
PROPRIETARY MEMBER CENTER IDENTIFICATION CODE: _____
SPONSOR IDENTIFICATION: _____

***** BASE24 *****
FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

SWITCH FIID

The FIID uniquely identifying the PLUS interchange as specified in the INTERCHANGE FIID field on screen 1 of the ICFE.

FIELD LENGTH: 4 alphanumeric characters or all '*'
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

TERMINAL LOGICAL NETWORK

The logical network of the originating terminal.

FIELD LENGTH: 4 alphanumeric characters or all '*'
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

TERMINAL FIID

The originating terminal's owner FIID.

FIELD LENGTH: 4 alphanumeric characters or all '*'
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

ACQUIRER INSTITUTION IDENTIFICATION

The terminal owner's routing/identification number.

FIELD LENGTH: 11 numeric characters or all '*'
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

ACQUIRER IDENTIFICATION

A unique acquirer identification value (Acquirer BIN) that is assigned by the network.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

PROPRIETARY MEMBER CENTER IDENTIFICATION CODE

The PLUS-assigned Proprietary Member Center (PMC) ID with which this terminal owner FIID is associated.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

SPONSOR IDENTIFICATION

An identification number that uniquely identifies the financial institution that sponsors an acquirer terminal.

FIELD LENGTH: 11 numeric characters

REQUIRED FIELD: No
DEFAULT VALUE: Spaces

Section 3 STARTUP AND CONTROL

This section provides information on the configuration of the PLUS ISO Interface process. It also describes the processing involved for initialization, network control commands, and timers.

PROCESS CONFIGURATION

The following paragraphs contain an explanation on how the PLUS ISO Interface process is configured. They provide information about the startup logic of the PLUS ISO Interface process and the method in which PLUS ISO Interface process log messages are logged.

STARTUP

Because of the length of time required for initialization, the PLUS ISO Interface process should be assigned a startup logic of AUTOMATIC in your network environment. AUTOMATIC means the PLUS ISO Interface process is started automatically when the XPNET node is initiated. This ensures that the PLUS ISO Interface process is ready when it receives its first transaction.

LOG MESSAGES

The PLUS ISO Interface process sends all of its log messages to the logging facility using a standard routing code of zero.

INITIALIZATION

The following steps are taken by the PLUS ISO Interface process when it is started. To initialize, the PLUS ISO Interface process:

1. Initializes the network.
2. Reads the name of its Logical Network Configuration File (LCONF). The startup message from XPNET contains the name of the LCONF. This name allows the PLUS ISO Interface process to access the LCONF for the assigns and params it requires for processing.
3. Opens the LCONF, reads the required assigns and params, and closes the LCONF. For a list of the required LCONF assigns and params and their descriptions, refer to section 2.
4. Reads the Log Message Configuration File (LMCF) for its modified log messages. The LMCF contains a record for each modified log message in the system. The PLUS ISO Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the PLUS ISO Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modifications.
5. Opens the Enhanced Interchange Configuration File (ICFE). If an error occurs, the PLUS ISO Interface process logs an error message and abends.
6. Using its process name, the PLUS ISO Interface process reads its own ICFE record.
7. Builds the Process Control Table (PCT). The PCT contains configuration and control information for the process. This information is read from the ICFE, and includes all of the controls required to process transactions.

8. Determines the posting dates for BASE24-atm and PLUS by comparing the current time to the BASE24-atm time taken from the LCONF.
9. Opens the Store-and-Forward File (SAF). If an error occurs, the PLUS ISO Interface process logs an error message and abends.
10. Opens the current and previous day's Interchange Log File (ILF). If an error occurs in opening the current ILF, the PLUS ISO Interface process logs an error message and abends.
11. Opens the PLUS Administrative Message (PAM) File.
12. Initializes the extended memory required for this process.
13. Opens the Key File (KEYF). If an error occurs, the PLUS ISO Interface process logs an error message and abends.
14. Using its process name, reads its own KEYF record, assigns Process Control Table (PCT) values, and then closes the KEYF.
15. Initializes the security devices.
16. Reads the Acquirer Information File (AIF) into memory, if the AIF exists.
17. Closes the ICFE.
18. If reports are required, starts a timer set to expire when the reports are to be run.
19. Set a cutover timer.
20. If the automatic logon option was selected, sends a logon message to PLUS.
21. Waits for a message from PLUS or a BASE24 process.

NETWORK CONTROL COMMANDS

Network control commands can be issued from the Network Control Supervisor's (NCS) screen or from the NCPcom prompt.

The commands are entered using the following syntax:

```
DELIVER PROCESS <process name>, "<Message Format>"
```

DEBUG

Message Format: DEBUG

Description: Places the interface process in DEBUG. This command should only be used in a TEST environment due to negative performance impacts.

DUMP

Message Format: DUMP

Description: Initiates a programmatic dump of the interface process. This command should only be used in a TEST environment or upon the request of ACI.

KEY CHANGE

Message Format: KEY B
KEY A
KEY I

Description: Initiates a key sequence. This command causes the PLUS ISO Interface process to send a network management message requesting a new working key to PLUS.
If "B" is specified in the message, the PLUS ISO Interface process sends two key change request messages to PLUS: one requesting a new acquirer working key and the other requesting a new issuer working key.
If "A" is specified in the message, the PLUS ISO Interface process sends a message to PLUS requesting only a new acquirer working key.
If "I" is specified in the message, the PLUS ISO Interface process sends a message to PLUS requesting only a new issuer working key.

KEY REPEAT

Message Format: KEY REPEAT B
KEY REPEAT A
KEY REPEAT I

Description: Initiates a key sequence. This command causes the PLUS ISO Interface process to send a network management message requesting the retransmittal of a working key to PLUS. If "B" is specified in the message, the PLUS ISO Interface process sends two retransmittal request messages to PLUS: one requesting a retransmittal of the acquirer working key and the other requesting a retransmittal of the issuer working key. If "A" is specified in the message, the PLUS ISO Interface process sends a message to PLUS requesting a retransmittal of only the acquirer working key. If "I" is specified in the message, the PLUS ISO Interface process sends a message to PLUS requesting a retransmittal of only the issuer working key.

LOGOFF

Message Format: LOGOFF
LOGOFF <station>

Description: Initiates a logoff sequence. This command causes the PLUS ISO Interface process to send a logoff message to the station specified in the message. If no additional parameter is specified in the message, the PLUS ISO Interface process sends a logoff message to PLUS on all stations.

LOGON

Message Format: LOGON

LOGON <station>

Description: Initiates a logon sequence. This command causes the PLUS ISO Interface process to send a logon message to the station specified in the message.
If no additional parameter is specified in the message, the PLUS ISO Interface process sends a logon message to PLUS on all stations.

MAINTENANCE

Message Format: MAINT <station> ON

MAINT <station> OFF

Description: This command is used to control whether the PLUS ISO Interface process responds to an interchange-initiated logon request.
If "ON" is specified in the message, the PLUS ISO Interface process does NOT respond to an interchange-initiated logon request.
If "OFF" is specified in the message, the PLUS ISO Interface process responds to an interchange-initiated logon request.

PERFORMANCE

Message Format: PERFORMANCE

Description: Requests a display of the performance statistics kept in memory by the interface process. These performance statistics are written to the log via a series of log messages generated by the process. The following is an example of a performance display:

```
*** <ATM> Performance Statistics Summary ***
Minutes in period: <#> <Outbound/Inbound> rqst limit: <#>
      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
```

STATUS

Message Format: STATUS

Description: Requests the current status of the interface process. The status information is written to the log via a series of log messages generated by the process. The following is an example of the status information:

```
LINK IS <UP/DOWN>
BASE24-atm Date:   <YYMMDD>
PLUS DATE:        <YYMMDD>
<#> Outbound Requests Pending of <#> Maximum
<#> Inbound Requests Pending of <#> Maximum
<#> Timers Active of <#> Maximum
ILF FILENAME: <ILF fname>
ILF %FULL <#>, Notify Interval <#>, Notify % <#>
SAF FILENAME: <SAF fname>
SAF %FULL <#>, Notify Interval <#>, Notify % <#>
SAF cycle in progress
PAM FILENAME: <PAM fname>
PAM %FULL <#>, Notify Interval <#>, Notify % <#>
Station <station> is <UP/DOWN> and
    Logged <ON/OFF> with <#> Timeouts
PLUS ISO Interface processing is through the
    SERIES 1
```

SWITCH LOG

Message Format: SWL <log destination>

Description: Switches the interface's primary log destination to the one specified in the command. This command should only be used in a TEST environment.

TRACE

Message Format: TRACE ON
TRACE OFF

Description: The interface process contains trace log messages that are optionally logged. The trace log messages should be requested only in a TEST environment due to negative performance impacts.
If "ON" is specified in the message, the interface starts the trace function and sends log messages to the current logging device.
If "OFF" is specified in the message, the interface stops the trace function.

WARMBOOT

Message Format: WARMBOOT
9501

Description: Warmboots the interface process. A message is logged to indicate whether the warmboot completes or aborts.

TIMERS

This section describes each timer used by the PLUS ISO Interface process. Each timer description includes the timer's function, and how and when the timer is set.

OUTBOUND REQUESTS TIMER

An outbound request timer is set each time a 0200, 0400, 0402, or 0600 message is sent to PLUS. The number of seconds used for this timer is taken from the OUTBOUND field on ICFE screen 8 for BASE24-atm transactions or ICFE screen 10 for BASE24-pos transactions. The outbound request timer is used to indicate whether a transaction sent to PLUS has timed out.

If the timer expires before a response is received from PLUS for the 0200 message, the request is denied and the appropriate performance and timeout counters are incremented. If the message type is 0400, 0402, or 0600, a message is displayed informing the user that the message timed out and should be sent again later. If the expiration of this timer causes the maximum allowed timeouts to be reached, the link to PLUS is marked down.

INBOUND REQUESTS TIMER

An inbound request timer is set each time a 0200 message is sent to a BASE24 process for authorization. The number of seconds used for this timer is taken from the INBOUND field on ICFE screen 8 for BASE24-atm transactions or ICFE screen 10 for BASE24-pos transactions. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorizing process, the request is denied. The appropriate performance and timeout counters are incremented.

STORE-AND-FORWARD TIMER

A store-and-forward timer is set each time a store-and-forward message is sent to PLUS. The number of seconds used for this timer is taken from the STORE AND FORWARD field on ICFE screen 8 for BASE24-atm transactions or ICFE screen 10 for BASE24-pos transactions. The store and forward timer is used to determine that a Store-and-Forward File (SAF) record sent to PLUS has timed out. If the timer expires before an acknowledgement is received from PLUS, the SAF timeout counter is incremented. If the SAF timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICFE screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the SAF message will be sent indefinitely until it is successfully processed.

CUTOVER TIMER

A cutover timer is set to handle cutting over the switch business date and creating an ILF for that date, if the business date has not already been changed.

NETWORK MANAGEMENT TIMER

A network management timer is set each time an 0800 message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS NETWORK MANAGEMENT field on ICFE screen 3. The network management timer is used to monitor the communications between BASE24 and PLUS. If the network management timer expires before a response is received from PLUS, the communications lines are checked and marked down, if necessary.

EXTENDED NETWORK MANAGEMENT TIMER

An extended network management timer is set each time a network management timer expires without a response being received from PLUS. The number of seconds used for this timer is taken from the TIMER LIMITS EXTENDED NETWORK field on ICFE screen 3. The network management message is repeated at the interval specified by the extended network management timer until an acknowledgement is received from PLUS or the communications line is marked up.

WAIT FOR TRAFFIC TIMER

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the TIMER LIMITS WAIT FOR TRAFFIC field on ICFE screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management echo test message is sent. The line is marked up or down, depending on the results of the echo test.

REPORT STARTUP TIMER

The report startup timer is used to start the SETL01 and STAT09 report programs. It is set at initialization and at interchange cutover. The value for this timer is taken from the SW-RPT-STARTUP param in the LCONF.

When the report startup timer expires, the obey file containing the commands to run the report programs are executed.

PERFORMANCE TIMER

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the TIMER LIMITS PERFORMANCE PERIOD field on ICFE screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the PLUS ISO Interface process:

1. Moves the performance statistics for interval four to interval five.
2. Moves the performance statistics for interval three to interval four.
3. Moves the performance statistics for interval two to interval three.
4. Moves the performance statistics for interval one to interval two.
5. Initializes the performance statistics for interval one to zero.
6. Starts a new performance timer.

Section 4 TRANSACTION HANDLING

INTRODUCTION

This section describes how the PLUS ISO Interface process handles various types of transactions. It describes the basic flow of transaction processing and provides transaction flow diagrams illustrating how messages to and from PLUS are handled.

The following transaction flows are described in this section. In most cases, these flows are described twice: once when the message flow is initiated by BASE24 and once when the message flow is initiated by PLUS.

- Requests
- Reversals/Overrides
- Timeouts
- Adjustments/Chargebacks
- Administrative
- Store-and-Forward
- Logon
- Logoff
- Echo Test
- Key Change Requests
- Settlement

NOTE: References in this section to transactions sent to PLUS for authorization implies that PLUS is sending the transaction to the appropriate authorizing entity (i.e., the PLUS Interchange does not perform authorization).

LOGGING TO THE ILF, SAF, AND PAM

Requests and reversals/overrides messages handled by the PLUS ISO Interface process are logged to the Interchange Log File (ILF). Reversals/overrides are also logged to the Store-and-Forward File (SAF). Incoming and outgoing administrative messages are logged to the PLUS Administrative Message (PAM) File.

TRACE NUMBERS

When the PLUS ISO Interface process receives an outbound transaction request, it assigns a trace number to the transaction. The trace number is sent to PLUS in Field 11 of the Standard External Message (SEM). Trace numbers are used to uniquely identify a transaction, and a new trace number is assigned to each new request or advice. Reversal transactions contain the trace number of the original transaction in Field 90 (Original Data Elements) of the message. Trace numbers begin at 000001 and are incremented by 1 until 999999 is reached. At this point, the trace number begins again at 000001.

Trace numbers must be unique, and the PLUS ISO Interface process keeps track of the range of trace numbers it is assigning by writing trace numbers periodically to a special record in the ILF (the first record in the file). The PLUS ISO Interface process updates the trace number in the ILF each time it processes the number of transactions set in the ILF MAX TRAN UPDATES field on ICFE screen 12. This way, it can keep track of the range of trace numbers it is assigning even if the process is stopped and restarted. At initialization, the PLUS ISO Interface process always reads the trace number from the ILF and increases that number by the number in the ILF MAX TRAN UPDATES field in order to arrive at the first new trace number to assign. Using this technique, the PLUS ISO Interface process can avoid creating duplicate trace numbers even in situations where the process is stopped and restarted.

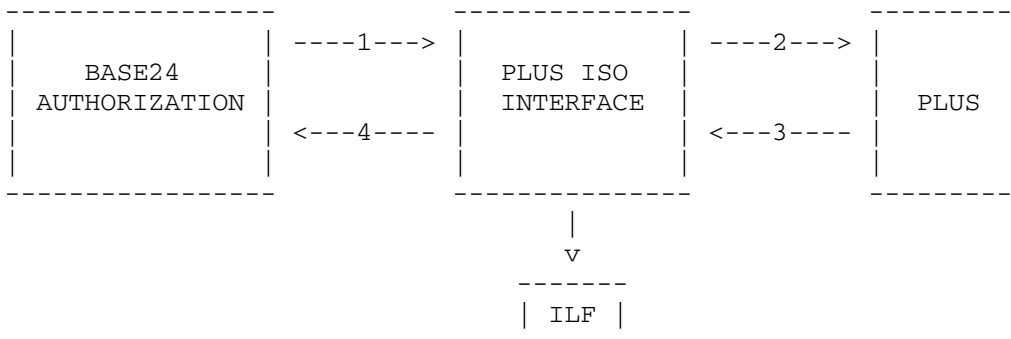
MESSAGE FLOWS FROM BASE24 TO PLUS

This subsection describes the message flows for the following transactions outbound from BASE24 to PLUS.

- Requests
- Reversals
- Timeouts
- Adjustments/Chargebacks
- Administrative
- Store-and-Forward

REQUEST MESSAGE TO PLUS

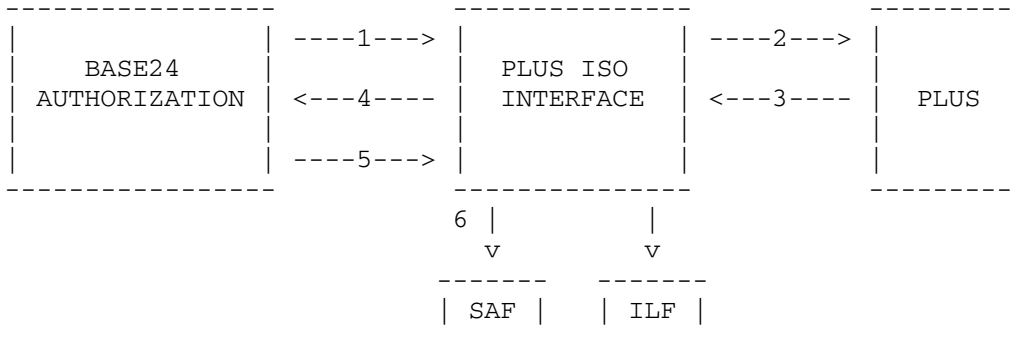
The diagram below illustrates the message routing that takes place when a transaction is initiated by the BASE24 network and must be sent to PLUS for authorization. The diagram steps are described below.



1. A BASE24 Authorization process sends the request initiated at an ATM or POS to the PLUS ISO Interface process in the 0200 internal message format.
NOTE: Authorization requests (message type 0100) are not initiated on a BASE24 network.
2. The PLUS ISO Interface process reformats the message into the 0200 external message format and sends the external message to PLUS for authorization. An outbound request timer is set to wait for a response from PLUS.
3. PLUS routes the transaction to the appropriate authorizer and receives the subsequent response. The response is formatted into the 0210 external message format and sent to the PLUS ISO Interface process.
4. The PLUS ISO Interface process deletes the outbound request timer and formats a 0210 internal response message. The response is returned to the original BASE24 Authorization process and completed at the ATM or POS. The transaction is logged to the Interchange Log File (ILF).

REVERSAL TO PLUS

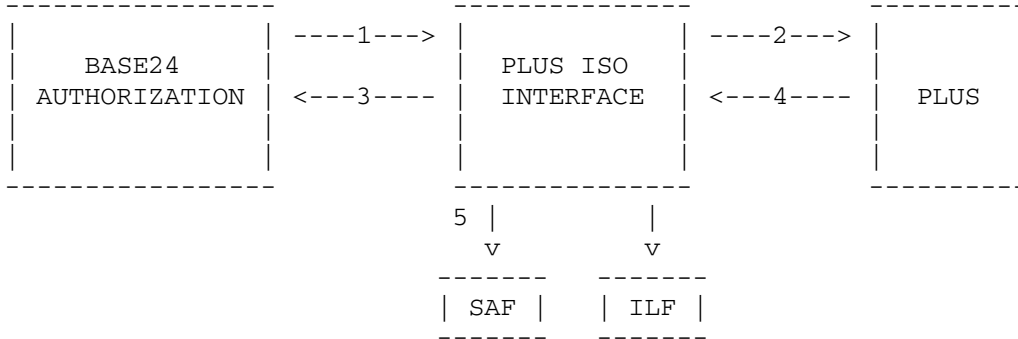
When an ATM or POS connected to the BASE24 network fails to complete a transaction as authorized, a reversal message must be sent to PLUS. The diagram below illustrates message routing when a reversal is sent to PLUS. The diagram steps are described below.



1. A BASE24 Authorization process sends the request initiated at an ATM to the PLUS ISO Interface process in the 0200 internal message format.
2. The PLUS ISO Interface process reformats the message into the 0200 external message format and sends the external message to PLUS for authorization. An outbound request timer is set to wait for a response from PLUS.
3. PLUS routes the transaction to the appropriate authorizer and receives the subsequent response. The response is formatted into the 0210 external message format and sent to the PLUS ISO Interface process.
4. The PLUS ISO Interface process deletes the outbound request timer and formats a 0210 internal response message. The response is returned to the original BASE24 Authorization process and completed at the ATM or POS. The transaction is logged to the Interchange Log File (ILF).
5. The BASE24 Authorization process returns a 0420 message to the PLUS ISO Interface process, indicating the transaction failed to complete as authorized. The PLUS ISO Interface process locates the ILF record for the original transaction using the primary key it assigned to the transaction. The PLUS ISO Interface process then updates the ILF record with the reversal information.
6. The PLUS ISO Interface process formats a 0420 external message. The external message is logged to the Store-and-Forward File (SAF) and sent to PLUS during SAF processing.

TIMEOUTS OF REQUESTS TO PLUS

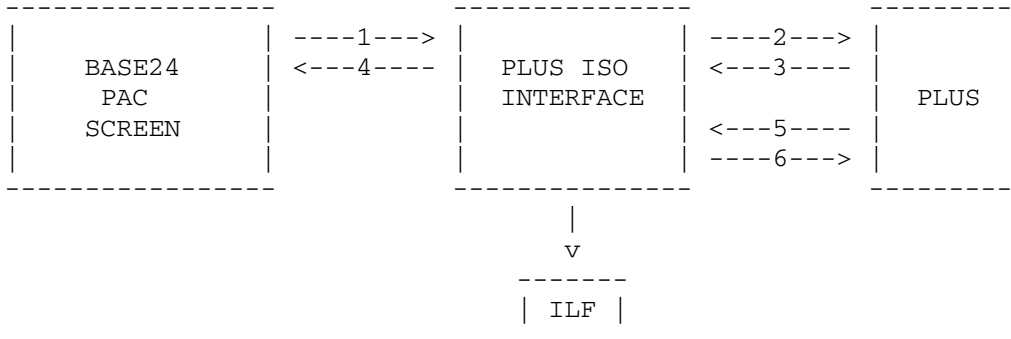
Each time the PLUS ISO Interface process sends a 0200 financial transaction request to PLUS for authorization, the PLUS ISO Interface process sets an outbound request timer. If the outbound request timer expires before a response is received from PLUS, the transaction times out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. A BASE24 Authorization process sends the request initiated at an ATM or POS to the PLUS ISO Interface process in the 0200 internal message format.
2. The PLUS ISO Interface process reformats the message into the 0200 external message format and sends the external message to PLUS for authorization. An outbound request timer is set to wait for a response from PLUS.
3. When the timer expires before a response is received from PLUS, the PLUS ISO Interface process formats a 0210 internal response message denying the transaction. The response is returned to the BASE24 Authorization process. The transaction is logged to Interchange Log File (ILF).
4. PLUS sends a 0210 late response to the PLUS ISO Interface process. If the transaction was denied by PLUS, the PLUS ISO Interface process simply drops the late response.
5. If the transaction has been approved by PLUS, the PLUS ISO Interface process formats a 0420 external reversal message and adds it to the Store-and-Forward File (SAF) to send to PLUS during SAF processing.

ADJUSTMENT/CHARGEBACK MESSAGES TO PLUS

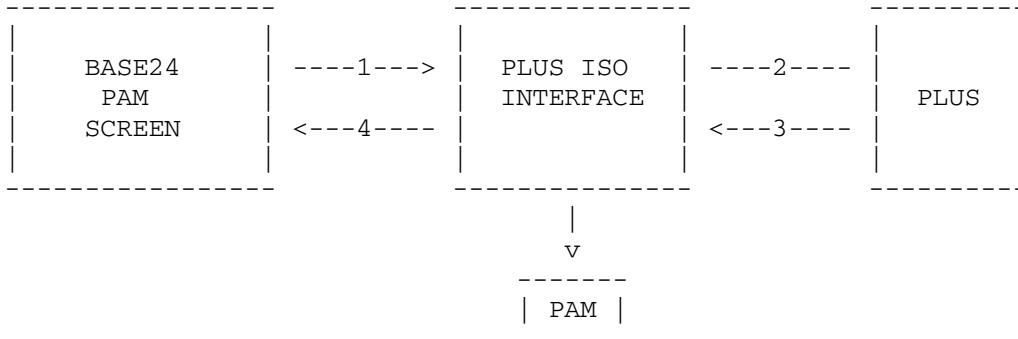
Acquirer Adjustment and Issuer Chargeback messages are entered from the BASE24 PAC screen and sent to PLUS by the PLUS ISO Interface process. The adjustment/chargeback is processed by PLUS and is not sent to another entity. The diagram below illustrates an adjustment/chargeback transaction. The steps in the diagram are described below.



1. The PLUS ISO Interface process receives an acquirer adjustment or issuer chargeback entered at the PAC screen.
2. The PLUS ISO Interface process reformats the message from PAC screen into either an acquirer adjustment (0400) or issuer chargeback (0402) and sends the external message to PLUS for authorization. An outbound request timer is set to wait for the response from PLUS. (If the timer expires before a response is received, a message is displayed indicating the message timed out and should be sent again.)
3. PLUS responds with an acquirer response (0410) or issuer response (0412) indicating PLUS received the message. The PLUS ISO Interface process deletes the outgoing request timer.
4. The PLUS ISO Interface process responds back to the PAC screen that PLUS responded to the request.
5. After the adjustment/chargeback request has been reviewed, PLUS responds back to the PLUS ISO Interface process. If the request is accepted, PLUS sends an acquirer adjustment advice (0420) or issuer chargeback advice (0422). The transaction is logged to the Interchange Log File (ILF). If the request is rejected, PLUS responds with an acquirer adjustment reject message (0480) or issuer chargeback reject message (0482). Although the 0480/0482 message is not processed, it is logged to the ILF.
6. Upon receipt of the response from PLUS, the PLUS ISO Interface process formats and sends an acknowledgement message. If the adjustment/chargeback is accepted by PLUS, the PLUS ISO Interface process responds with an acquirer adjustment acknowledgement (0430) or issuer chargeback acknowledgement (0432). If the adjustment/chargeback is rejected by PLUS, the PLUS ISO Interface responds with an acquirer adjustment reject acknowledgement (0490) or issuer chargeback reject acknowledgement (0492).

ADMINISTRATIVE MESSAGES TO PLUS

Administrative messages are entered from a BASE24 PAM screen and sent to PLUS by the PLUS ISO Interface process. The administrative message may be processed by PLUS, or PLUS may forward the message to another entity. The diagram below illustrates an administrative message flow. The steps in the diagram are described below.



1. The PLUS ISO Interface process receives an administrative message entered at the PAM screen.
2. The PLUS ISO Interface process formats an administrative message (0600) and sends the external message to PLUS. An outbound request timer is set to wait for the response from PLUS. (If the timer expires before a response is received, a message is displayed indicating the message timed out and should be sent again.)
3. PLUS responds with an administrative response (0610) indicating PLUS received the message. The PLUS ISO Interface process deletes the outbound request timer. The administrative message is written to the PLUS Administrative Message (PAM) File.
4. The PLUS ISO Interface process responds back to the PAM screen that PLUS responded to the request.

STORE-AND-FORWARD MESSAGES TO PLUS

Non-interactive messages intended for PLUS are stored in the Store-and-Forward File (SAF) and sent to PLUS during the SAF processing cycle. After the processing of a message is completed, the PLUS ISO Interface process checks to determine whether a SAF record should be sent. If the link to PLUS is up, the SAF contains records, and a SAF record is not in the process of being sent to PLUS, the PLUS ISO Interface process retrieves the first record from the SAF and sends it to PLUS. Only one message from the SAF is sent at a time. The general processing flow for SAF messages is the same as for normal messages. To PLUS, the messages appear to be happening as they are sent. Therefore, PLUS cannot distinguish them from normal messages. The PLUS ISO Interface process uses different timers that identify the messages as SAF records. The following message types can be logged to the SAF:

- 0420 Reversal/Override messages
- 0720 Acquirer Fee Advice messages
- 0722 Issuer Fee Advice messages

PLUS sends acknowledgements of these messages to the PLUS ISO Interface process. A 0430 message is an acknowledgement to the reversal/override message, a 0730 message is an acknowledgement to the acquirer fee advice message and a 0732 message is an acknowledgement to the issuer fee advice message. When the PLUS ISO Interface process receives the acknowledgement, the SAF timer is deleted and the SAF record is deleted from the SAF. The acknowledgement message from PLUS is dropped.

MESSAGE FLOWS FROM PLUS TO BASE24

This subsection describes the message flows for the following transactions inbound from PLUS to BASE24.

- Requests
- Reversals
- Timeouts
- Adjustments/Chargebacks
- Administrative messages

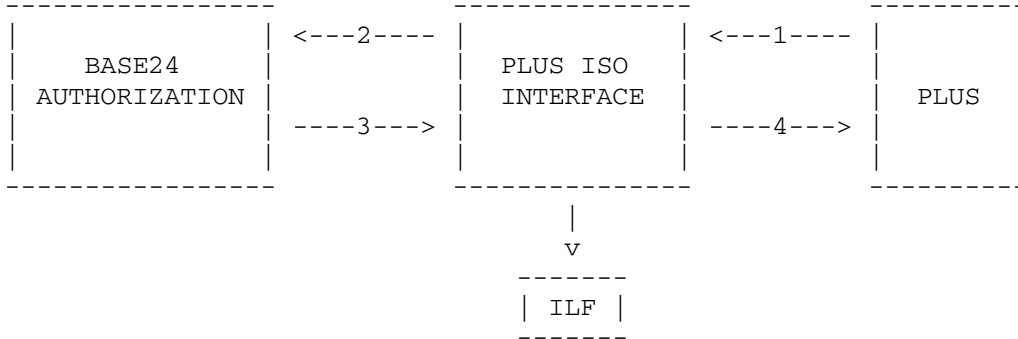
Note: When a reversal request is rejected with the code '0406', the originating member should not create an adjustment because the PLUS SYSTEM switch will automatically create an adjustment advice and forward it to both the acquirer and issuer for the same day settlement. In the event that a reversal request is rejected with any other reject code, the member must still manually create an online adjustment request and forward it to the PLUS SYSTEM switch. The PLUS ISO Interface logs a message indicating whether or not the PLUS SYSTEM switch will automatically send an adjustment advice to the acquirer and issuer for the same day settlement.

REQUEST MESSAGE FROM PLUS

The diagram below illustrates message routing when a transaction is sent from PLUS to BASE24 for authorization.

NOTE: The external message type sent from PLUS can either be 0100 or 0200. A 0100 external message is an authorization request acquired by VISA and is mapped to a 0200 internal message.

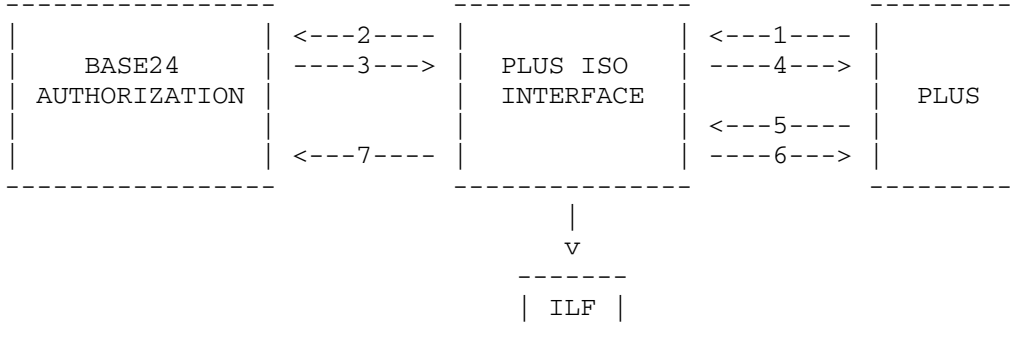
The steps in the diagram are described below.



1. PLUS sends the request to the PLUS ISO Interface process in the 0100 or 0200 external message format.
2. The PLUS ISO Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24 Authorization process for authorization. An inbound request timer is started to wait for the response.
3. The BASE24 Authorization process sends the response to the PLUS ISO Interface process in the 0210 internal message format.
4. The PLUS ISO Interface process deletes the inbound request timer and formats a 0110 or 0210 external response message. The response is returned to PLUS for completion. The transaction is logged to the Interchange Log File (ILF).

REVERSAL FROM PLUS

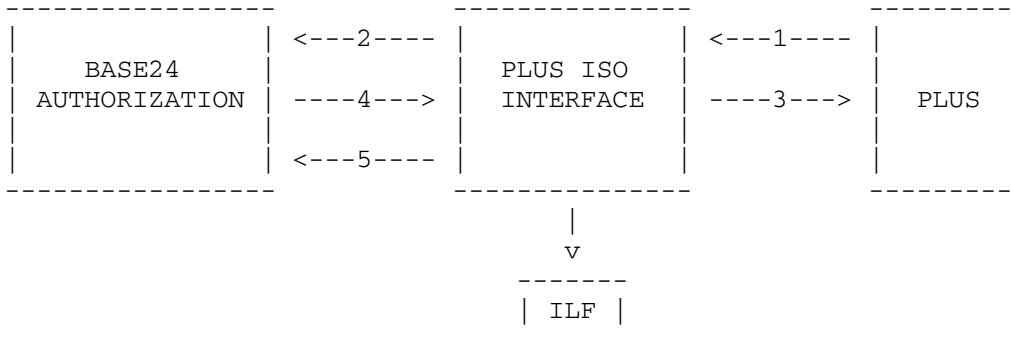
When an ATM device connected to the transaction acquirer fails to complete a transaction as authorized, a reversal message must be sent to BASE24. The steps in the diagram are described below.



1. PLUS sends the request to the PLUS ISO Interface process in the 0100 or 0200 external message format.
2. The PLUS ISO Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24 Authorization process for authorization. An inbound request timer is started to wait for the response.
3. The BASE24 Authorization process sends the response to the PLUS ISO Interface process in the 0210 internal message format.
4. The PLUS ISO Interface process deletes the inbound request timer and formats a 0110 or 0210 external response message. The response is returned to PLUS for completion. The transaction is logged to the Interchange Log File (ILF).
5. PLUS returns a 0420 message to the PLUS ISO Interface process, indicating the transaction failed to complete as authorized.
6. The PLUS ISO Interface process sends a 0430 external reversal acknowledgement to PLUS.
7. The PLUS ISO Interface process updates the ILF record for the original transaction and formats a 0420 internal message. The internal reversal is sent to a BASE24 Authorization process.

TIMEOUTS OF REQUEST MESSAGES FROM PLUS

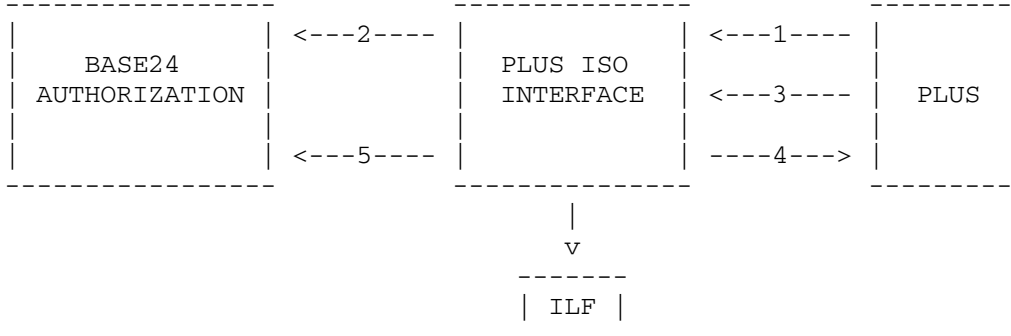
Each time the PLUS ISO Interface process sends a financial transaction to a BASE24 Authorization process for authorization, the PLUS ISO Interface process sets an inbound request timer. If the inbound request timer expires before a response is received, the transaction times out. The diagram below illustrates message routing that takes place when a financial transaction times out. The steps in the diagram are described below.



1. PLUS sends the request to the PLUS ISO Interface process in the 0100 or 0200 external message format.
2. The PLUS ISO Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24 Authorization process for authorization. An inbound request timer is set to wait for a response from the BASE24 Authorization process.
3. When the inbound request timer expires, the PLUS ISO Interface process formats a 0110 or 0210 external response message denying the transaction, and returns the response message to PLUS. The transaction is logged to the ILF.
4. The BASE24 Authorization process sends a late 0210 response to the PLUS ISO Interface process. If the transaction has been denied by the BASE24 Authorization process, the PLUS ISO Interface process simply drops the late response because denied transactions are not reversed.
5. If the transaction has been approved, the PLUS ISO Interface process formats a 0420 internal reversal message and sends the reversal to the BASE24 Authorization process.

TIMEOUTS AT PLUS

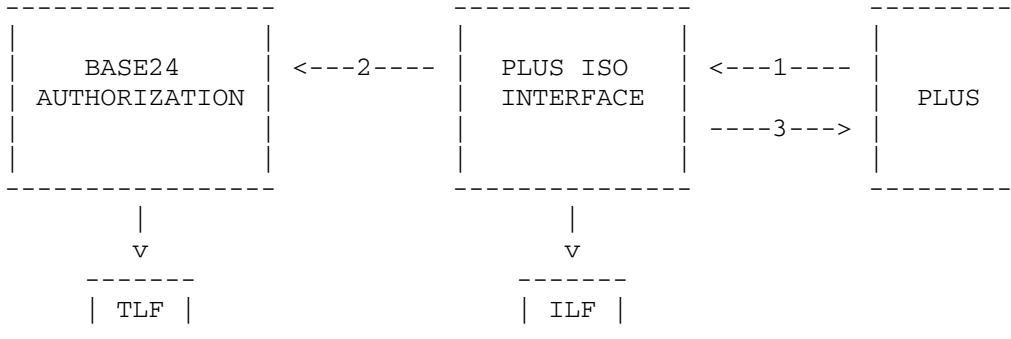
If the PLUS ISO Interface process sends a financial transaction to a BASE24 Authorization process for authorization and PLUS times out before receiving a response from BASE24, the transaction times out at PLUS. The diagram below illustrates the message routing that takes place when a financial transaction times out at PLUS. The steps in the diagram are described below.



1. PLUS sends the request to the PLUS ISO Interface process in the 0100 or 0200 external message format.
2. The PLUS ISO Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24 Authorization process for authorization. An inbound request timer is set to wait for a response from the BASE24 Authorization process.
3. PLUS times out and sends a 0120/0220 time out advice to the PLUS ISO Interface process. If the BASE24 inbound request timer still exists, it is deleted and a late response from the BASE24 Authorization process will be reversed.
4. The PLUS ISO Interface process sends a 0130/0230 external acknowledgement message to PLUS.
5. If the timer does not exist, the Interchange Log File (ILF) is searched for the transaction. If the transaction is not found in the ILF, the 0120/0220 message is dropped. If the ILF record is found and the transaction was approved, the ILF record is updated and an internal 0420 reversal message is formatted and sent to the BASE24 Authorization process.

ADJUSTMENT/CHARGEBACK MESSAGES FROM PLUS

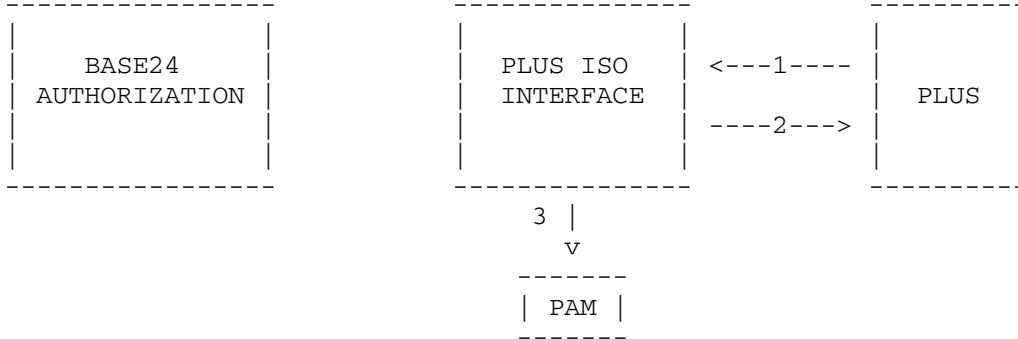
The issuer receives notification of acquirer adjustment transactions accepted by PLUS. Likewise, the acquirer receives notification of issuer chargeback transactions accepted by PLUS. The diagram below illustrates the message routing when adjustment/chargeback notifications are received by BASE24. The steps in the diagram are described below.



1. PLUS sends notification of an acquirer adjustment (0420) or issuer chargeback (0422). The PLUS ISO Interface process logs the transaction to the Interchange Log File (ILF).
2. The PLUS ISO Interface process reformats the message into the 5400 internal message format and sends the internal message to a BASE24 Authorization process in order to be logged to the TLF.
3. The PLUS ISO Interface process sends an external adjustment acknowledgement (0430) or chargeback acknowledgement (0432) to PLUS.

ADMINISTRATIVE MESSAGES FROM PLUS

Administrative messages can be sent from PLUS to BASE24. The diagram below illustrates message routing when an administrative message is sent to BASE24. The steps in the diagram are described below.



1. PLUS sends an administrative message advice (0620) to the PLUS ISO Interface process.
2. The PLUS ISO Interface process sends an external administrative acknowledgement message (0630) to PLUS.
3. The PLUS ISO Interface process logs the transaction to the PLUS Administrative Message File (PAM).

NETWORK MANAGEMENT MESSAGE FLOWS

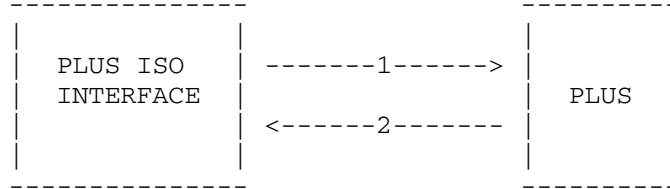
Network management messages are used to manage the operational status of the communications lines between BASE24 and PLUS. The PLUS ISO Interface process supports the following types of network management messages.

- Logon messages
- Echo test messages
- Logoff messages
- Key management messages

These messages are sent as 0800 messages and require 0810 messages in response. This section describes the message flows that occur when network management messages are sent.

LOGON/ECHO TEST INITIATED BY BASE24

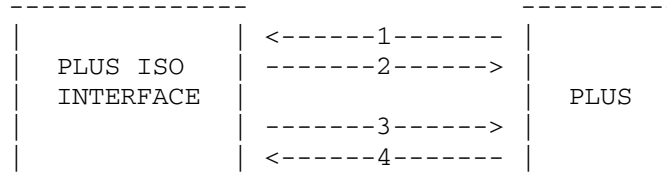
The diagram below illustrates the message flow between the PLUS ISO Interface process and PLUS when BASE24 initiates a logon or echo test message. The steps in the diagram are described below.



1. To initiate a logon, the PLUS ISO Interface process sends an 0800 logon request with a network management code of 091 to PLUS. This message is used to establish communications between BASE24 and PLUS. When a wait-for-traffic timer expires two consecutive times with no line traffic in between, the PLUS ISO Interface process sends an 0800 echo test message with a network management code of 301 to PLUS. This message is used to ensure the link is still up between the PLUS ISO Interface process and PLUS.
2. PLUS responds with an 0810 message. If the response code is set to 00, the response indicates the logon/echo test has been accepted. If the response code is set to 12, the response indicates the logon/echo test was rejected.

ECHO TEST INITIATED BY PLUS

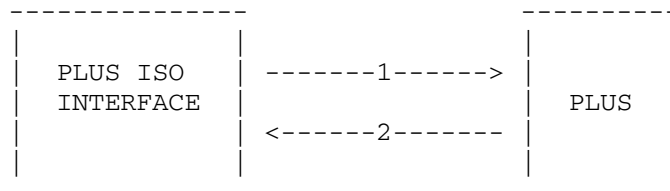
The diagram below illustrates the message flow between the PLUS ISO Interface process and PLUS when PLUS initiates an echo test message. In order for PLUS to logon to BASE24, PLUS must first send an echo test, and then BASE24 must send a logon request. The steps in the diagram are described below.



1. PLUS sends an 0800 echo test message with a network management code of 301 to the PLUS ISO Interface process. This message is used to ensure the link is still up between the PLUS ISO Interface process and PLUS.
2. The PLUS ISO Interface process sends an 0810 echo test response to PLUS.
3. The PLUS ISO Interface process sends an 0800 logon request with a network management code of 091 to PLUS. This message is used to establish communications between BASE24 and PLUS.
4. PLUS responds with an 0810 message. If the response code is set to 00, the response indicates the logon was accepted. If the response code is set to 12, the response indicates the logon was rejected.

LOGOFF INITIATED BY BASE24

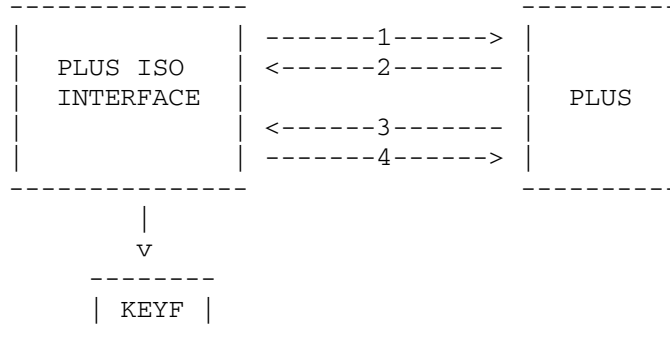
The diagram below illustrates the message flow between the PLUS ISO Interface process and PLUS when BASE24 initiates a logoff message. The steps in the diagram are described below.



1. To initiate a logoff, the PLUS ISO Interface process sends an 0800 logoff request with a network management code of 092 to PLUS. This message is used to terminate communications between BASE24 and PLUS.
2. PLUS responds with an 0810 message. If the response code is set to 00, the response indicates the logoff was accepted. If the response code is set to 12, the response indicates the logoff was rejected.

KEY CHANGE REQUEST INITIATED BY BASE24

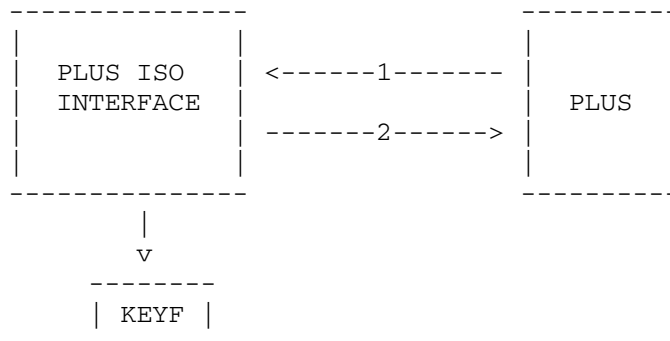
The diagram below illustrates the message flow between the PLUS ISO Interface process and PLUS when BASE24 initiates a key change request or a key retransmit request. The steps in the diagram are described below.



1. The PLUS ISO Interface process sends an 0800 message to PLUS with a network management code set to 161 for a key change request, or 162 for a key retransmit request.
2. PLUS responds with an 0810 message. If the response code is set to 00, the response indicates the key request was accepted. If the response code is set to 12, the response indicates the key request was rejected. If the request has been accepted and the request has been for a key retransmit, the 0810 message from PLUS also contains the key. The Key File (KEYF) is updated with the new key information. The flow is completed at this point for a key retransmit.
3. If the request was for a key change, PLUS sends an 0800 message containing the new key.
4. The PLUS ISO Interface process responds with an 0810 message indicating the key was received. If the key was successfully translated, the response code is 00. Otherwise, the response code is 12. The KEYF is updated with the new key information.

KEY CHANGE REQUEST INITIATED BY PLUS

The diagram below illustrates the message flow between the PLUS ISO Interface process and PLUS when PLUS initiates a key change. The steps in the diagram are described below.



1. PLUS sends an 0800 message containing the new key to the PLUS ISO Interface process.
2. The PLUS ISO Interface process responds with an 0810 message indicating the key was received. If the key was successfully translated, the response code is 00. Otherwise, the response code is 12. The Key File (KEYF) is updated with the new key information.

Section 5 CUTOVER AND REPORTING

INTRODUCTION

This section describes the PLUS ISO Interface process's cutover processing and report generation.

CUTOVER PROCESSING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports are provided to aid in reconciling the BASE24 network with PLUS. PLUS initiates cutover seven days a week at 14:00 MST (2:00 p.m.).

An ILF cutover timer will be inserted during initialization and warmboot. The PLUS ISO Interface will perform cutover when the cutover timer expires or by comparing the settlement date in the external message on request and responses.

REPORTING

BASE24's Interchange Interface processes each produce two reports of interchange transaction activity: the SETL01 Clearings Statement Report and the STAT09 Transaction Detail Report. These reports can be used in conjunction with BASE24's other reports to reconcile the differences between the BASE24 processing day and the interchange processing day.

For each network which PLUS is acting as a gateway (i.e. VISA, AFFN, CIRRUS, MASTERCARD, DISCOVER, and AMERICAN EXPRESS), separate pages are generated with totals.

A brief description is provided below for both the SETL01 and STAT09 reports. For additional information on these reports, refer to the BASE24-atm or BASE24-pos settlement and reporting manuals.

SETL01 CLEARINGS STATEMENT REPORT

The SETL01 Clearings Statement Report reflects the debit, credit, and net position of the interchange to BASE24 in U.S. currency. This report contains the totals of all approved transactions logged to the Interchange Log File (ILF).

The first section of the report provides transaction totals for the BASE24 processing day. The second section of the report provides totals for transactions that occurred during the BASE24 processing day, but not during the interchange's processing day. The third section of the report provides totals for transactions that occurred during the interchange's processing day, but not during the BASE24 processing day. Totals from each of these sections are used to compute adjusted BASE24 processing day totals that can be reconciled with the interchange's processing day totals.

STAT09 TRANSACTION DETAIL REPORT

The STAT09 Transaction Detail Report provides a complete listing of all interchange activity processed through BASE24. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the Interchange Interface process during its processing day. The information provided includes the cardholder number, transaction type, transaction time, posting date, and transaction amount. If a non currency dispense transaction has occurred, the NCD description and quantity will also be reported.

STARTING REPORTS

The interface process is responsible for starting the interchange reports. It does so by creating an obey file to start the report programs and then running the obey file at a specified time.

The interface process gets the information it needs for creating and running the obey file from the LCONF. If the report assigns and params are not included in the LCONF, the interface reports will not be run. These assigns and params are documented in an earlier section of the manual.

The interface process sets a reports timer to expire based on the SW-RPT-STARTUP param in the LCONF. When the timer expires, the interface process creates and runs an obey file containing the commands necessary to run the report programs from Tandem's TACL.

Under normal circumstances, the report programs start and stop without operator intervention. However, if the interface process does not start the programs, they can be started manually from a TACL prompt by running the report obey file created by the Interface process. If the report programsabend or fail, they can be restarted in the same way.

Section 6 BASE24 CRT OPERATIONS

INTRODUCTION

BASE24 supports several PLUS-specific operations that are performed from BASE24 CRT screens. These operations include:

- Creating and sending administrative messages to PLUS
- Perusing administrative messages sent to or received from PLUS
- Creating and sending acquirer adjustment transactions to PLUS
- Creating and sending issuer chargeback transactions to PLUS
- Creating and sending acquirer fee advice transactions to PLUS
- Creating and sending issuer fee advice transactions to PLUS
- Peruse the Maintenance Log File
- Configure up to 16 station pairs on which messages are sent to/from PLUS

PLUS OPERATIONS SCREENS

There are two sets of BASE24 screens used to perform PLUS-specific operations. These are:

- PLUS Administrative File (PAM) Screens - the PAM screens are used to create and send administrative messages to PLUS and also to peruse administrative messages stored in the PLUS Administrative Message File (PAM).
- PLUS Adjustment/Chargeback (PAC) Screen - the PAC screen is used to create and send acquirer adjustment and issuer chargeback transactions to PLUS. It also sends Interlink adjustments and chargebacks.
- Station Name File (SNF) Screen - the SNF screen is used to configure up to 16 pairs of stations on which message are sent to and received from the PLUS Interchange.
- PLUS Maintenance Log File (MLF) Screen - the MLF screen is used to allow issuer file update requests to be sent to PLUS. Responses received from PLUS are logged to the MLF which can then be perused.

Section 6 describes how to use the PAM, PAC, SNF and MLF screens. The information provided is intended for persons familiar with BASE24 CRT access and other BASE24 operations (e.g., files maintenance, network control, device control). Additional information on BASE24 CRT access is available in the "BASE24 CRT Access Manual."

ACCESSING THE PLUS OPERATION SCREENS

The PLUS Menu screen is accessed from screen 12 of the Interchange Configuration File (ICFE). From this screen, the PAM and PAC screens can be accessed. Provided an operator is assigned the necessary security clearance for the ICFE, the following procedures can be followed to access the PAM and PAC screens. Please refer to the "BASE24 CRT Access Manual" for an explanation of the BASE24 CRT Access system and a complete set of procedures for accessing it.

1. Complete a successful logon to BASE24 via the BASE24 Logon screen by completing the fields displayed on the screen and pressing F1. The Virtual Menu screen is then displayed.
2. Select the SWITCH FUNCTIONS option from the Virtual Menu screen and press F1.
3. Select ICFE from the Switch Functions Menu screen and press F1. Interchange Configuration File (ICFE) screen 1 is then displayed. Access to the ICFE maintenance screens can also be gained by using the FILE DESTINATION field located at the bottom of all BASE24 screens.
4. Enter PISO in the SWITCH TYPE field on ICFE screen 1, or read an existing record that has PISO as its SWITCH TYPE.
5. Go to ICFE screen 12 by pressing F11. Access to screen 12 can also be achieved by entering 12 in the NEW PAGE field located at the bottom of any ICFE screen and pressing F7.

BASE24 CRT Operations

6. Press F8 to access the PLUS Menu (shown below).

BASE24-SWI ICFE FILE MAINTENANCE LLLL YY/MM/DD HH:MM 01 OF 01
INTERCHANGE FIID: PISO PROCESS: P1A^PISO

- _ PLUS ADMINISTRATIVE MESSAGES (PAM)
- _ PLUS ADJUSTMENT/CHARGEBACK (PAC)
- _ ACQUIRER INFORMATION FILE (AIF)
- _ STATION NAME FILE (SNF)
- _ PLUS FILE MAINTENANCE LOG FILE (MLF)
- _ PLUS ISO/INTERLINK SERVICES FILE (PISF)

***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F1-ENTER

The PAM, PAC, SNF and MLF screens are described in detail in this section. The Acquirer Information File (AIF) screen is a file maintenance screen for the AIF and is described in section 2. The PLUS ISO/Interlink Services File (PISF) screen is a file maintenance screen for the PISF and is described in section 2.

To access the PAM screens, place the cursor to the left of the PLUS ADMINISTRATIVE MESSAGES (PAM) entry and press F1.

To access the PAC screen, place the cursor to the left of the PLUS ADJUSTMENT/CHARGEBACK (PAC) entry and press F1.

To access the SNF screen, place the cursor to the left of the STATION NAME FILE (SNF) entry and press F1.

To access the MLF screen, place the cursor to the left of the PLUS FILE MAINTENANCE LOG FILE (MLF) entry and press F1.

PLUS ADMINISTRATIVE MESSAGE FILE (PAM) SCREENS

The PLUS Administrative Message File (PAM) contains administrative messages sent to or received from PLUS by BASE24. The PAM is accessed via a set of BASE24 screens, which have two main purposes. The first is to provide a method of sending administrative messages to PLUS. The second is to allow for the perusal of administrative messages contained in the PAM.

PAM MESSAGE TYPES

The PAM can contain records for two different types of administrative messages, numbered 0 and 5. These administrative message types are described below:

- MESSAGE TYPE 0 (Free-Form Text Message) - a free-form text message to PLUS or to any PMC.
- MESSAGE TYPE 5 (Errored Transaction) - a message that was sent to PLUS that PLUS found in error.

PAM SCREENS

A different PAM screen exists for each of the different messages types (0 and 5) found in the PAM, and each of these screens is accessed from a PAM Menu screen. This section describes the PAM Menu screen and each of the other PAM screens.

The PAM screen for message type 0 can be used to send the corresponding administrative message to PLUS or peruse the corresponding message type in the PAM. The PAM screen for message type 5 can only be used to peruse the type 5 messages in the PAM.

FUNCTION KEYS

The PAM screens require the use of a special set of function key assignments. These assignments are described below for Tandem 6520/6530 terminals.

- F1 VALIDATE DATA - The F1 function key is used as a selection key from the PAM Menu screen. When the cursor is positioned next to a message type, and F1 function key is pressed, the default screen for the requested message type is displayed. The F1 function key also checks the data currently displayed on a PAM screen for errors. If the data is error-free, a "DATA O.K." response will appear on the bottom of the screen. If the data is not valid, the field in error will be highlighted and an error message will be displayed on the bottom of the screen. The F1 function key also displays default value for fields which have not been edited.
- F4 DELETE RECORD - The F4 function key is used to delete a record from the PAM. To delete a record, the record must be read using the F5 or F6 function key, and the F4 function key must be pressed twice. If a record is not read first, an error message will be displayed on the bottom of the screen. When the F4 function key is pressed, an informational message will be displayed at the bottom of the screen informing the user that the F4 function key must be pressed again to delete the displayed record.
- F5 READ NEXT OUTBOUND RECORD - The F5 function key is used to display the first outbound record in the PAM. Additional outbound records will be displayed by continuing to press the F5 function key. The outbound records are displayed sequentially from the PAM; therefore, the displayed records will jump from message type to message type. When all outbound messages have been read, an end-of-file (EOF) message will be displayed at the bottom of the screen. If the F5 function key is pressed again, the first outbound record in the PAM will be displayed.

- F6 READ NEXT INBOUND RECORD - The F6 function key is used to display the first inbound record in the PAM. Additional inbound records will be displayed by continuing to press the F6 function key. The inbound records are displayed sequentially from the PAM; therefore, the displayed records will jump from message type to message type. When all inbound messages have been read, an end-of-file (EOF) message will be displayed at the bottom of the screen. If the F6 function key is pressed again, the first inbound record in the PAM will be displayed.
- F7 SEND TO NETWORK - The F7 function key is used to send the displayed information on the PAM screen to the PLUS ISO Interchange Interface process. A data check is performed prior to the message being sent. If the data is not valid, the field in error will be highlighted and an error message will be displayed at the bottom of the screen.
- F8 CLEAR SCREEN - The F8 function key is used to clear all field data on the displayed PAM screen and replace it with default values.
- F9 GET NEXT MESSAGE TYPE - The F9 function key is used to display the NEXT message types default screen. If the "new msg type" field contains a valid value (0 or 5), the F9 function key will display the requested message type default screen.
- F10 PRINT SCREEN - The F10 function key is used to print the currently displayed screen to the designated spooler location (BASE24 Logon screen default value is \$S.#HOLD). When the F10 function key is pressed, a message including the spooler location will be displayed at the bottom of the screen.
- F11 GET PREVIOUS MESSAGE TYPE - The F11 function key will display the PREVIOUS message types default screen. If the "new msg type" field contains a valid value (0 or 5), the F11 function key will display the requested message type default screen.
- F12 HELP SCREEN - The F12 function key is used to display the help screen for the PAM screens.
- F16 EXIT OR GO TO FILE - The F16 function key is used to display the PLUS Administrative Message Menu. The F16 function key will also display a requested file providing a valid file acronym has been entered in the "FILE DESTINATION" field. If an invalid acronym is entered, an error message will be displayed at the bottom of the screen.
- SF1 LOGOFF - The shifted F16 function key is used to display the Logon screen. The shifted F16 function key will also log the user off BASE24 from the Logon screen.

NOTE: Administrative message type 5 differs from type 0; it is for perusal of inbound messages only. The only valid function keys on this screen are F4, F5, F6, F10, F12, F16, and SF16. These keys have the same functionality as described above.

PLUS ADMINISTRATIVE MESSAGE MENU SCREEN

The PLUS Administrative Message Menu screen allows the selection of an administrative message type in order to peruse PAM records or send an administrative message to PLUS. The PLUS Administrative Message screen is shown below. Selection is made by using the tab key, arrow keys, or the return key to position the cursor next to the desired message type, then pressing F1.

BASE24-SWI PLUS ADMIN INTERFACE LLLL MM/DD/YY HH:MM 01 OF 01

ADMINISTRATIVE MESSAGE TYPES

- (0) Free Form Text Message
- (5) Errored Transaction

***** BASE24 *****

FILE DESTINATION:	NEW MSG TYPE:
F1-ENTER DATA	F12-HELP

HEADER FIELDS

All PAM screens display the following fields at the top of each screen in addition to the standard BASE24 header fields. These fields are common to all message types:

```
-----
BASE24-SWI    PLUS ADMIN INTERFACE    LLLL    YY/MM/DD  HH:MM  01 OF 01
MSG TYPE: _ (XXXXXXXXXXXXXXXXX)          ORIG PMC: ____
DEST PMC: _____ SWI-TIM: 000000000 B24-TIM: _____
```

<Administrative type specific screen information>

```
***** BASE24 *****
                FILE DESTINATION:          NEW MSG TYPE:
F7-SEND TO NET          F12-HELP
-----
```

MSG TYPE

The value is set in this field when a message choice is made from the PLUS ADMIN (PAM) menu, or a value is entered in the NEW MSG TYPE field located at the bottom of the PLUS ADMIN (PAM) menu. Valid values are 0 and 5.

```
FIELD LENGTH:      1 numeric character
REQUIRED FIELD:    System-protected
DEFAULT VALUE:     None
```

ORIG PMC

The originating institution ID code. This code is assigned by PLUS to identify different institutions. This should be set to the code of the institution that originates the message.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

DEST PMC

The destination institution ID code. This code is assigned by PLUS to identify different institutions. This should be set to the final destination of the message.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

SWI-TIM

The PLUS timestamp (HHMMSSNNN) displayed in inbound messages.

FIELD LENGTH: 9 numeric characters
REQUIRED FIELD: System-protected
DEFAULT VALUE: 000000000

B24-TIM

The BASE24 timestamp (MM/DD/YY HH:MM:SS) displayed in inbound messages.

FIELD LENGTH: 17 alphanumeric characters
REQUIRED FIELD: System-protected
DEFAULT VALUE: Spaces

MESSAGE TYPE 0 SCREEN

The PLUS Administrative Message Type 0 screen allows entry of a free-form text message. It is used to send text messages from the ORIG-PMC to the DEST-PMC. The PLUS Administrative Message Type 0 screen is shown below followed by a description of its fields.

```
-----
BASE24-SWI  PLUS ADMIN INTERFACE    LLLL  YY/MM/DD  HH:MM  01 OF 01
MSG TYPE: 0 (FREE FORM TEXT)          ORIG PMC:
DEST PMC:          SWI-TIM: 000000000  B24-TIM:
```

FREE FORM TEXT MESSAGE:

TRACE DATA:

```
***** BASE24 *****
                FILE DESTINATION:          NEW MSG TYPE:
F7-SEND TO NET          F12-HELP
-----
```

FREE FORM TEXT MESSAGE

A user-defined free-form text message.

FIELD LENGTH: 100 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

TRACE DATA

Optional user data used to help trace messages between PLUS and BASE24.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

MESSAGE TYPE 5 SCREEN

The PLUS Administrative Message Type 5 screen allows perusal of data contained in an errored transaction. The PLUS Administrative Message Type 5 screen is shown below followed by a description of its fields.

```
-----
BASE24-SWI    PLUS ADMIN INTERFACE      LLLL  YY/MM/DD  HH:MM  01 OF 01
MSG TYPE: 5 (ERRORED TRANSACTION)      ORIG PMC:
DEST PMC:                SWI-TIM: 000000000  B24-TIM:
```

FORMAT RECORD:

REJECT CODE: REJECT DESCRIPTION:

```
***** BASE24 *****
FILE DESTINATION:      NEW MSG TYPE:
F12-HELP
```

FORMAT RECORD

The rejected transaction data.

```
FIELD LENGTH:      400 alphanumeric characters
REQUIRED FIELD:    System-protected
DEFAULT VALUE:     Spaces
```

REJECT CODE

A code indicating the reason for rejection. Valid values are:

0000	= Unspecified Reason
0001-0128	= Field Position Format Error
0201	= Undefined Originating ID
0202	= Undefined Forwarding ID
0203	= Undefined Destination ID
0204	= Undefined Receiving ID
0205	= Invalid Network ID
0206	= Unroutable PAN
0207	= Invalid PIN Block Format
0208	= Acquirer PMC Not Signed On
0209	= Invalid Function
0401	= Invalid PSI Reference Information
0402	= Transaction For Previous Settlement Date
0403	= Original Transaction Not Found
0404	= Invalid Original Data Elements
0405	= Duplicate Transaction
0901	= Invalid Message Type Identifier
0902	= Invalid Bit Map

FIELD LENGTH: 4 alphanumeric characters
REQUIRED FIELD: System-protected
DEFAULT VALUE: Spaces

REJECT DESCRIPTION

The descriptive message indicating the reason for rejection.

FIELD LENGTH: 15 alphanumeric characters
REQUIRED FIELD: System-protected
DEFAULT VALUE: Spaces

PLUS ADJUSTMENT/CHARGEBACK (PAC) SCREEN

The PLUS Adjustment/Chargeback (PAC) screen is used to send an acquirer adjustment or issuer chargeback message to the PLUS ISO Interface process. An acquirer adjustment message is used when the transaction acquirer wants to inform the card issuer of the actual completion amount of a transaction. An issuer chargeback is sent by the card issuer to the transaction acquirer to charge the acquirer for the transaction.

When a reversal request is rejected with the code '0406', the originating member should not create an adjustment because the PLUS SYSTEM switch will automatically create an adjustment advice and forward it to both the acquirer and issuer for the same day settlement.

FUNCTION KEYS

The PAC screen requires the use of a special set of function key assignments. These assignments are described below for Tandem 6520/6530 terminals.

- F1 VALIDATE DATA - The F1 function key is used to check the data currently displayed on the PAC screen for errors. If the data is error-free, a "DATA O.K." response will appear on the bottom of the screen. If the data is not valid, the field in error will be highlighted and an error message will be displayed on the bottom of the screen. The F1 function key will also display default values for fields which have not been edited.
- F7 SEND TO NETWORK - The F7 function key is used to send the displayed information on the PAC screen to the PLUS ISO Interchange Interface process. A data check is performed prior to the message being sent. If the data is not valid, the field in error will be highlighted and an error message will be displayed at the bottom of the screen.
- F8 CLEAR SCREEN - The F8 function key is used to clear all field data on the displayed PAC screen and replace it with default values.
- F10 PRINT SCREEN - The F10 function key is used to print the currently displayed screen to the designated spooler location (BASE24 Logon screen default value is \$S.#HOLD). When the F10 function key is pressed, a message including the spooler location will be displayed at the bottom of the screen.
- F12 HELP SCREEN - The F12 function key is used to display the help screen for the PAC screen.
- F16 EXIT OR GO TO FILE - The F16 function key is used to display the PLUS Menu. The F16 function key will also display a requested file providing a valid file acronym has been entered. If an invalid acronym is entered, an error message will be displayed at the bottom of the screen.
- SF16 LOGOFF - The shifted F16 function key will display the Logon screen from any file or menu. The shifted F16 function key will also log the user off BASE24 from the Logon screen.

All other function keys have no meaning on the PAC screen. If an invalid key is struck, an error message will appear in the bottom line of the screen.

PLUS ADJUSTMENT/CHARGEBACK SCREEN

The PLUS Adjustment/Chargeback (PAC) screen contains all of the information necessary to send either an acquirer adjustment or an issuer chargeback message to the PLUS ISO Interface process. The PLUS Adjustment/Chargeback screen is shown below followed by a description of its fields.

```
-----
BASE24-SWI    PLUS ADJ/CHARGEBACK    R513    97/04/11  14:17
MSG TYPE:    ORIGINATING PMC:    DEST PMC:    ORIGINATING INST:
ORIG TRAN INFO:  PAN:    MERCH TYP:
PROC CODE:    TRAN AMT:    TRAN FEE:
TRAN CUR CDE:    SETL AMT:    SETL FEE:
TRACE #:    SWI DAT/TIM:    COND CDE:    ENTRY CAP:
REIMB ATTRIB:    TERM ID:    LOCAL TIME:    LOCAL DATE:
CRD ACCT ID:    CRD ACCEPT NAM/LOC:
CRD ACCT CITY/ST:    CRD ACCEPT CNTRY: US TRM CNTY:
TRM ZIP:    ACQ INST:    ISS INST:
PSI TIME:    PSI DAY:    PSI DATE:    PSI LOG:
SETL RATE:    SETL DATE:    NET IND:    ACQ CNTRY CDE:
REF NUM:    ACQ TRC DATA:    DOC INFO IND:
EXCEPTION REF NUM:    MBR MSG TXT:
CORRECTION INFO:    REASON CODE:    CONTACT NAM:
PHONE NUM:    TRAN AMT:    TRAN FEE:
SETL AMT:    SETL FEE:
INTERLINK INFO:    TRACE NUM:    TRANSMISSION DATE/TIME:
GFC DATE:    GFC TEXT:
-----
```

***** BASE24 *****

FILE DESTINATION:

F7-SEND TO NET

F12-HELP

MSG TYPE

A code indicating the message type being sent or displayed. Valid values are:

A = Acquirer adjustment
C = Issuer chargeback
F = Acquirer fee advice
I = Issuer fee advice
R = Resubmissions (Interlink acquirers only)

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: Space

ORIGINATING PMC

The originating institution ID code. This is set to the PLUS assigned PMC for the institution sending the adjustment/chargeback.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000

DEST PMC

The destination institution ID code. This is set to the PLUS assigned PMC for the institution receiving the adjustment/chargeback.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000

ORIGINATING INST

The originating institution ID code of the original transaction.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000

PAN

The Primary Account Number (PAN) identifying the card. Leading and embedded spaces are not allowed.

FIELD LENGTH: 19 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

MERCH TYP

The merchant type (SIC code) from the original transaction.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 6011

PROC CODE

Processing code.

Valid Values:

<u>Position 1 and 2</u>	<u>Function</u>
02	Debit Adjustment
22	Credit Adjustment
18	Debit Adjustment of Deposit or Return
28	Credit Adjustment of Deposit or Return
19	Debit Fee
29	Credit Fee

<u>Position 3 and 4</u>	<u>From Account</u>
00	No Account Selected
10	Saving
20	Checking
30	Credit Card

<u>Position 5 and 6</u>	<u>To Account</u>
00	Not Applicable

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

TRAN AMT

The transaction amount from the original transaction. This amount represents the currency of the original transaction. Due to differences between international currencies, this number must be entered with three digits following the decimal point. Extraneous digits are truncated (e.g., 1.234 becomes 1.23 in U.S. Dollars). Valid values are 0.00 to 9999999.999.

FIELD LENGTH: 10 numeric characters plus decimal position
REQUIRED FIELD: No (Required only if different from SETL AMT)
DEFAULT VALUE: 0.000

TRAN FEE

The transaction fee amount of the original transaction. This amount represents the currency of the original transaction. Due to differences between international currencies, this number must be entered with three digits following the decimal point. Extraneous digits are truncated (e.g., 1.234 becomes 1.23 in U.S. Dollars). Valid values are -9999999.999. to 9999999.999.

FIELD LENGTH: 10 numeric characters, signed plus decimal position
REQUIRED FIELD: No (Required if different from SETL FEE)
DEFAULT VALUE: 0.000

TRAN CUR CDE

Transaction currency code from the original transaction.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 840

SETL AMT

The settlement amount from the original transaction. This is a dollar amount, and must include a decimal point and two decimal places. This amount is equivalent to TRAN AMT, taking conversion rates into account, and is always in U.S. Dollars. Valid values are 0.00 to 9999999.99.

NOTE: If TRAN CRNCY CDE equals 840 (U.S. Dollars), then this field must have the same amount as TRAN AMT.

FIELD LENGTH: 9 numeric characters plus decimal position
REQUIRED FIELD: Yes
DEFAULT VALUE: 0.00

SETL FEE

The settlement fee amount of the original transaction. This is a dollar value, and must include a decimal point and two decimal places. This amount is equivalent to TRAN FEE AMT, taking conversion rates into account, and is always in U.S. Dollars. Valid values are -9999999.999. to 9999999.999.

NOTE: If TRAN CRNCY CDE equals 840 (U.S. Dollars), then this field must have the same amount as TRAN FEE AMT.

FIELD LENGTH: 9 numeric characters, signed plus decimal position
REQUIRED FIELD: No
DEFAULT VALUE: 0.00

TRACE

The system trace audit number of the original transaction.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

SWI DAT/TIM

Transmission date (MMDD) and Transmission time (hhmmss) from the original transaction.

FIELD LENGTH: 10 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000

COND CDE

A code indicating the transaction conditions at the point of sale. Valid values are:

13 = Representation of item
17 = Chargeback
21 = Manual reversal (adjustment)
54 = Reversal of chargeback

FIELD LENGTH: 2
REQUIRED FIELD: Yes
DEFAULT VALUE: 00

ENTRY CAP

The entry capability of the terminal.

Position 1: Type of terminal

0 = Unspecified
2 = Unattended Terminal
4 = Electronic Cash Register
7 = Telephone Device
8 = Reserved

Position 2: Capability of terminal

0 = Unknown
1 = Terminal not used
2 = Magnetic stripe read capability
3 = Bar Code read capability
4 = OCR read capability
5 = Integrated circuit card read capability
9 = Terminal does not read card data

Position 3-4: Online Reason Code

00 = None
01 = Cardholder data manually entered
02 = Acquirer terminal option
03 = Account range
04 = Issuer service code
05 = Transaction type required online authorization
07 = Issuer option
10 = Risk limit A
11 = Risk limit B
12 = Risk limit C
13 = Risk limit D

20 = Random selection
21 = Expiration date
22 = Excessive use
30 = Online PIN verification
31 = Response to host "enter PIN" request
60 = Invalid data
61 = Offline card check failed
70 = Check guarantee

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REIMB ATTRIB

The reimbursement attribute indicating which reimbursement program a specific merchant is qualified to receive. Valid values are:

W = Pre-existing qualified Interlink merchant
X = Pre-existing Interlink merchant
Y = Qualified Interlink merchant
Z = Standard Interlink merchant

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes, if NETWORK IND is 06.
DEFAULT VALUE: None

TERM ID

The card acceptor terminal ID from the original transaction.

FIELD LENGTH: 8 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

LOCAL TIME

The local time, in HHMMSS format, from the original transaction.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 000000

LOCAL DATE

The local date, in MMDD format, from the original transaction.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

CRD ACCT ID

The card acceptor ID from the original transaction.

FIELD LENGTH: 15 alphanumeric characters
REQUIRED FIELD: Yes, if MSG TYPE is A,C or R
DEFAULT VALUE: Spaces

CRD ACCEPT NAM/LOC

Physical location of the ATM from the original transaction.

For fee advice transactions, the load completion signature (if available) is entered in the first 16 positions and the acquirer BIN is entered in positions 17 through 22.

FIELD LENGTH: 23 alphanumeric characters
REQUIRED FIELD: Yes, if MSG TYPE type is A,C or R
DEFAULT VALUE: Spaces

CRD ACCPT CITY/ST

City and State where the ATM is physically located from the original transaction.

FIELD LENGTH: 13 alphanumeric characters (city) 2 alphanumeric characters (state)
REQUIRED FIELD: Yes (State required only if CARD ACCEPT CNTRY is 'US' or blanks and MSG TYPE is A,C or R)
DEFAULT VALUE: Spaces

CRD ACCEPT CNTRY

Country where the ATM is physically located from the original transaction.

FIELD LENGTH: 2 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 'US'

TRM CNTY

County code where the ATM is physically located from the original transaction.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

TRM ZIP

ZIP code where the ATM is physically located from the original transaction.

FIELD LENGTH: 9 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

ACQ INST

Acquiring institution ID code from the original transaction.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

ISS INST

Issuing institution ID code from the original transaction.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

PSI TIME

The PSI (PLUS System, Inc.) timestamp (HHMMSSNNN) of the original transaction. This field may not be transmitted as all zeros.

FIELD LENGTH: 9 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000000

PSI DAY

PSI business day from the original transaction.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

PSI DATE

PSI date (MMDDYY) from the original transaction.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: Yes, if MSG TYPE is A,C or R
DEFAULT VALUE: 000000

PSI LOG

PSI transaction log address from the original transaction.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 00000000

SETL RATE

The settlement conversion rate. The leftmost digit denotes the number of positions the decimal is moved from the right (e.g., 69972522 = 9.972522). This amount is multiplied by the TRAN AMT to obtain the SETL AMT.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00000000

SETL DATE

Settlement date (MMDD) from the original transaction.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

NET IND

The PSI network indicator of the original transaction. Valid values are 00 through 15.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 00

ACQ CNTRY CDE

The code of the country in which the original transaction acquirer is located.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 840

REF NUM

The retrieval reference number from the original transaction

FIELD LENGTH: 12 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000000000

ACQ TRC DATA

The trace data from the original transaction.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

DOC INFO IND

A code that identifies the status of supporting documentation.
Valid values are:

blank = No supporting documentation required.
0 = No supporting documentation provided.
1 = Supporting documentation to follow.
2 = Invalid trace information and no supporting documentation
required or received.
3 = Invalid trace information and supporting documentation
was received.
4 = No supporting documentation received.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

EXCEPTION REF NUM

A number optionally used by acquirer for tracking adjustments
for control purposes and optionally used by issuers for matching
chargebacks between internal systems and Visa/Plus. Acquirers
are required to retain the value from a chargeback and return the
value in subsequent representments.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Zeros

MEMBER MESSAGE TEXT

Unformatted text describing an adjustment, chargeback, or
representment.

FIELD LENGTH: 33 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

CORRECTION INFO

REASON CODE

The correction reason code. Valid values for adjustments are:

2002 = ATM Non-Dispense - Credit to Cardholder
2004 = Acquirer Error - Debit to Cardholder
2006 = Reversal of Previous Credit Adjustment
2008 = Reversal of Previous Debit Adjustment
2020 = Response to Cardholder Dispute Chargeback
2300 = Deposit keying error, Credit cardholder
2301 = Deposit keying error, Debit cardholder
2302 = Deposit missing envelope
2303 = Deposited empty envelope
2304 = Deposited item non-negotiable
2305 = Deposited item up-presentable
2306 = Deposited amount differs from entered amount
2307 = Deposited foreign item
2308 = Deposited altered item
2309 = Deposited item missing written amount
2310 = Deposited item missing written date
2311 = Deposited post-dated item
2312 = Deposited stale-dated item
2313 = Deposited item missing endorsement
2314 = Deposited item missing maker's signature
2350 = Deposit not settled - Credit cardholder
2351 = Deposit did not complete - Debit cardholder
2352 = Deposited item returned - NSF
2353 = Deposited item returned - Stop payment
2354 = Deposited item returned - Account closed
2355 = Deposited item returned - No account
2356 = Deposited item returned - Refer to maker
2357 = Deposited item returned - Third party item
2358 = Deposited item has incorrect endorsement
2359 = Deposited item needs conditional endorse
2360 = Deposited item has invalid payee data
2361 = Deposit duplicate posting
2362 = Deposit cancelled by cardholder
2370 = Deposit problem occurred, Please call
2401 = Adjustment within timeframes (Interlink only)
2402 = Adjustment not within settlement limits (Interlink only)
2403 = Good faith adjustments (Interlink only)
2483 = Cardholder has the Credit Return Receipt
2484 = Cardholder does not have the Credit Return Receipt
2485 = Cardholder has the Credit Return Receipt
2486 = Cardholder does not have the Credit Return Receipt
5201 = Paper Sales Draft Submission
5205 = Paper Sales Draft Resubmission

Valid values for chargebacks are:

2480 = Invalid/Unpostable Adjustment (Interlink)
2493 = Special Merchant (Interlink)
2494 = Unauthorized Use (Interlink)

2496 = Duplicate Processing (Interlink)
2498 = Processing Error (Interlink)
2606 = Reversal of Previous Credit Chargeback
2608 = Reversal of Previous Debit Chargeback
2650 = No document received for deposit return time, CR CH
2651 = Deposit performed, not processed, CR CH
2662 = Counterfeit Transaction
2670 = Deposit problem occurred, please call
2674 = Late Presentment, Acct Closed or Insufficient Funds
2676 = Invalid Data, Currency or Tran Code

2682 = Duplicate Processing
2690 = Non-Receipt of Cash
Valid values for representations are:

2481 = Adjustment Already Issued (Interlink)
2482 = Invalid Chargeback (Interlink)
2491 = Missort (Interlink)
2495 = Documentation Not Received (Interlink)

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Zeroes

CONTACT NAM

Free form information containing name of the requester.

FIELD LENGTH: 25 alphanumeric characters
REQUIRED FIELD: Yes, if MSG TYPE is A,C or R
DEFAULT VALUE: Spaces

PHONE NUM

Free form information containing phone number of the requester.

FIELD LENGTH: 25 alphanumeric characters
REQUIRED FIELD: Yes, if MSG TYPE is A,C or R
DEFAULT VALUE: Spaces

TRAN AMT

The actual completion amount of the transaction. This field specifies the amount to be credited to or debited from the Cardholder's account. This amount represents the currency of the original transaction. Due to differences between international currencies, this number must be entered with three digits following the decimal point. Extraneous digits are truncated (e.g., 1.234 becomes 1.23 in U.S. Dollars). Valid values are 0.00 to 9999999.999.

FIELD LENGTH: 10 numeric characters plus decimal position
REQUIRED FIELD: No (Required if different from correction SETL AMT)
DEFAULT VALUE: 0.000

TRAN FEE

The completion amount of the transaction fee. This amount represents the currency of the original transaction. Due to differences between international currencies, this number must be entered with three digits following the decimal point. Extraneous digits are truncated (e.g., 1.234 becomes 1.23 in U.S. Dollars). Must be negative if positions 1 and 2 of the PROC CDE is '22' or '28'. Must be positive if positions 1 and 2 of the PROC CDE is '02' or '18'. Valid values are -9999999.999 to 9999999.999.

FIELD LENGTH: 10 numeric characters, signed plus decimal position
REQUIRED FIELD: No (Required if different from correction SETL FEE)
DEFAULT VALUE: 0.000

SETL AMT

The actual completion settlement amount from the original transaction. This is a dollar value, and must include a decimal point and two decimal places. This amount is equivalent to REPL TRAN AMT, taking conversion rates into account, and is always in U.S. Dollars. Valid values are -9999999.99 to 9999999.99.

NOTE: If TRAN CRNCY CDE equals 840 (U.S. Dollars), then this field must have the same amount as correction TRAN AMT.

FIELD LENGTH: 9 numeric characters plus decimal position
REQUIRED FIELD: Yes
DEFAULT VALUE: 0.00

SETL FEE

The completion settlement amount of the transaction fee. This is a dollar value, and must include a decimal point and two decimal places. This amount is equivalent to REPL TRAN FEE, taking conversion rates into account, and is always in U.S. Dollars. Must be negative if positions 1 and 2 of the PROC CDE is '22' or '28'. Must be positive if positions 1 and 2 of the PROC CDE is '02' or '18'. Valid values are -9999999.99 to 9999999.99.

NOTE: If TRAN CRNCY CDE equals 840 (U.S. Dollars), then this field must have the same amount as correction TRAN FEE.

FIELD LENGTH: 9 numeric characters, signed plus decimal position
REQUIRED FIELD: Yes
DEFAULT VALUE: 0.00

PLUS MAINTENANCE LOG FILE SCREENS

PLUS Maintenance Log File Screens have been added to allow issuer file update requests to be sent and the MLF to be perused. Responses received by the PLUS ISO Interface process are logged to the MLF. MLF screen 1 is used to send file update request to PLUS. The following function keys are used for screen 1:

- F2 = Send file inquiry request
- F3 = Send file add request
- F4 = Send file delete request
- F5 = Send file update request

MLF screen 2 is used to peruse the MLF. Screen 2 will display multiple records on the screen at one time. To delete a record from the MLF, position the cursor in front of the record to be deleted and press F4. The following function keys are used for screen 2:

- F1 = Read first record
- F4 = Delete record
- F6 = Read next record

MLF screen 1 and a description of its fields are shown below:

BASE24-SWI PISO MAINT LOG FILE lgnt yy/mm/dd hh:mm 01 OF 02

PLUS FILE MAINTENANCE DATA ENTRY SCREEN

SYMBOLIC NAME: (OR RN NAME AND PROCESS NUMBER)
RN NAME: PROCESS NUMBER:

ACCOUNT NUMBER: CARD ISSUER COUNTRY CODE:
ORIGINAL INST ID CODE: ACTION DATE: (YYMMDD)
FILE SECURITY CODE: FILE NAME:

DEST INST ID CODE:
PLUS: VISA ATM: MASTERCARD: AFFN: CIRRUS:
INTERLINK: DISCOVER: AMEX: VISA DEBIT: VISA BASEII:
PIN VERIFICATION INFORMATION: ALGO ID: KEY INDEX: PVV:
EXCEPTION INFORMATION: ACTION CODE: REGION CODE:
AMOUNT LIMIT: COUNT LIMIT:

SYMBOLIC NAME

The symbolic name of the PLUS ISO Interface process to which the File Update Requests must be routed.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

RN NAME

The XPNET node name used in conjunction with the process number to route File update requests when a valid symbolic name has not been entered.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

PROCESS NUMBER

The process number of the PLUS ISO Interface process used in conjunction with the XPNET node name to route Negative File update requests when a valid symbolic name has not been entered.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

ACCOUNT NUMBER

The cardholder number as encoded on Track II of the magnetic stripe of the card.

FIELD LENGTH: 28 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

CARD ISSUER COUNTRY CODE

A code indicating the country the card was issued.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes, if the ACCOUNT NUMBER starts with 59.
DEFAULT VALUE: 840

ORIGINATING INST ID CODE

The institution ID code that originated the file update request.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

ACTION DATE

The date, in YYMMDD format, that the record should be purged from the interchanges file.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

FILE SECURITY CODE

A identification code set by the originator of the file update request. Used to trace the file update request back to the requesting operator.

FIELD LENGTH: 2 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

FILE NAME

The name of the file to be updated. Valid values are:

C4 = Visa Debit PIN Verification File
E4 = Visa Debit Exception File
E3 = Visa Base I Exception File and Visa Debit Exception File

FIELD LENGTH: 2 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

DEST INST ID CODE

The institution ID code of the destination of the file update request.

FIELD LENGTH: 11 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: Spaces

PLUS

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

VISA ATM

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

MASTERCARD

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

AFFN

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

CIRRUS

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

INTERLINK

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

DISCOVER

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

AMEX

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

VISA DEBIT

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

VISA BASEII

A flag indicating if this record is valid for the indicated network.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

ALGO ID

A code that indicates the method used to verify the PIN. Valid values are:

01 = Visa PIN method
04 = IBM PIN offset

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: Yes, if FILE NAME is C4.
DEFAULT VALUE: 00

KEY INDEX

Indicates the PIN verification key pair used by the card issuer to generate the PVV.

FIELD LENGTH: 1 numeric characters
REQUIRED FIELD: Yes, if FILE NAME is C4.
DEFAULT VALUE: 0

PVV

The four digit PIN Verification Value or PIN Offset Value.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes, if FILE NAME is C4.
DEFAULT VALUE: Spaces

ACTION CODE

The response code to be used when Visa stands-in on the issuer's behalf. Valid values are:

04 = Pick up card (no fraud)
05 = Do not honor
07 = Pick up card, special condition (fraud)
11 = Approval for VIP
41 = Lost card, pick up (fraud)
43 = Stolen card, pick up (fraud)

FIELD LENGTH: 2 alphanumeric characters
REQUIRED FIELD: Yes, if FILE NAME is E4 or E3.
DEFAULT VALUE: Spaces

REGION CODE

A code defining the Card Recovery Bulletin regions that this request should be distributed.

FIELD LENGTH: 9 alphanumeric characters
REQUIRED FIELD: Yes, if FILE NAME is E4 or E3.
DEFAULT VALUE: Spaces

AMOUNT LIMIT

The dollar amount that can be approved for this cardholder in any one day and overrides the issuer's normal cardholder total daily spending limit.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 000000

COUNT LIMIT

The number of transaction that can be approved for this cardholder in any one day and overrides the issuer's normal cardholder total daily count limit.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

STATION NAME FILE (SNF)

The Station Name File contains one record for each PLUS ISO Interface process. Each process contains up to 16 station pairs on which messages are sent to and received from the PLUS Interchange.

The SNF is read at initialization and on warmboots to obtain the station names to which the PLUS ISO Interface is connected to.

ACCESSING THE SNF SCREEN

To access the SNF screen, an operator must move to screen 12 of the ICFE for the PLUS ISO Interface. From there the operator presses the F8 function key. This moves the operator to screen 1 of the PLUS Menu screen. From there the operator places the cursor on Station Name File and press the F1 function key.

interchange. This name must match the name of a station added in your network environment for the PLUS ISO Interface process and must be paired with a receive station.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

RECEIVE STATION

The symbolic name of the station on which messages will be received from the interchange. This name must match the name of a station added in your network environment for the PLUS ISO Interface process and must be paired with a send station.

NOTE: PLUS can receive messages on either LU but will only send on one LU. Therefore the BASE24 receive station must match PLUS's send LU, or the PLUS ISO Interface will not recognize the incoming message.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

TYPE

This value indicates the type of station pair. Valid values are:

A = Acquirer
I = Issuer
B = Both (Acquirer and Issuer)

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: B

SWITCH DATA

Reserved for future use.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

SWITCH INFO

Reserved for future use.

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: Spaces

ACI Worldwide Inc.

Appendix A RDT FILE

RDT FILE

This information is an example of the compile commands used by Tandem's SNAX-HLS product. The SNAX/HLS operating environment presumes that Resource Definition Table (RDT) exists. The RDT accessed by SNAX/HLS is created by compiling an edit file. The commands shown below can be used as input to the SNAX/HLS compiler to create the RDTDATA file. For further information regarding the RDTDATA, RDTTEXT and HLSRDT, see the Tandem "SNAX/HLS Configuration and Control Manual."

```
reset BIND
Bind BPISOLU1 BIND IMAGE FOR PISO DIALUP TESTING
set bind bind-type                NONNEGOTIABLE
set bind fm-profile                4
set bind ts-profile                4
set bind pri-chain-use            MULTIPLE-RU
set bind pri-rq-mode              IMMEDIATE
set bind pri-chain-protocol       ANY
set bind pri-two-phase-commit     FALSE
set bind pri-compression-ind      FALSE
set bind pri-eb-send              FLASE
set bind sec-chain-use            MULTIPLE-RU
set bind sec-rq-mode              IMMEDIATE
set bind sec-chain-protocol       ANY
set bind sec-two-phase-commit     FALSE
set bind sec-compression-ind      FALSE
set bind sec-eb-send              FALSE
set bind no-segments              FALSE
set bind fm-header-use            FALSE
set bind brkt-reset              INB
set bind brkt-term                CONDITIONAL
set bind alt-code-use             FALSE
set bind seqno-use                FALSE
set bind bis-sent                 FALSE
set bind bind-queuing             FALSE
set bind send-recv                FDUX
set bind recovery-mode            LOSER
set bind contention-winner        SECONDARY
set bind hdx-ff-reset-sender      SECONDARY
set bind pri-send-pacing-size 0
set bind sec-send-pacing-size 0
set bind sec-max-ru                1024
set bind pri-recv-pacing-size 0
set bind sec-recv-pacing-size 0
set bind pri-max-ru                1024
set bind lu-session-type          0
set bind ps-usage                  000000000000000000000000
set bind user-data                 0
set bind plu-name                  PISP
add BIND                BPISOLU1
!-----
reset PROFILE
Profile PPISOLU1 PROFILE FOR PLUS ISO DIALUP TESTING
```

```
set profile bind-name          BPISOLU1
set profile init-name          IPISO
set profile open-mode          ACQUIRE
set profile hs-polarity        SECONDARY
set profile want-sscp-text     FALSE
set profile response-type      3
set profile pipeline-lu-ind    FALSE
set profile translate          FALSE
set profile lmo                FALSE
set profile max-requests       3
set profile want-bind-data     NONE
set profile user-responses     YES
set profile want-flow-control  FALSE
set profile encode-eri         FALSE
set profile single-fmh         FALSE
set profile omit-modem-wait    FALSE
add PROFILE                    PPISOLU1
add INIT                       IPISO
```

The above information is for BASE24's support of the Secondary Logical Unit (SLU P) protocol. To configure the RDT File for BASE24's support of the Primary Logical Unit (PLU) protocol the following Profile param should be used in place of the one defined above.

```
set profile hs-polarity        PRIMARY
```

CREATING AN RDT FILE

An RDTDATA file can be created by the following set of commands.

1. Edit the source file using the following command:
 > EDIT RDTTEXT
 (Perform changes)
 *EXIT
2. Compile the RDTTEXT file using the following command:
 > RUN HLSRDT / in RDTTEXT, out \$S.#LP / RDTDATA

OPERATING SNAX/HLS

The following set of commands is used to start the new HLSOBJ. For further details, see the section on operating SNAX/HLS of the Tandem "SNAX/HLS Configuration and Control Manual" for more details.

```
> stop $hlxs
> hlsobj/in rdtdata, out $sudo, nowait, name $hlxs, pri 140/
  abend^on^error on, max^lus 32, initial^trace 15
```

CFI OBEY FILE

The following is a list of Communication Management Interface (CFI) commands to start the SNA sessions. For further information on CFI, see Tandem's "Communication Management Interface Operator Reference Manual".

The following CFI file should be used for BASE24's support of the Secondary Logical Unit (SLU P) Protocol.

```
allow all errors
assume line $snap1
abort lu under
abort pu under
abort
delete lu under
delete pu under
delete
add line,characteraset ascii,duplex full, recsize 1036, station secondary
add pu $snap1.#pu1,type(13,2),recsize 1034,maxlus 2,address %h80, window 7
add lu $snap1.#lu1,type(14,2),puname #pu1,recsize 1034, protocol snalu, address
    2, & characteraset ascii
add lu $snap1.#lu2,type(14,2),puname #pu1,recsize 1034, protocol snalu, address
    3, & characteraset ascii
start
start pu under
start lu under
```

Note - The value of addresses mentioned above were set to match the PLUS ISO Test System. These might need to be changed depending upon the configuration.

The following CMI file should be used for BASE24's support of the Primary Logical Unit (PLU) Protocol.

```
allow all errors
assume line $snap1
abort lu under
abort pu under
abort
delete lu under
delete pu under
delete
add line,character set ascii, recsize 1036,flagfill on,duplex full
add pu $snap1.#pu1,type(13,2),recsize 1034,maxlus 2,address %h80, window 7
add lu $snap1.#lu1,type(14,2),puname #pu1,recsize 1034, protocol snalu, address
    2, & character set ascii
add lu $snap1.#lu2,type(14,2),puname #pu1,recsize 1034, protocol snalu, address
    3, & character set ascii
start
start pu under
start lu under
```

Note - The value of addresses mentioned above were set to match the PLUS ISO test system. These might need to be changed depending upon the configuration.

Appendix B TOKENS

INTRODUCTION

The table below identifies the various tokens used by the PLUS ISO Interface along with their token IDs. For more information on these tokens, refer to the "BASE24 Tokens Manual".

ID	Token Name	Product
B8	Transaction Profile token	BASE24-BASE
B9	Transaction Description token	BASE24-BASE
BA	Acquirer Routing token	BASE24-BASE
BM	Transaction Subtype Token	BASE24-BASE
C4	Point of Service Data	BASE24-POS
CL	Healthcare Token	BASE24-POS

The table below identifies the various tokens used by the PLUS ISO Interface along with their token IDs. Each of these tokens is described on the following pages.

ID	Token Name	Product
B0	Switch Token (Acquirer)	BASE24-BASE
B1	Switch Token (Issuer)	BASE24-BASE

SWITCH TOKEN (TOKEN B0 AND TOKEN B1)

Described below are the fields in the PLUS Switch Token which redefines the Switch Token (swi-tnk).

<u>Position</u>	<u>Level</u>	<u>Field Name and Description</u>	<u>Data Type</u>
1-175		PISO-SWI-TKN-DATA	
1	02	LGTH Indicates the length of the switch token.	BINARY 16 SIGNED
3	02	FIID The FIID for the specific switch. For PLUS, the FIID will be "PISO".	PIC X(4).
7	02	VERSION-ID Indicates the version of the switch token. At this time the version is 06.	PIC X(2).
9	02	NETWORK-ID Bit-63 (sem.pri^rsrvd4^prvt.network^data.network^id) The Network ID identifies the originating or an interim network.	PIC X(4).
13	02	FRWD-INST-ID Bit-33 (sem.frwd^inst^id.num) This is the forwarding institution identification code. If backend PMCs are used, this identifies the acquiring agent directly attached to PSI.	PIC 9(11).
24	02	PSI-REF-INFO Bit-62 (sem.pri^rsrvd3^prvt.reference^info) This field contains a reference code set by PSI to uniquely identify a transaction.	PIC X(40).
64	02	RCV-INST-ID Bit-100 (sem.rcv^inst.id^num) This field contains the receiving institution ID code. If backend PMCs are used, this identifies the authorizing agent processor directly attached to PSI.	PIC 9(11).
75	02	TRAN-FEE Bit-29 (sem.set1^fee.amt or sem.tran^fee.amt) or Bit-28 This field contains the fee associated with this transaction in the currency of TRAN-AMT.	PIC 9(8).
83	02	RESP-CDE Bit-39 (sem.resp^cde) This field contains the response code associated with this transaction. This token field will not contain significant data in acquirer tokens, B0, when BASE24 is the authorizer.	PIC X(2).
85	02	ENTRY-MDE Bit-22 (sem.entry^mde) The PISO value for point of service entry mode.	PIC 9(3).

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88	02	TRACE-NUM Bit-11 (sem.trace^num) This field contains the message trace number associated with the transaction.	PIC 9(6).
94	02	CRD-ISS-INST-CNTRY-CDE Bit-20 (sem.crd^iss^inst^cntry^cde) This field contains the issuing institution's country code.	PIC 9(3).
97	02	ACQ-INST-CNTRY-CDE Bit-19 (sem.acq^inst^cntry^cde) This field contains the acquiring institution's country code.	PIC 9(3).
100	02	DOC-INFO-IND Subfield of Bit-48 (sem.add^data^prvt.ind07.doc^info^ind) This field contains a code that identifies the status of supporting documentation.	PIC X.
101	02	EXCPT-REF-NUM Subfield of Bit-48 (sem.add^data^prvt.ind07.excpt^ref^num) This field contains a number optionally used by acquirer for tracking adjustments for control purposes and optionally used by issuers for matching chargebacks between internal systems and Visa/Plus.	PIC 9(6).
107	02	MBR-MSG-TXT Subfield of Bit-48 (sem.add^data^prvt.ind07.mbr^msg^txt) This field contains unformatted text describing an adjustment, chargeback, or representation.	PIC X(50).
157	02	ISA-FEE Subfield of Bit-61 (sem.pri^rsrvd2^prvt.process^fee.intl) This field contains international service assessment (ISA) fee.	PIC 9.
158	02	FEE-PGM Subfield of Bit-61 or Bit-125 (sem.pri^rsrvd2^prvt.process^fee.fee^pgm or sem.secndry^rsrvd6^prvt.ilnk^info.fee^pgm) This field contains fee program.	PIC X(3).
161	02	MCCR-FEE Subfield of Bit-125 (sem.secndry^rsrvd6^prvt.ilnk^info.mccr^fee) This field contains the MasterCard Currency Conversion Rate Fee.	PIC X(12).
173	02	CRD-LVL-PROD-ID-VAL Subfield of Bit-125 This field contains the Card Level Product Identifier Value.	PIC X(2).

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